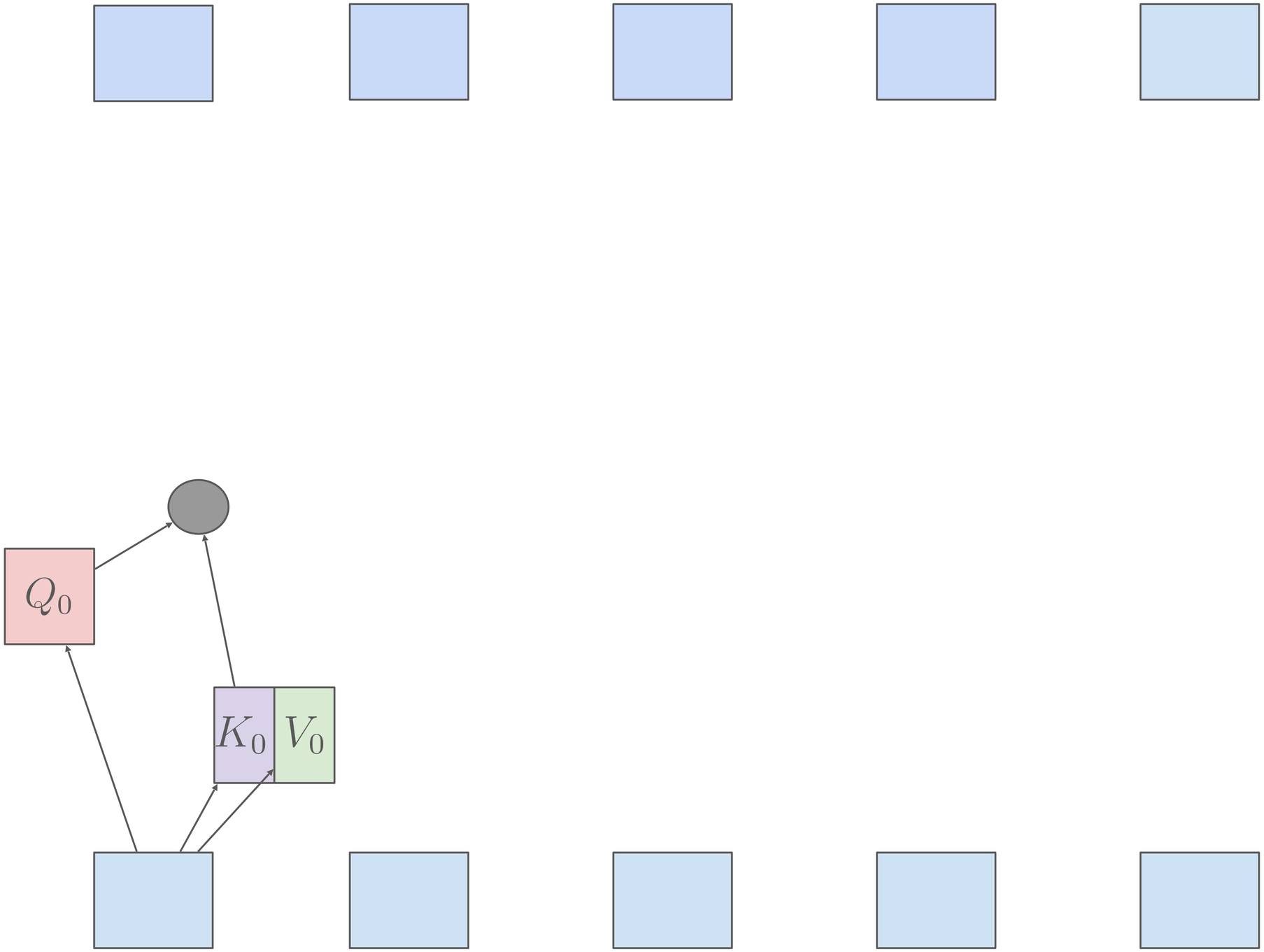
Introduction to Deep Learning

Caltech CS/EE/CMS 148a - Winter 2026

Lecture 13 - Transformers


Q_0
K_0
V_0

Where do Q, K, V come from?

I like white chocolate

Where do Q, K, V come from?

x_1

I like white chocolate

Where do Q, K, V come from?

x_1 x_2

I like white chocolate

Where do Q, K, V come from?

x_1	x_2	x_3	
I	like	white	chocolate

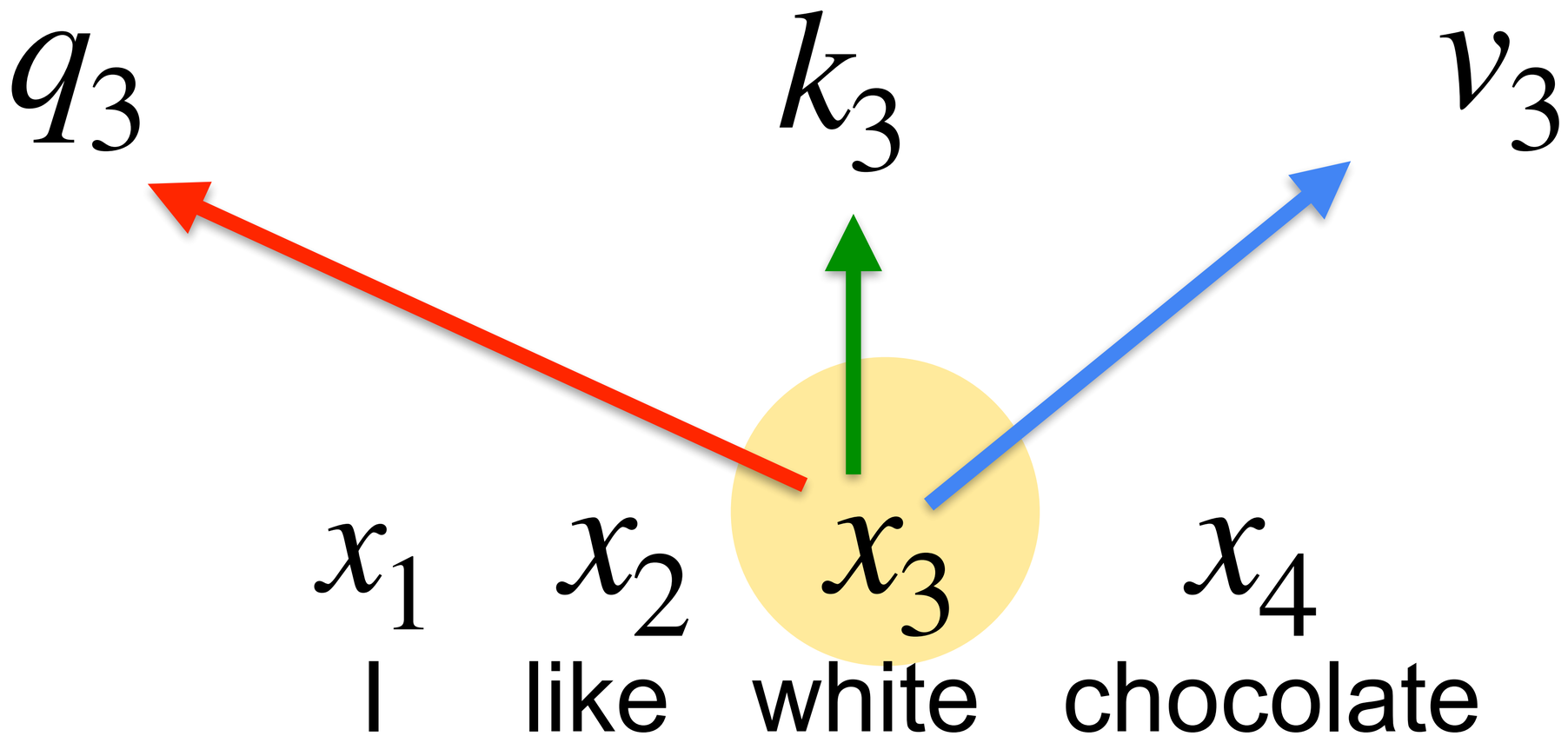
Where do Q, K, V come from?

x_1	x_2	x_3	x_4
I	like	white	chocolate

Where do Q, K, V come from?

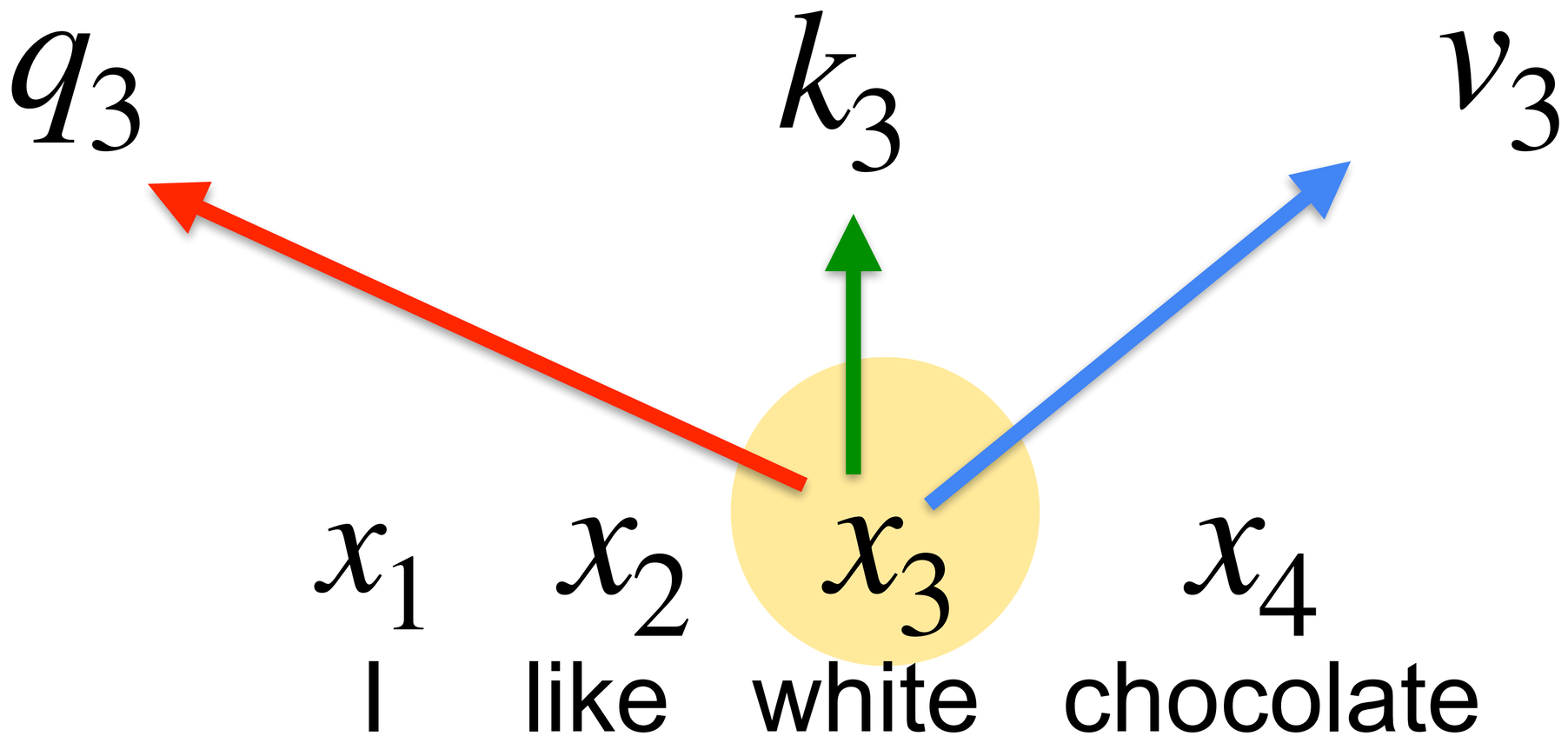
x_1	x_2	x_3	x_4
I	like	white	chocolate

Where do Q, K, V come from?



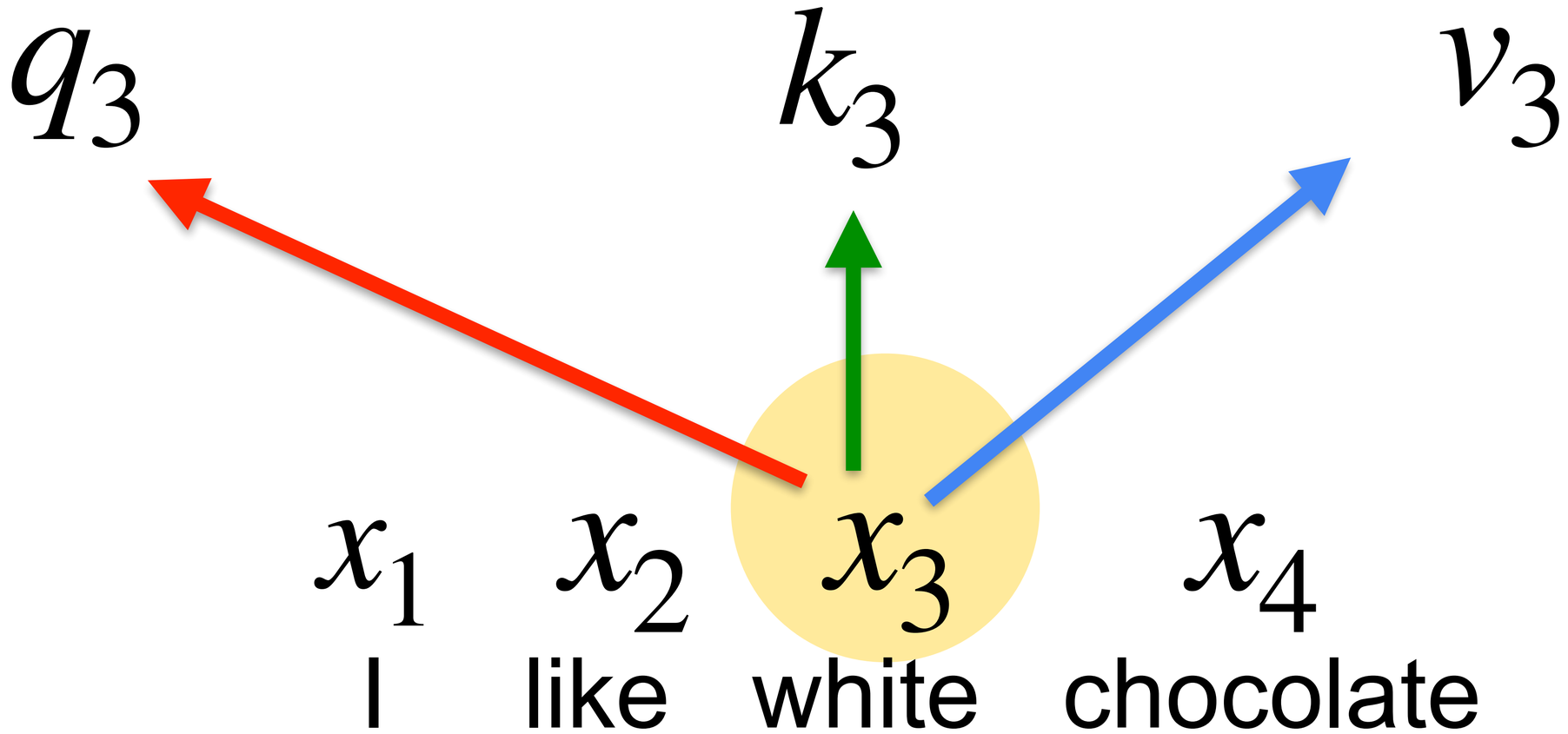
Where do Q, K, V come from?

$$q = W_Q x$$



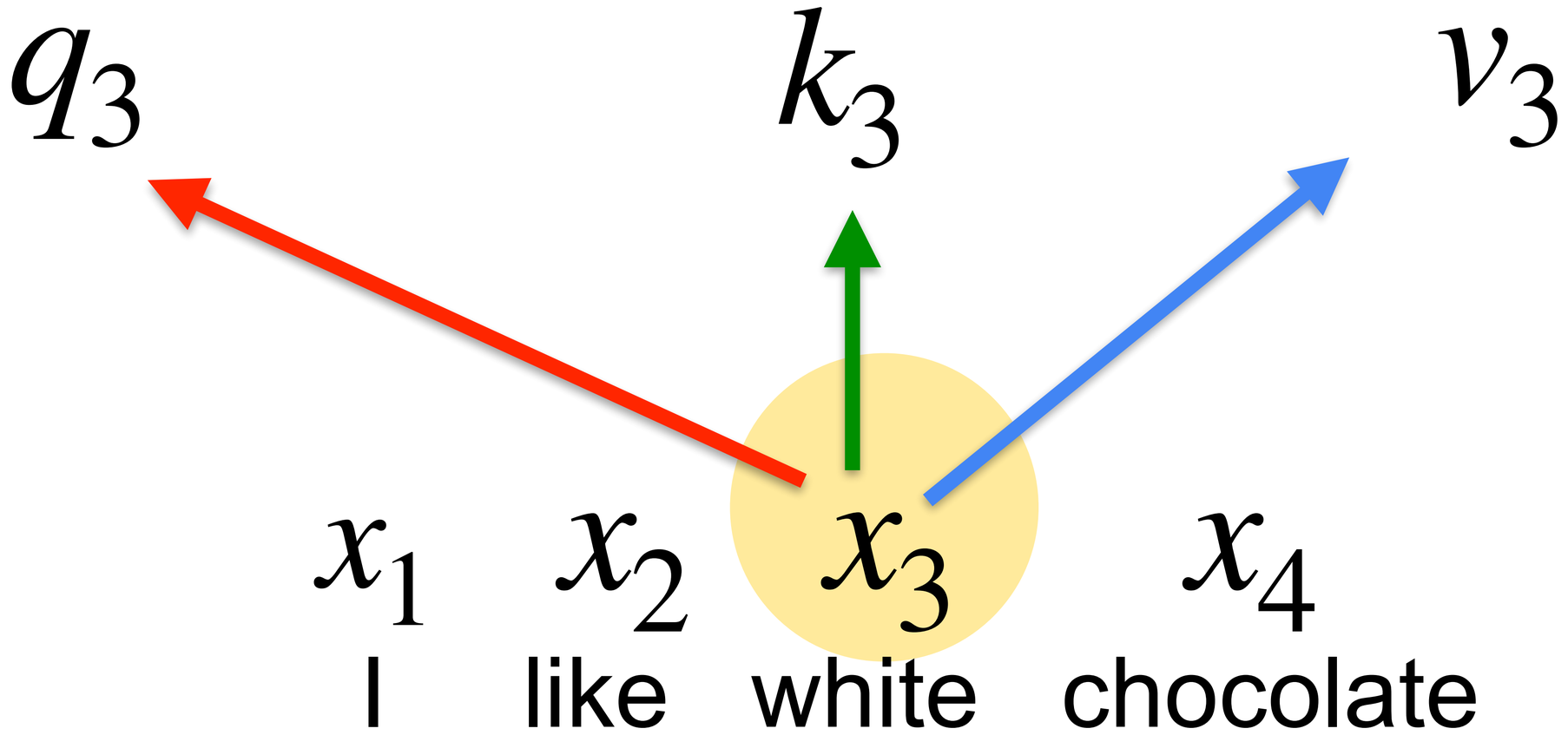
Where do Q, K, V come from?

$$q = W_Q x \qquad k = W_K x$$



Where do Q, K, V come from?

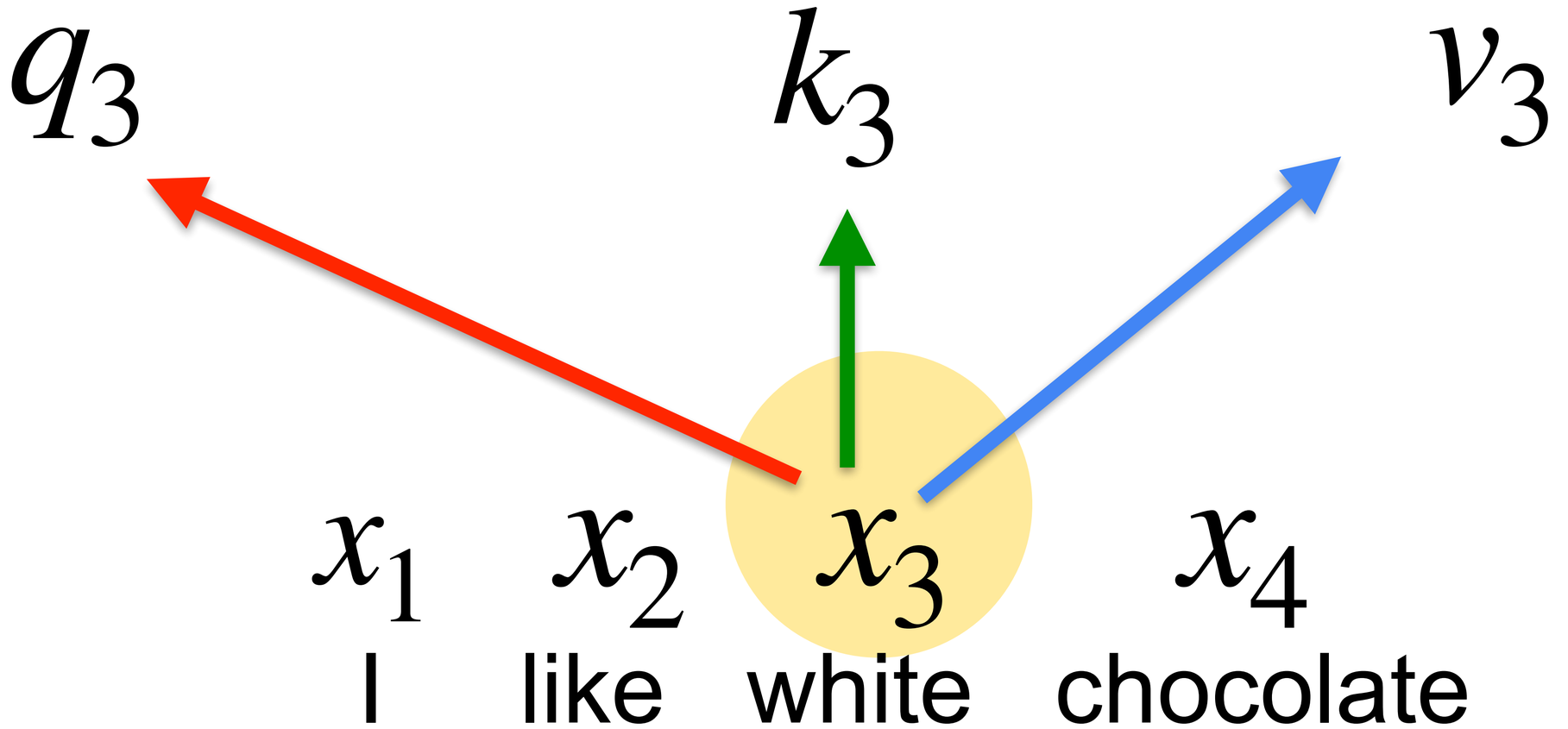
$$q = W_Q x \qquad k = W_K x \qquad v = W_V x$$

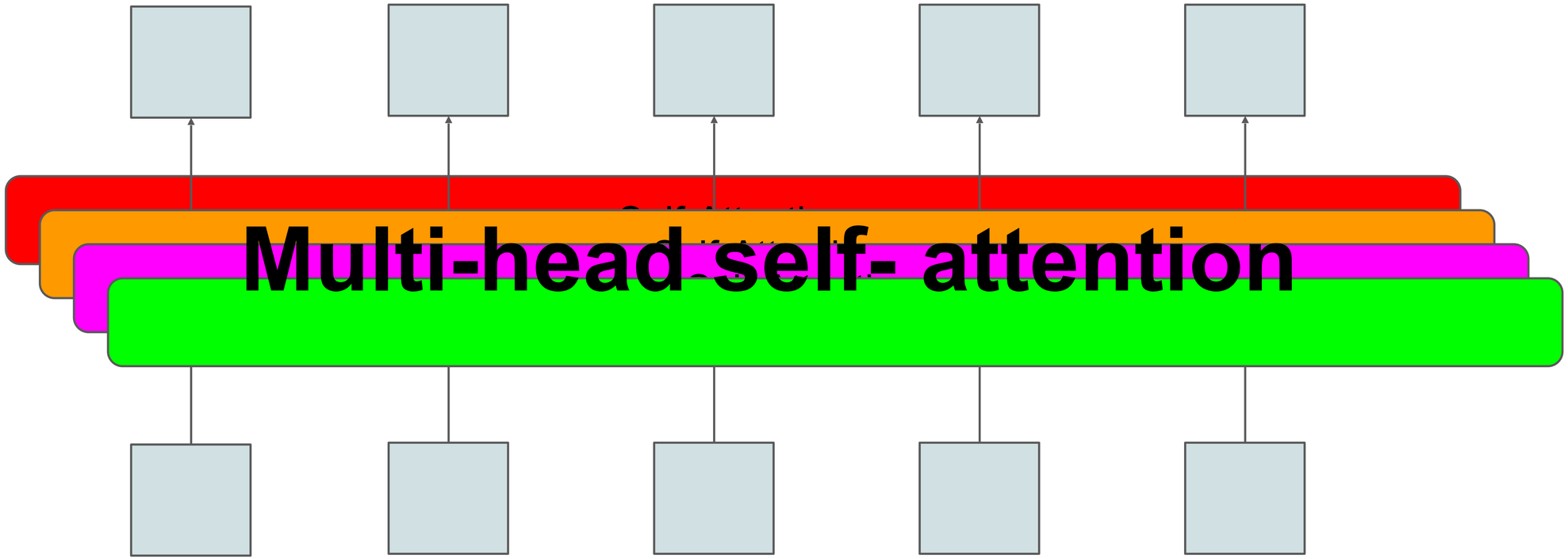


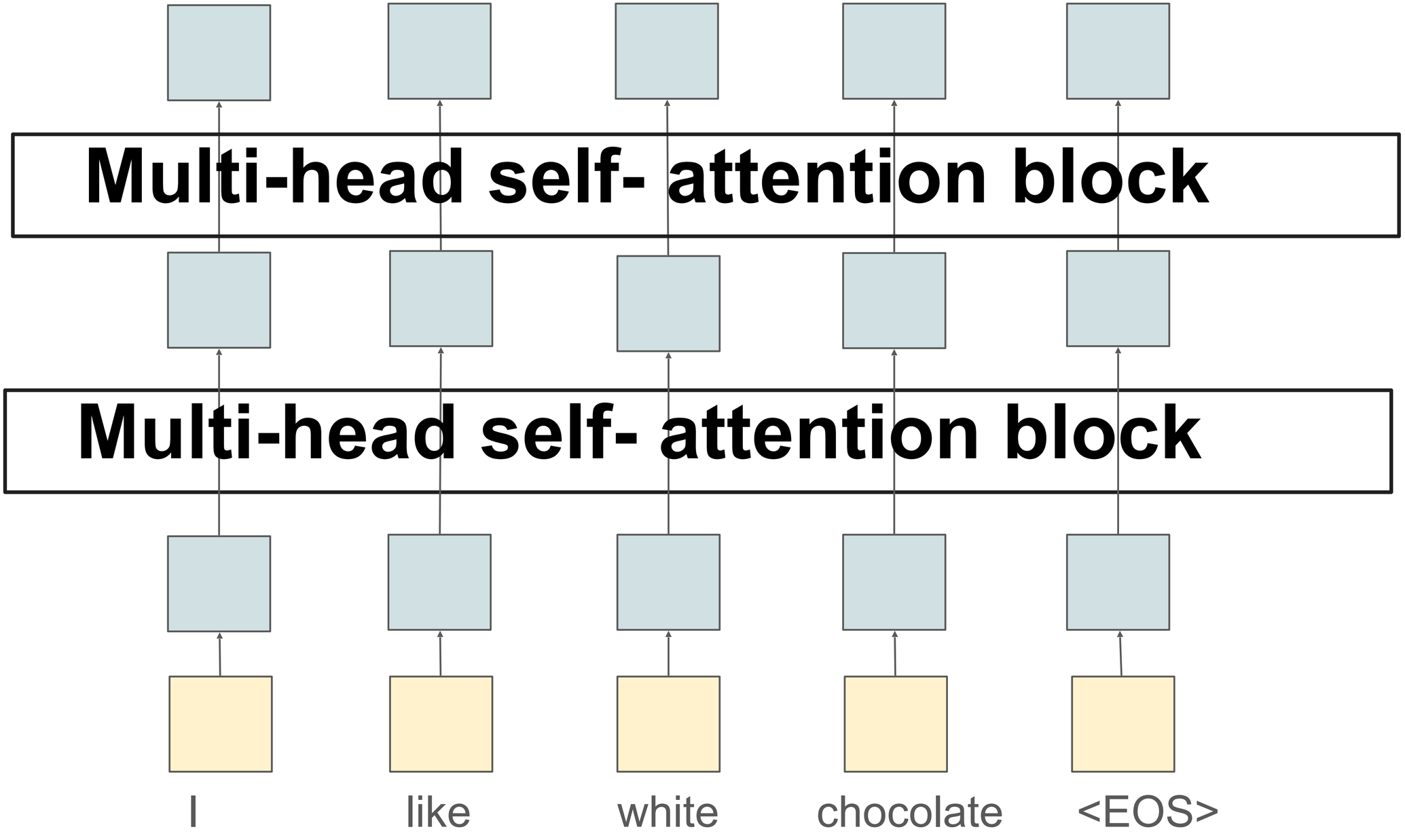
Where do Q, K, V come from?

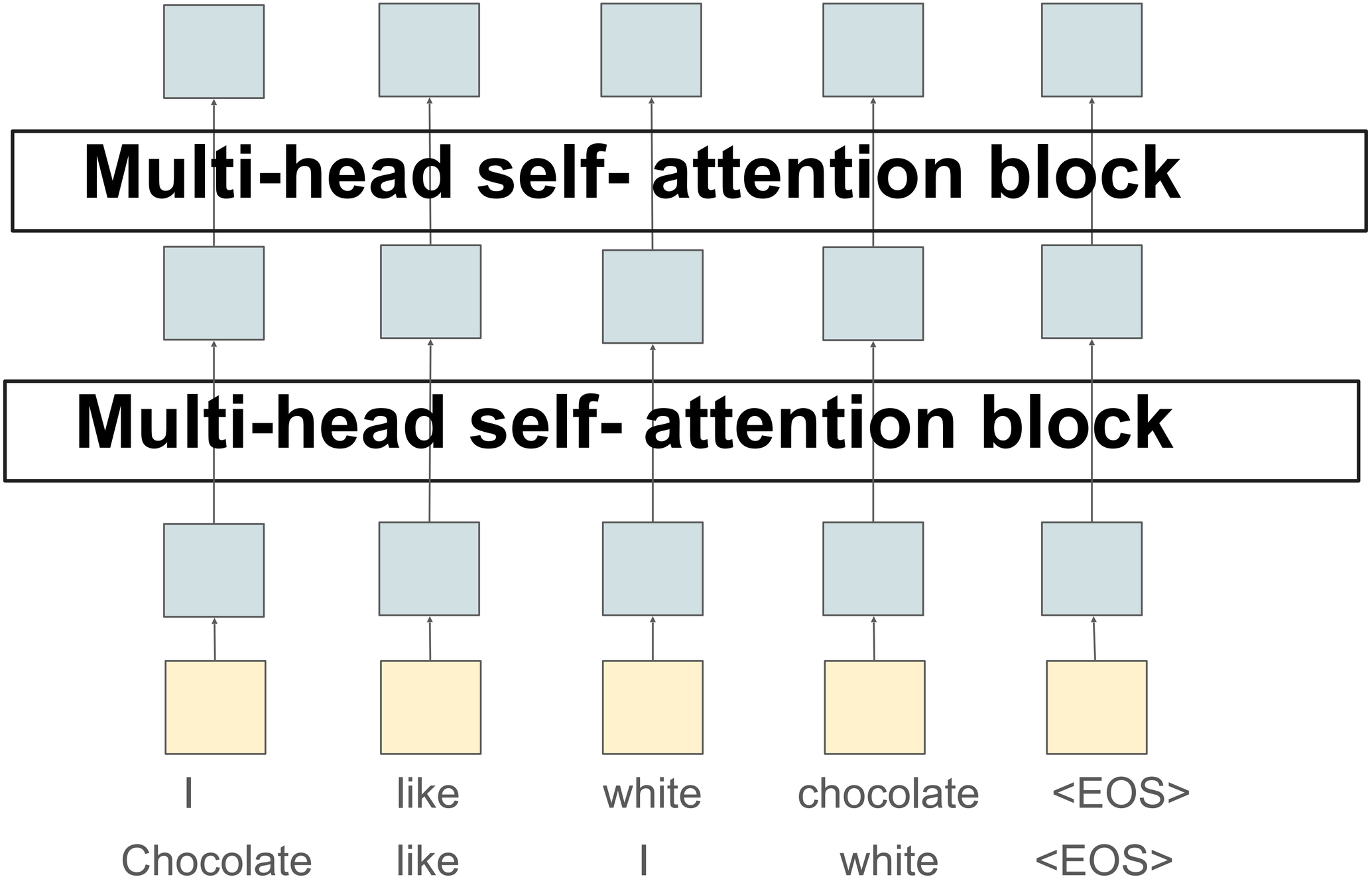
Learning

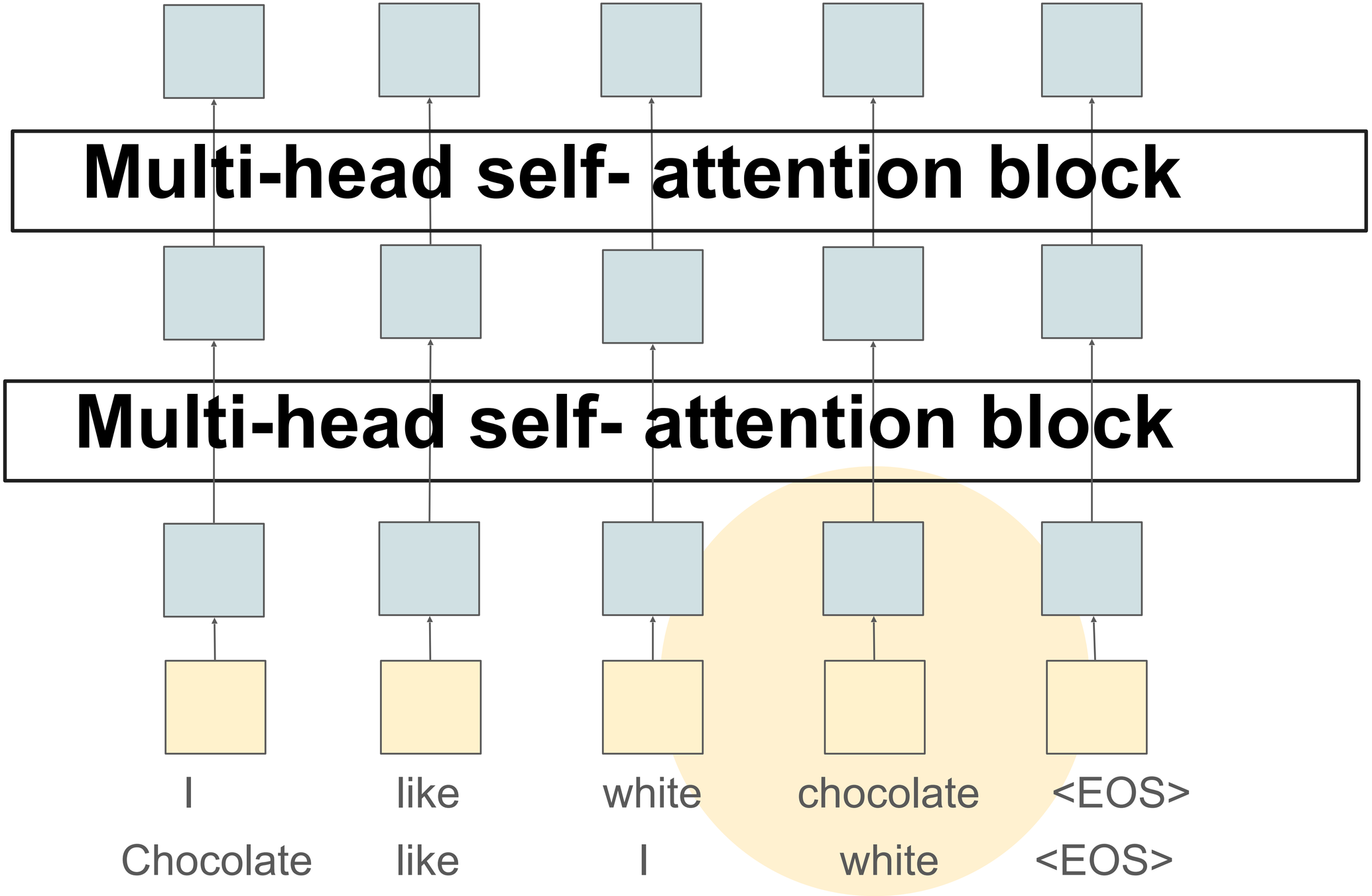
$$q = W_Q x \qquad k = W_K x \qquad v = W_V x$$




Multi-head self- attention


Multi-head self- attention block
Multi-head self- attention block
I
like
white
chocolate
<EOS>


Multi-head self- attention block
Multi-head self- attention block
I
like
white
chocolate
<EOS>
Chocolate
like
I
white
<EOS>


Multi-head self- attention block
Multi-head self- attention block
I
like
white
chocolate
<EOS>
Chocolate
like
I
white
<EOS>

Tokenization

Byte pair encoding

Vocabulary size = 100

[We', 'loo', 'ke', 'd', '!', 'Then', 'we', 'saw', 'hi', 'm', 's', 't', 'e', 'p', 'in', 'o', 'n', 'the', 'm', 'at', '!', 'We', 'loo', 'ke', 'd', '!', 'And', 'we', 'saw', 'hi', 'm', '!', 'The', 'Cat', 'in', 'the', 'Hat', '!', 'And', 'he', 'sa', 'id', 'to', 'u', 's', ',', '"', 'W', 'h', 'y', 'do', 'you', 's', 'it', 'ther', 'e', 'l', 'i', 'ke', 'that', '?', '"', '"', 'I', 'know', 'it', 'is', 'we', 't', 'And', 'the', 'sun', 'is', 'not', 'sun', 'ny', '.', 'B', 'u', 't', 'we', 'c', 'a', 'n', 'h', 'a', 'v', 'e', 'L', 'ot', 's', 'of', 'good', 'fun', 'that', 'is', 'fun', 'ny', '!', '"', '"', 'I', 'know', 'some', 'good', 'g', 'a', 'me', 's', 'we', 'c', 'ou', 'l', 'd', 'p', 'l', 'ay', ',"', 'Said', 'the', 'c', 'at', '.', '"', 'I', 'know', 'some', 'n', 'e', 'w', 'trick', 's', ',"', 'Said', 'the', 'Cat', 'in', 'the', 'Hat', '.', '"', 'A', 'l', 'ot', 'of', 'good', 'trick', 's', '.', 'I', 'w', 'ill', 's', 'h', 'o', 'w', 'the', 'm', 'to', 'you', '.', 'Y', 'ou', 'r', 'M', 'other', 'W', 'ill', 'not', 'm', 'i', 'nd', 'at', 'a', 'll', 'i', 'f', 'I', 'do', '.', '"', 'Then', 'Sa', 'll', 'y', 'a', 'nd', 'I', 'D', 'id', 'not', 'know', 'w', 'hat', 'to', 'sa', 'y', '.', 'O', 'u', 'r', 'm', 'other', 'w', 'a', 's', 'ou', 't', 'of', 'the', 'h', 'ou', 's', 'e', 'F', 'o', 'r', 'the', 'd', 'ay', '.']

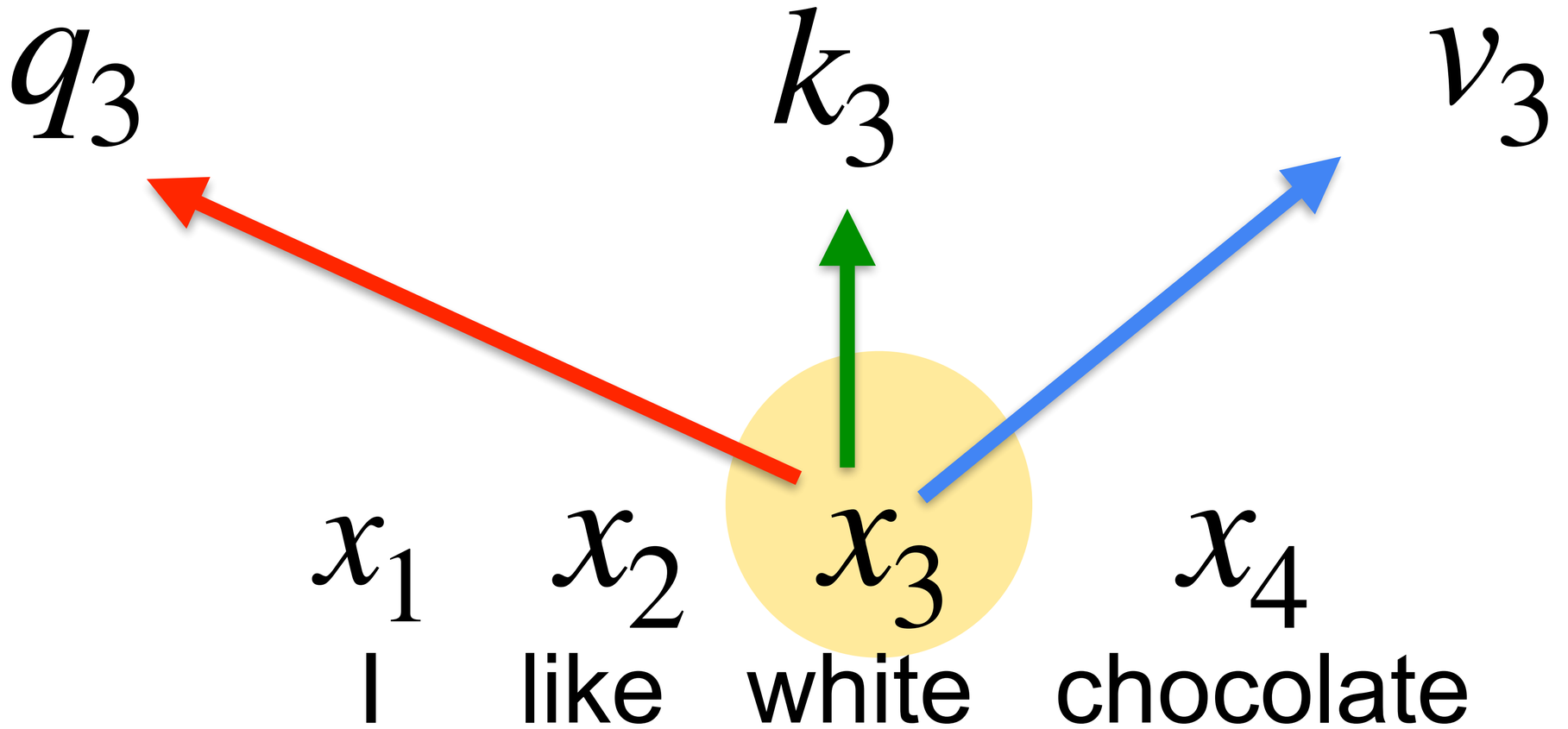
Vocabulary size = 150

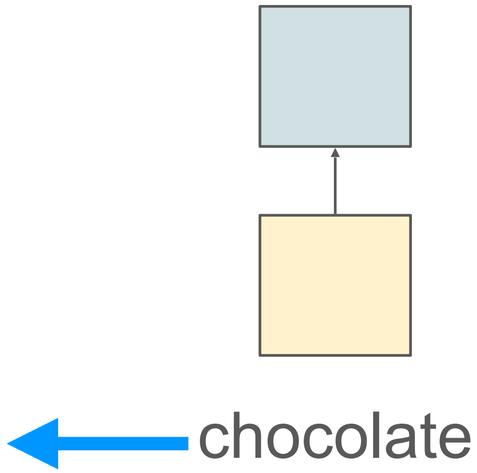
["We', 'looked', '!', 'Then', 'we', 'saw', 'him', 'st', 'ep', 'in', 'on', 'the', 'mat', '!', 'We', 'looked', '!', 'And', 'we', 'saw', 'him', '!', 'The', 'Cat', 'in', 'the', 'Hat', '!', 'And', 'he', 'sa', 'id', 'to', 'us', ',', '"', 'Wh', 'y', 'do', 'you', 'sit', 'ther', 'e', 'like', 'that', '?"', '"', 'I', 'know', 'it', 'is', 'we', 't', 'And', 'the', 'sun', 'is', 'not', 'sun', 'ny', '.', 'Bu', 't', 'we', 'can', 'hav', 'e', 'Lot', 's', 'of', 'good', 'fun', 'that', 'is', 'fun', 'ny', '!"', '"', 'I', 'know', 'some', 'good', 'game', 's', 'we', 'cou', 'ld', 'play', ',"', 'Said', 'the', 'cat', '.', '"', 'I', 'know', 'some', 'new', 'tricks', ',"', 'Said', 'the', 'Cat', 'in', 'the', 'Hat', '.', '"', 'A', 'lot', 'of', 'good', 'tricks', '.', 'I', 'w', 'ill', 'sho', 'w', 'the', 'm', 'to', 'you', '.', 'You', 'r', 'Mother', 'Will', 'not', 'mind', 'at', 'all', 'if', 'I', 'do', '."', 'Then', 'Sa', 'll', 'y', 'and', 'I', 'Did', 'not', 'know', 'w', 'hat', 'to', 'sa', 'y', '.', 'Ou', 'r', 'mother', 'w', 'as', 'ou', 't', 'of', 'the', 'hou', 'se', 'Fo', 'r', 'the', 'day', '.']

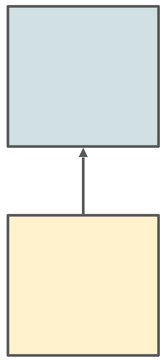
Where do Q, K, V come from?

Learning

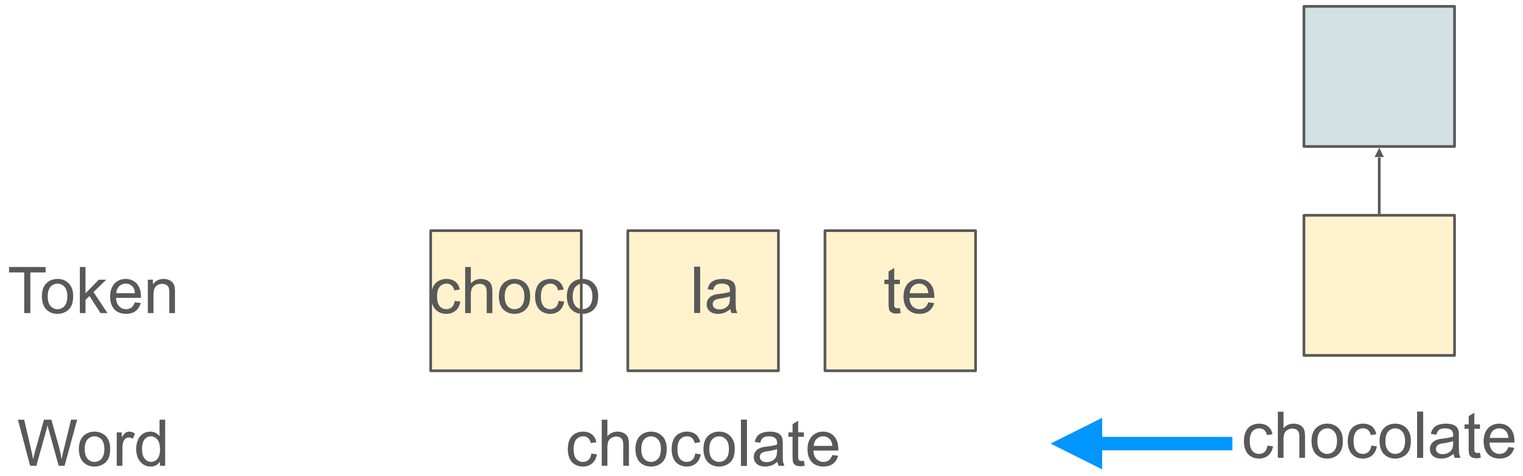
$$q = W_Q x \qquad k = W_K x \qquad v = W_V x$$

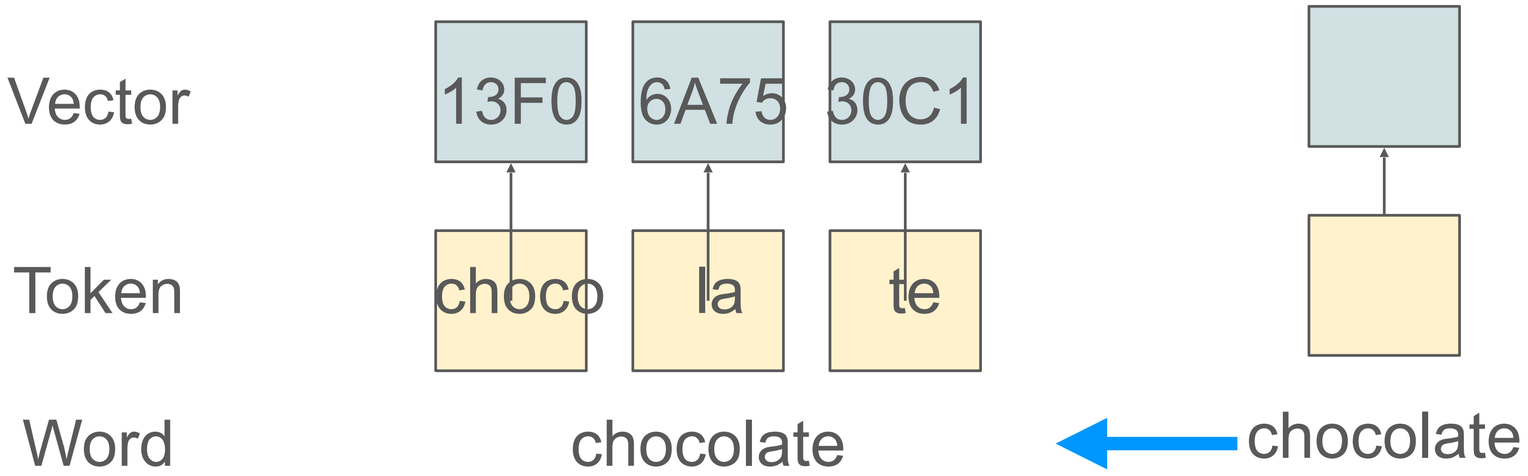


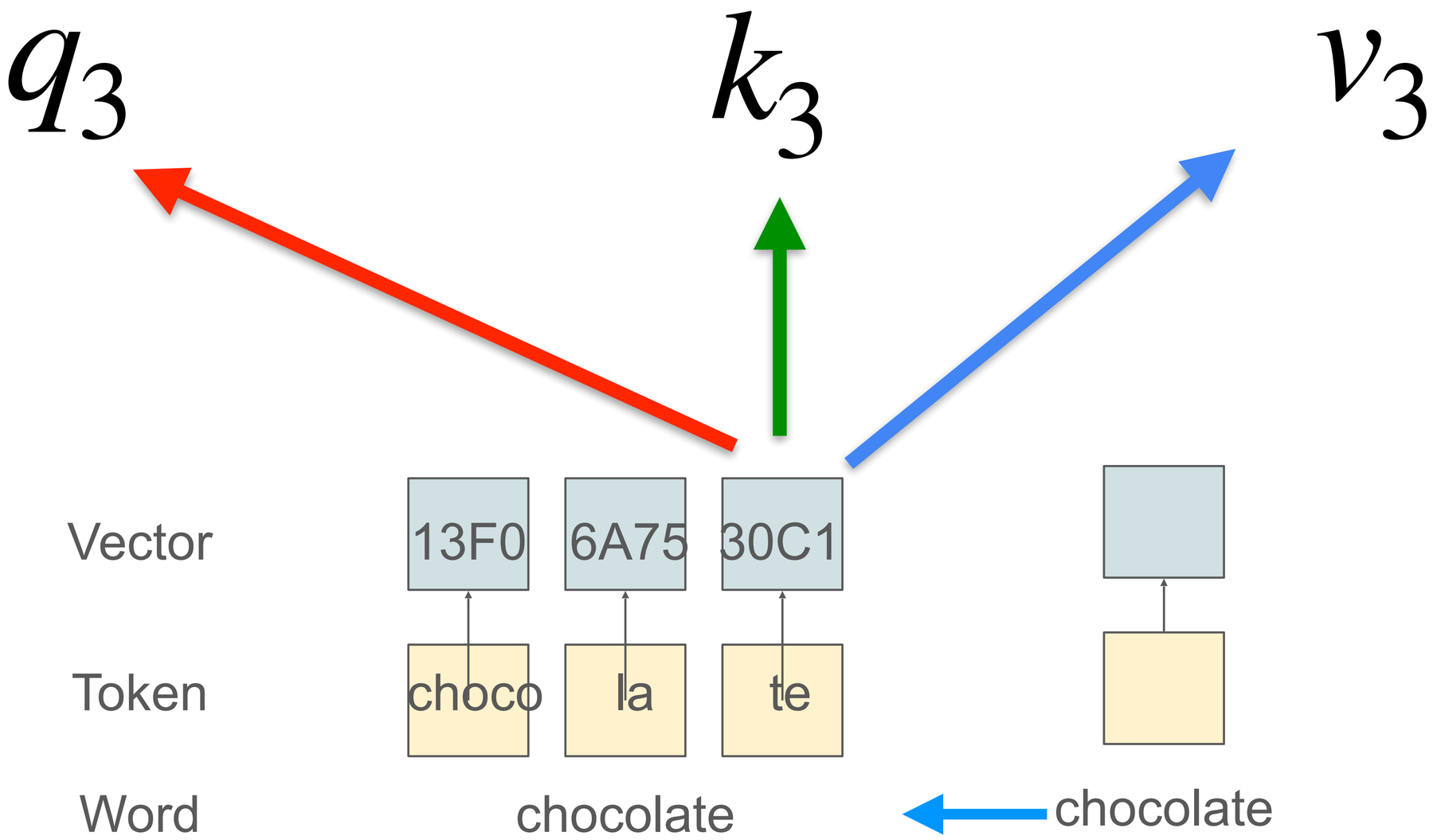

chocolate



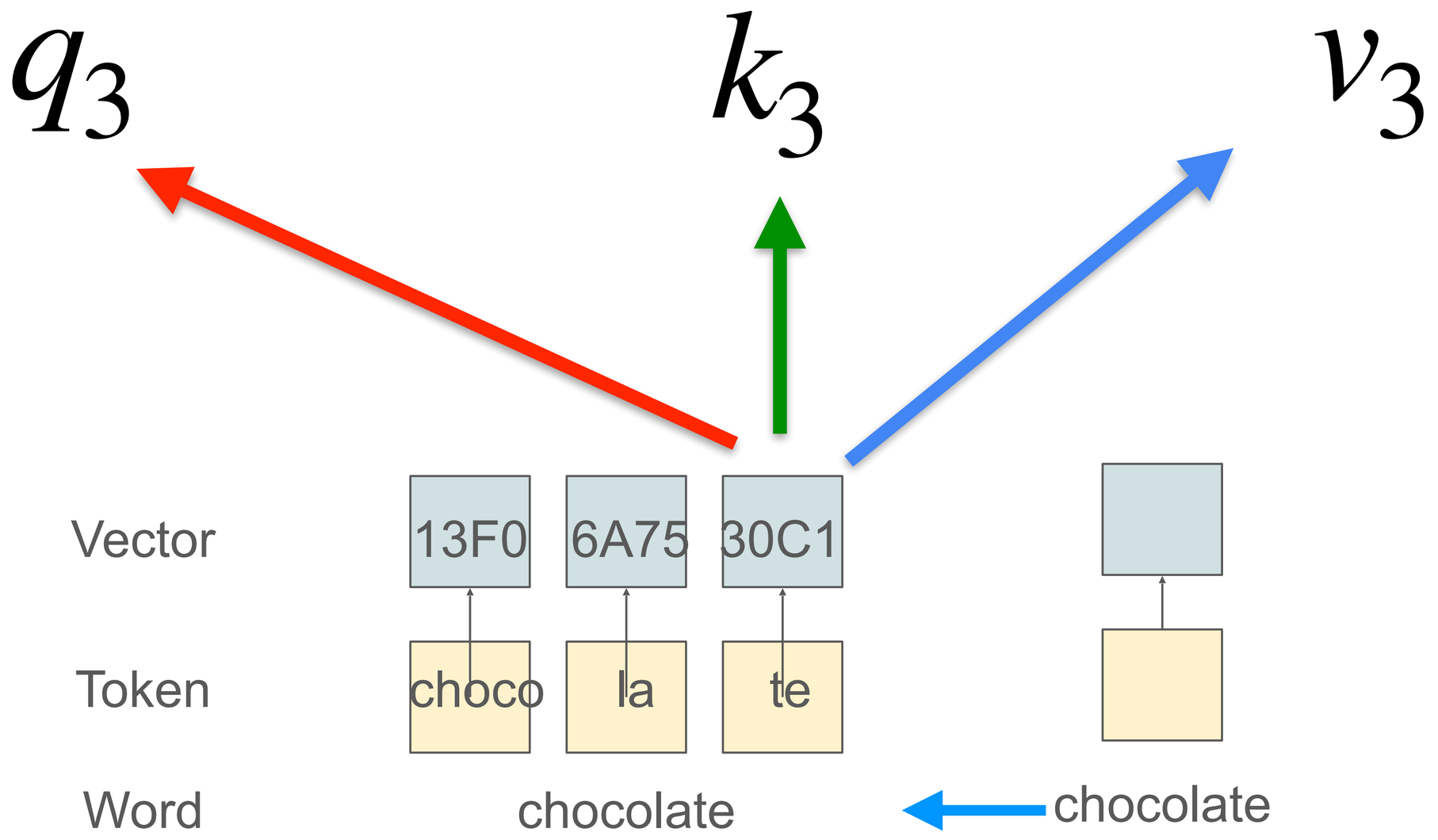
Word chocolate ← chocolate

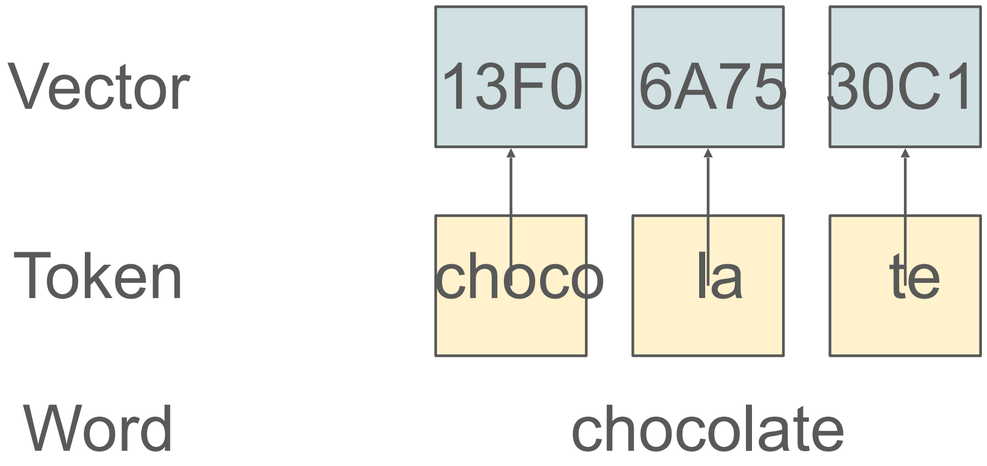

Token
choco
la
te
Word
chocolate
chocolate


Vector
13F0
6A75
30C1
Token
choco
la
te
Word
chocolate
chocolate


q_3
k_3
v_3
Vector
13F0
6A75
30C1
Token
choco
la
te
Word
chocolate
chocolate

$$q = W_Q x \qquad k = W_K x \qquad v = W_V x$$




Vector
13F0
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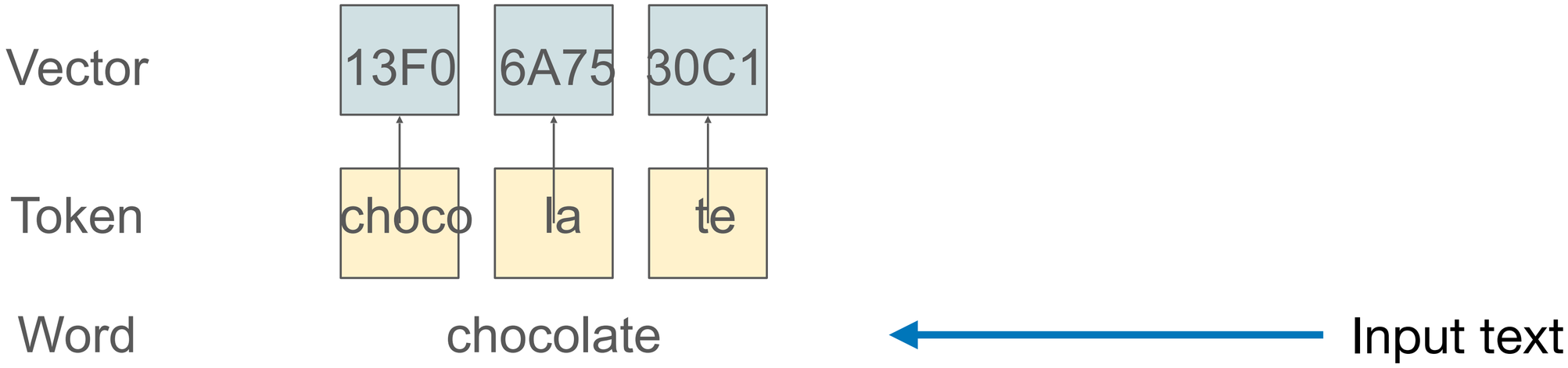
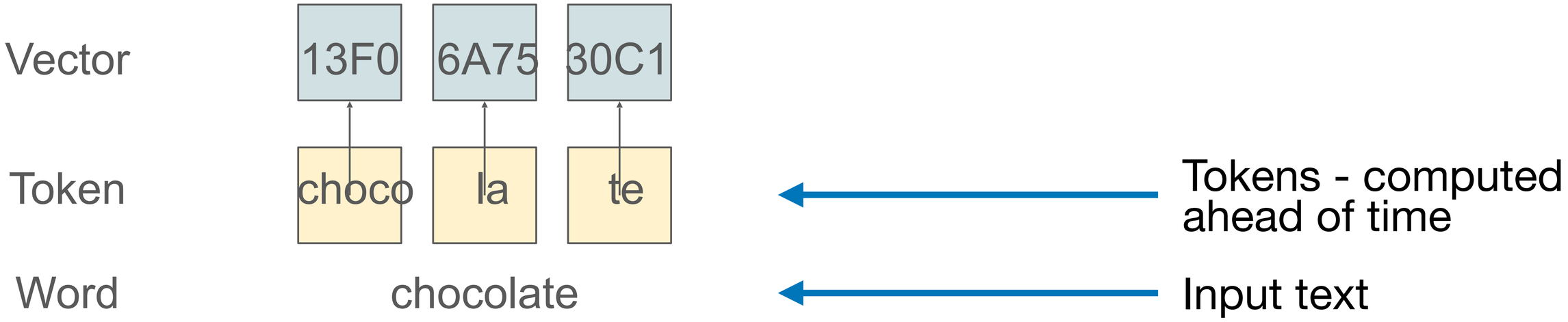
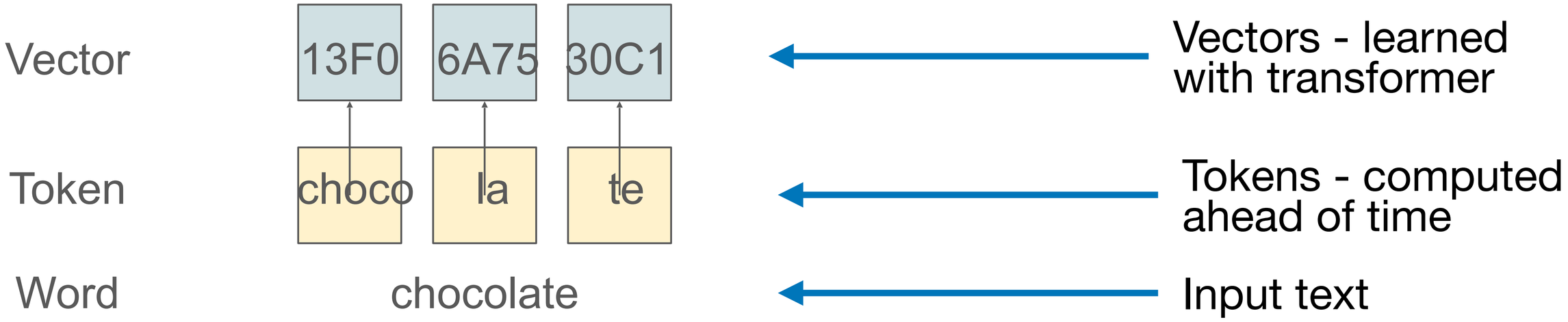
30C1
Token
choco
la
te
Word
chocolate


Vector
13F0
6A75
30C1
Token
choco
la
te
Word
chocolate
Input text


Vector
13F0
6A75
30C1
Token
choco
la
te
Word
chocolate
Tokens - computed ahead of time
Input text

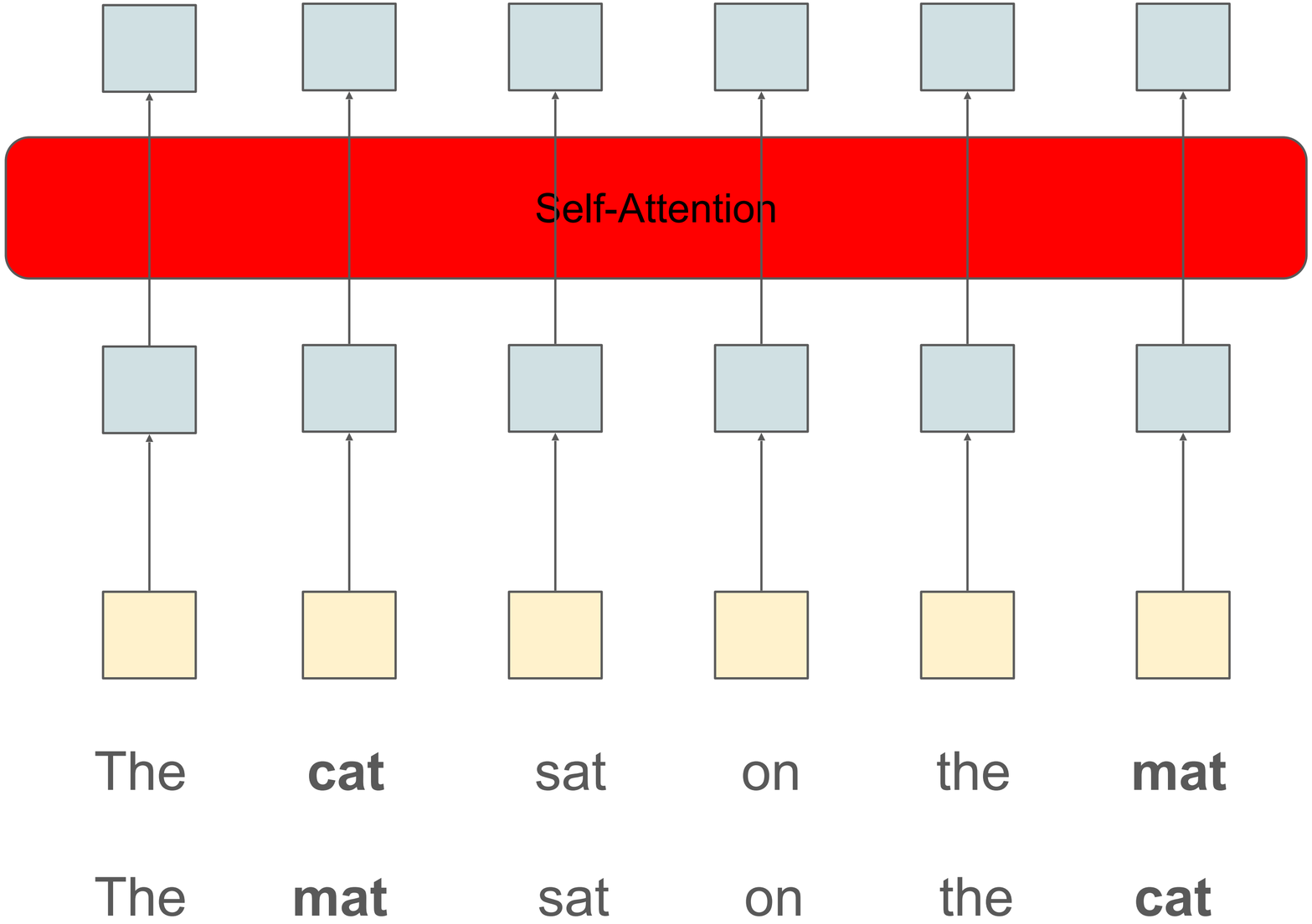

Vector
13F0
6A75
30C1
Vectors - learned with transformer
Token
choco
la
te
Tokens - computed ahead of time
Word
chocolate
Input text

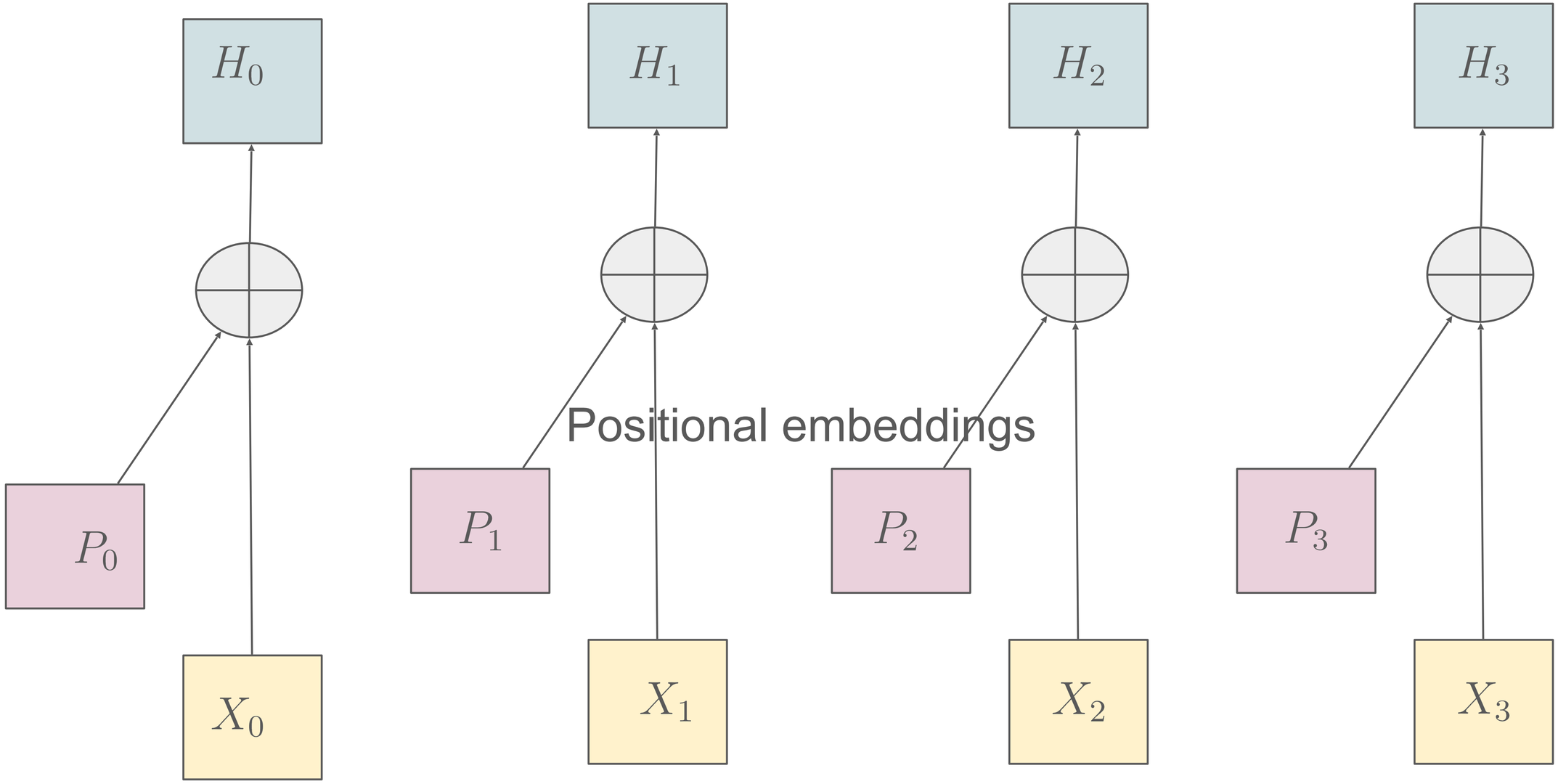
One more thing!!

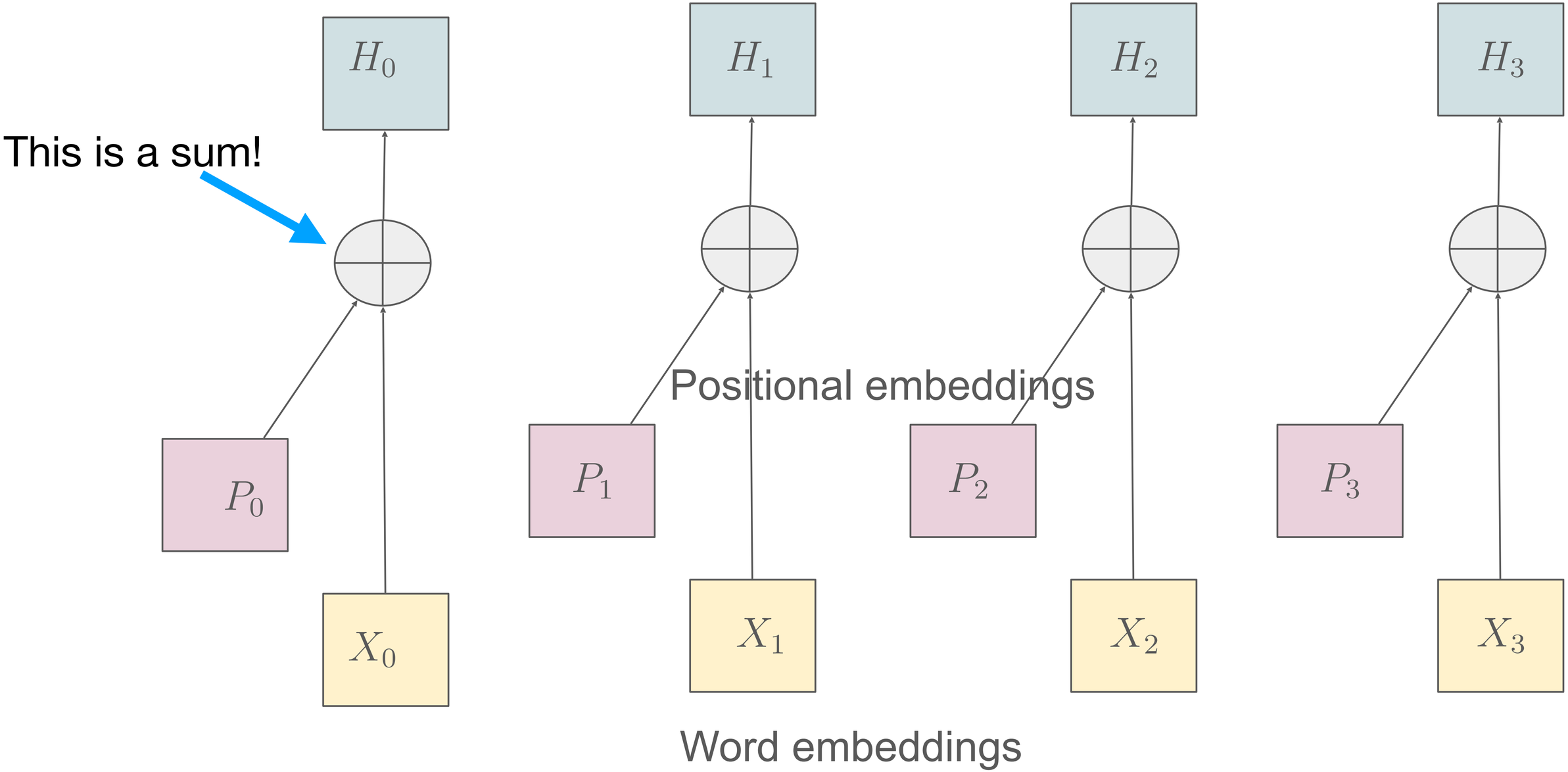
One more thing!!

One more thing!!

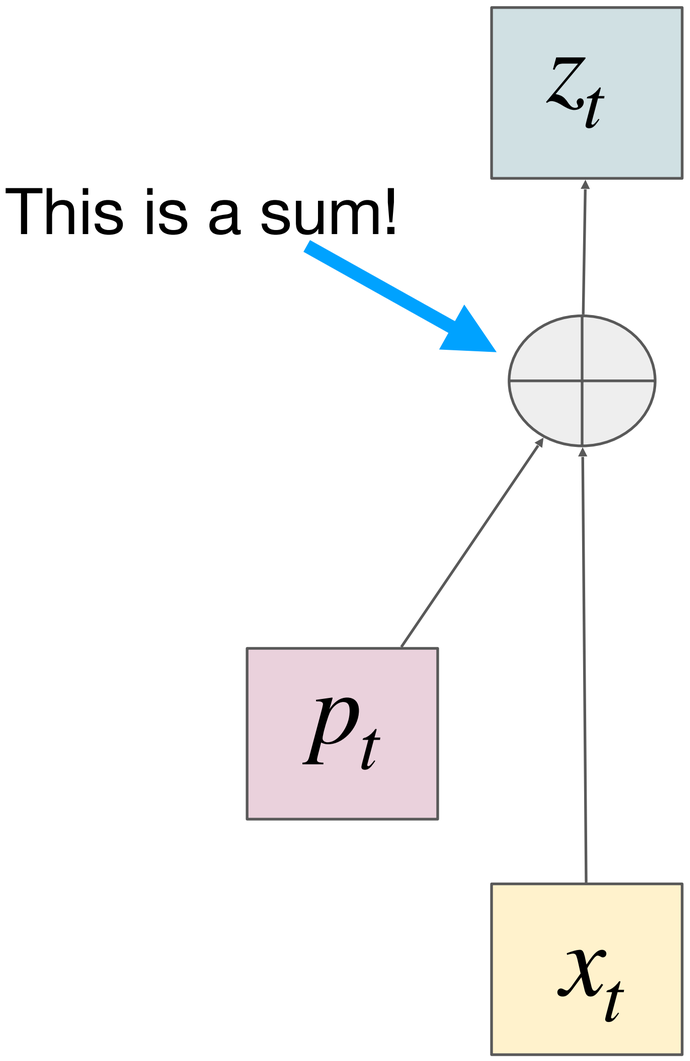
Positional encoding


Self-Attention
The cat sat on the mat
The mat sat on the cat

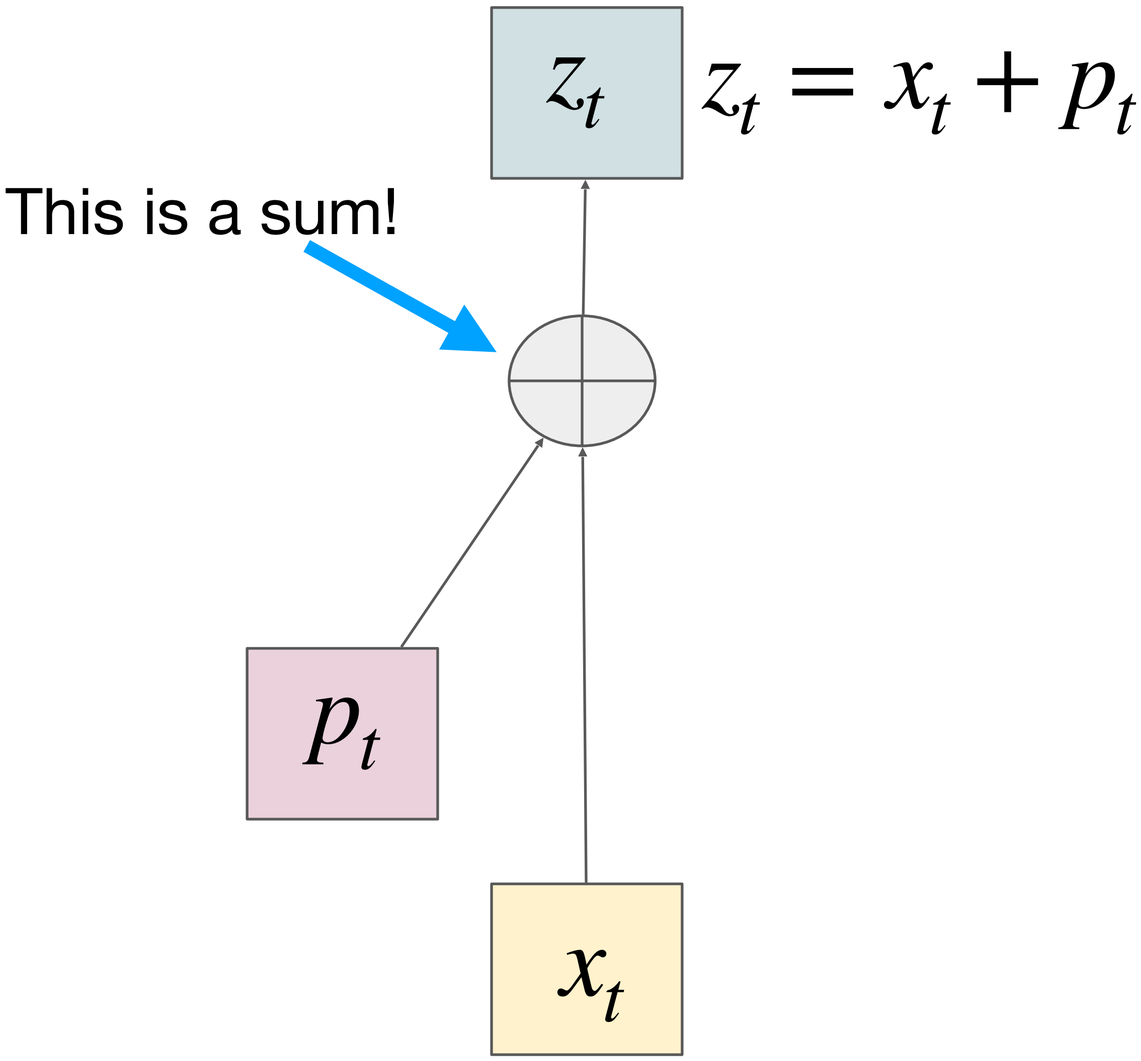

H_0
H_1
H_2
H_3
Positional embeddings
P_0
P_1
P_2
P_3
X_0
X_1
X_2
X_3
Word embeddings


This is a sum!
H_0
H_1
H_2
H_3
Positional embeddings
P_0
P_1
P_2
P_3
X_0
X_1
X_2
X_3
Word embeddings

Why a sum?

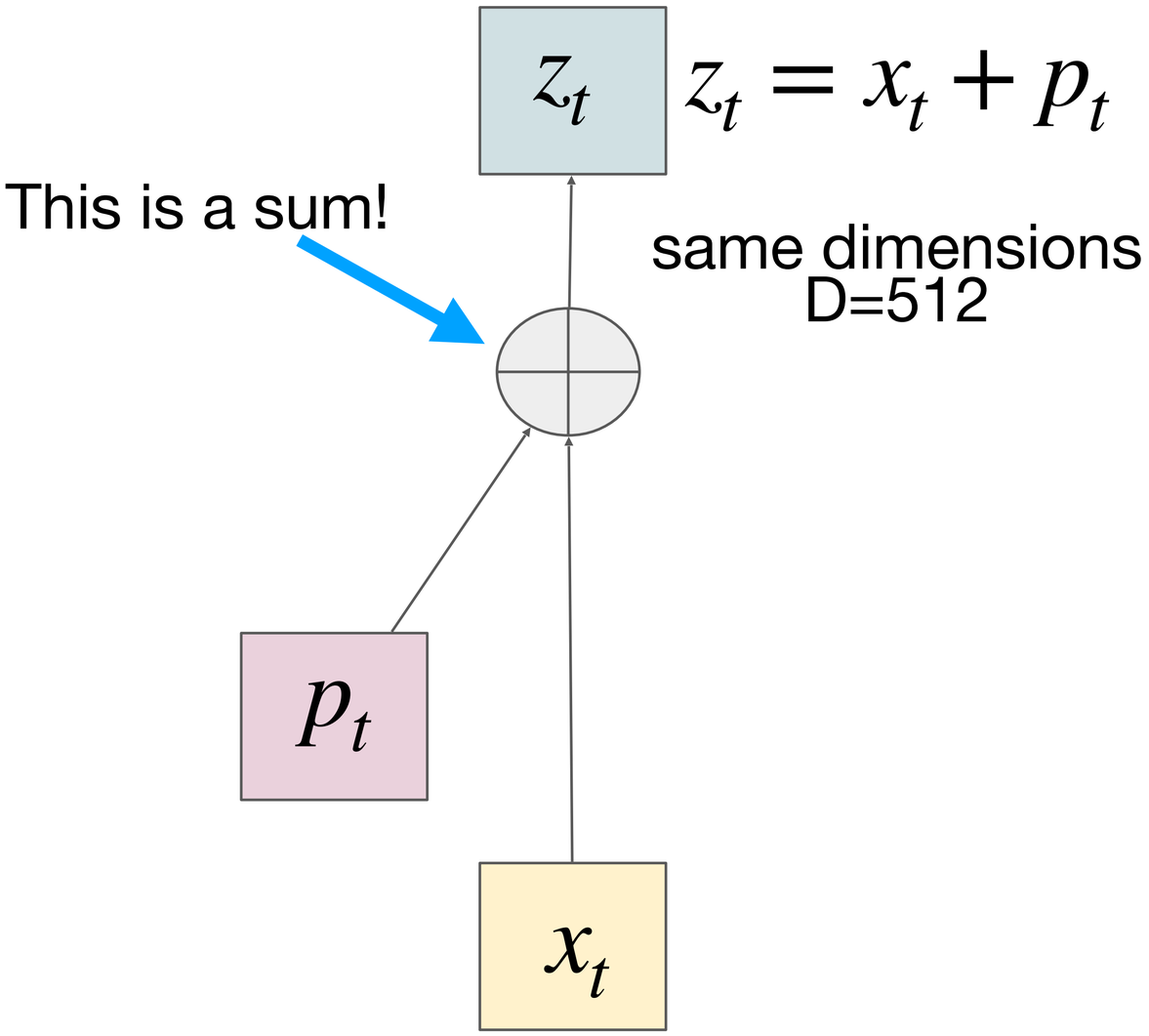


Why a sum?



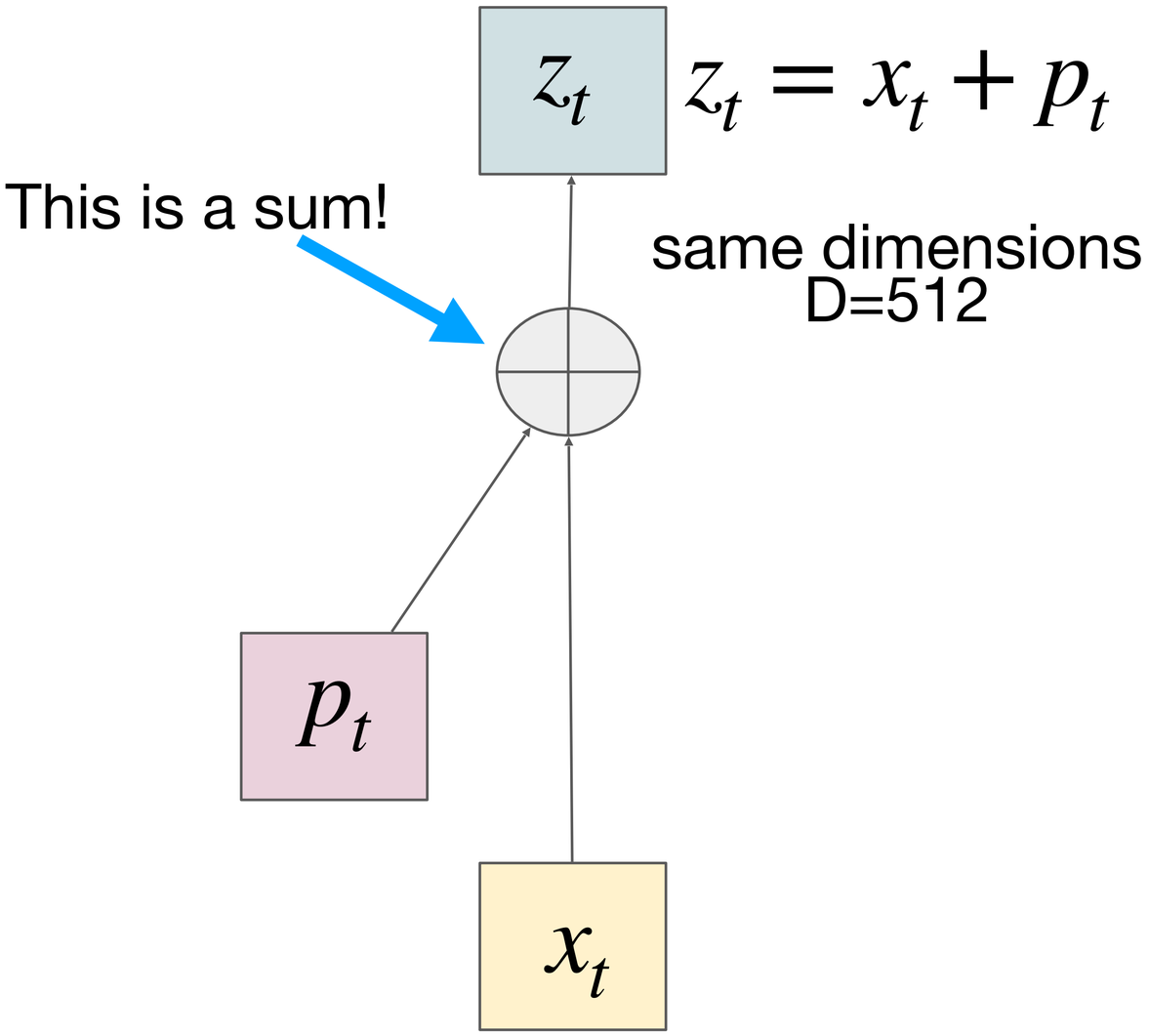
$$z_t = x_t + p_t$$

Why a sum?



$$z_t = x_t + p_t$$

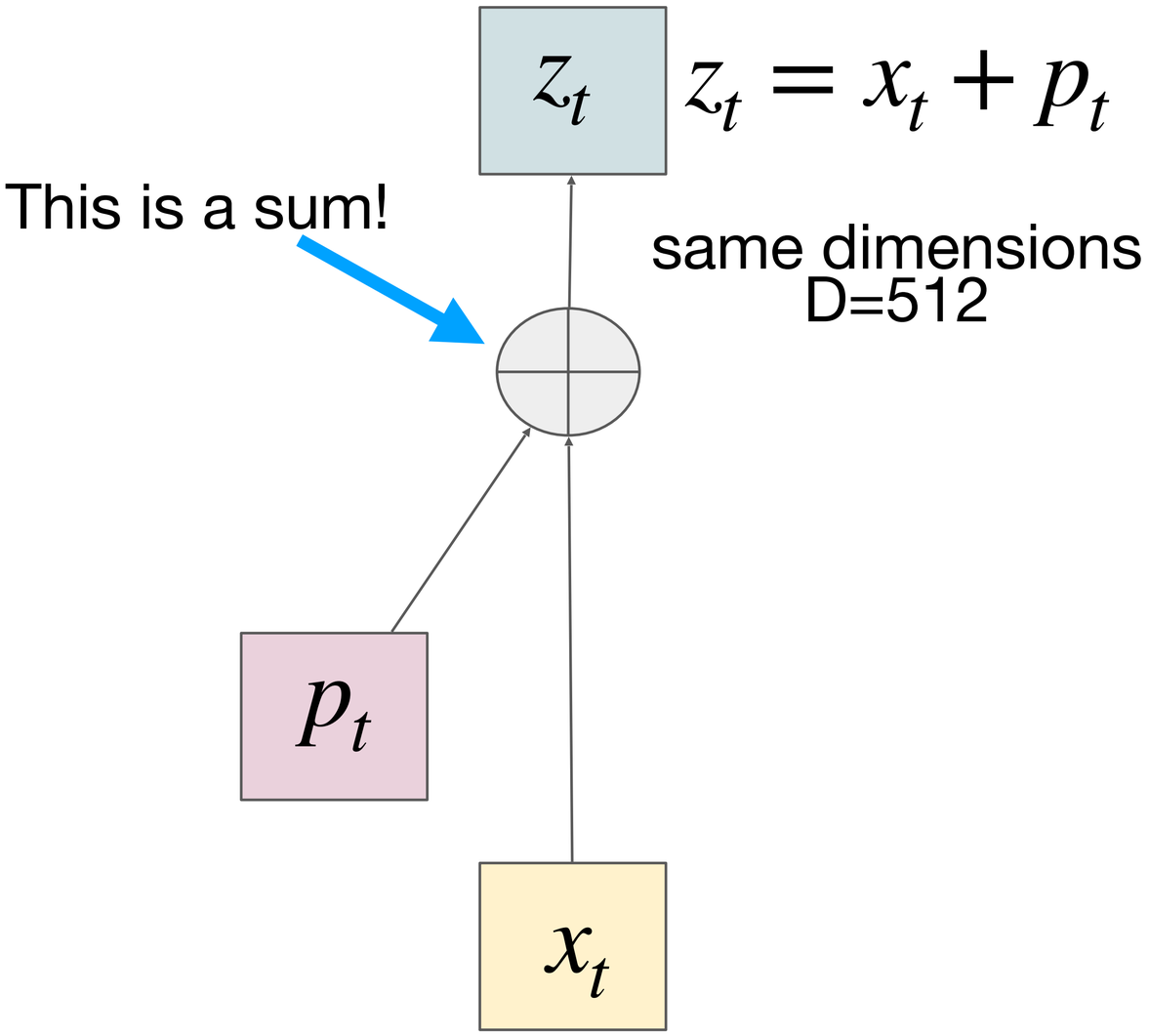
Why a sum?



$$z_t = x_t + p_t$$

$$q_t = W_Q z_t = W_Q x_t + W_Q p_t$$

Why a sum?

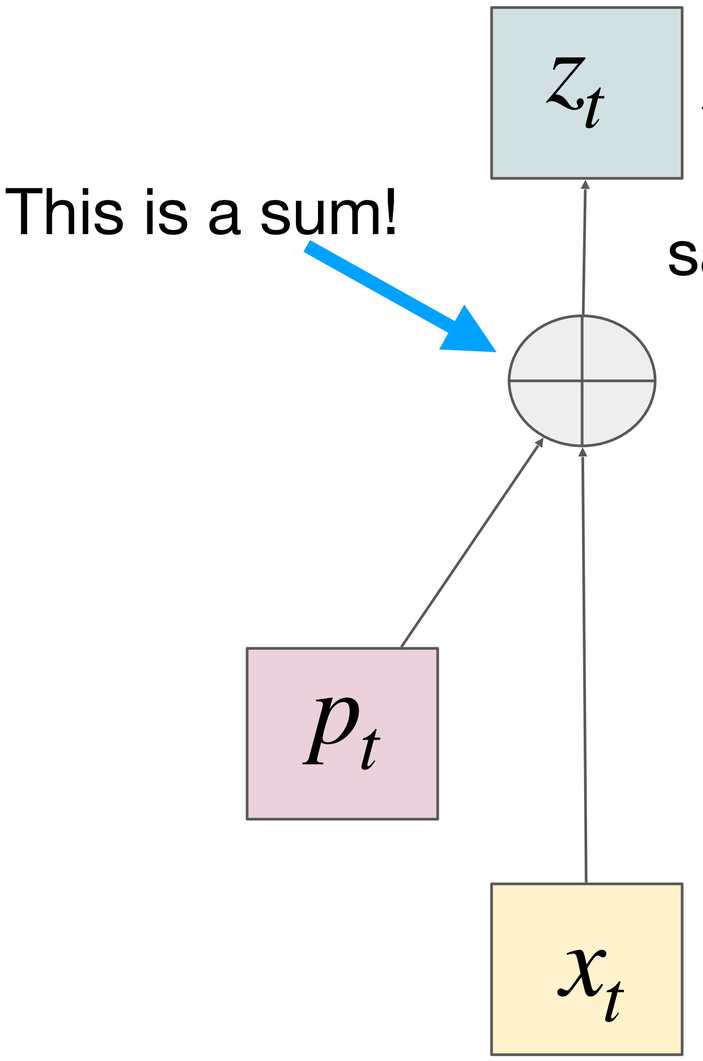


$$z_t = x_t + p_t$$

$$q_t = W_Q z_t = W_Q x_t + W_Q p_t$$

$$k_s = W_K z_s = W_K x_s + W_K p_s$$

Why a sum?



$$z_t = x_t + p_t$$

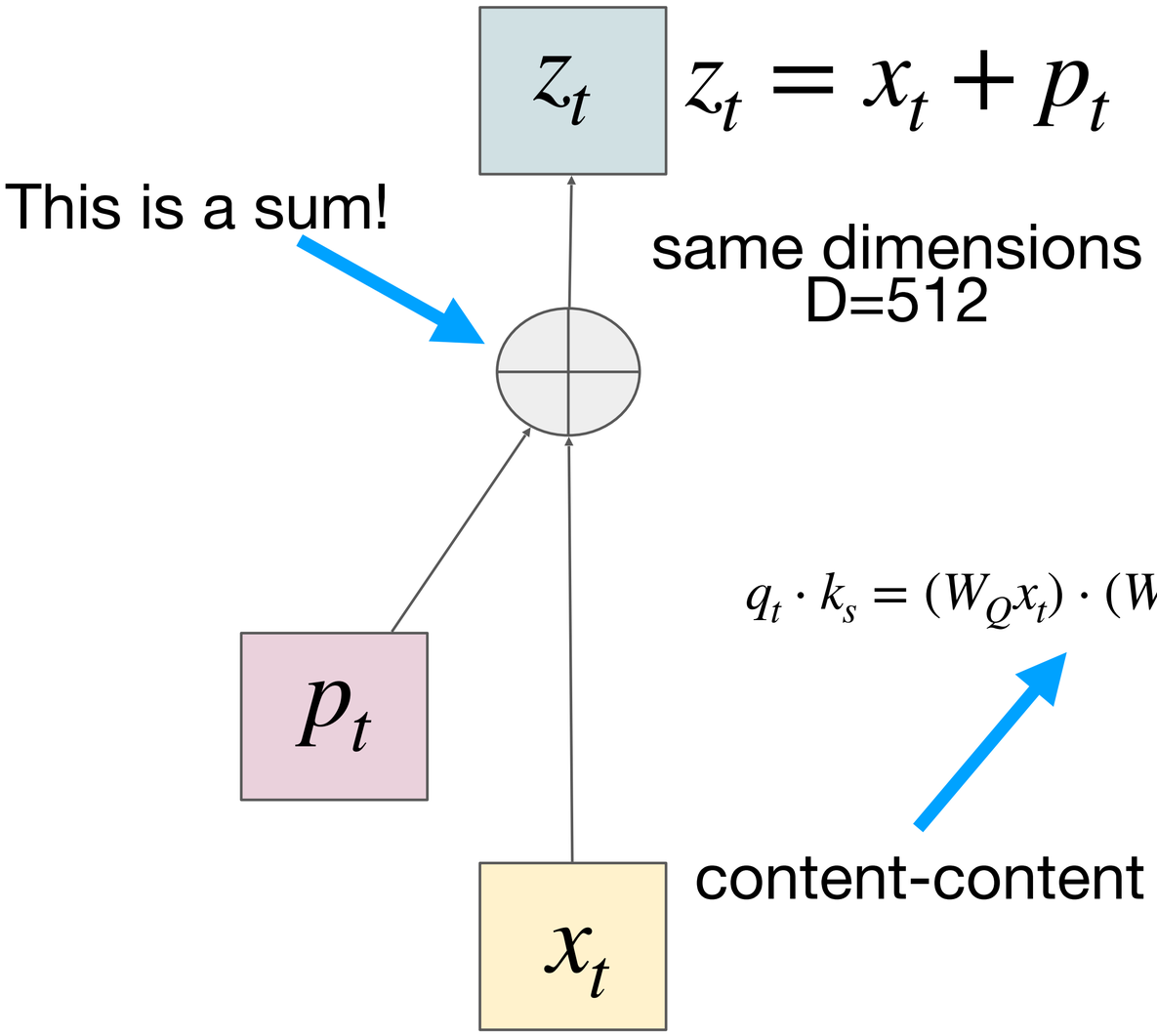
same dimensions
D=512

$$q_t = W_Q z_t = W_Q x_t + W_Q p_t$$

$$k_s = W_K z_s = W_K x_s + W_K p_s$$

$$q_t \cdot k_s = (W_Q x_t) \cdot (W_K x_s) + (W_Q x_t) \cdot (W_K p_s) + (W_Q p_t) \cdot (W_K x_s) + (W_Q p_t) \cdot (W_K p_s)$$

Why a sum?



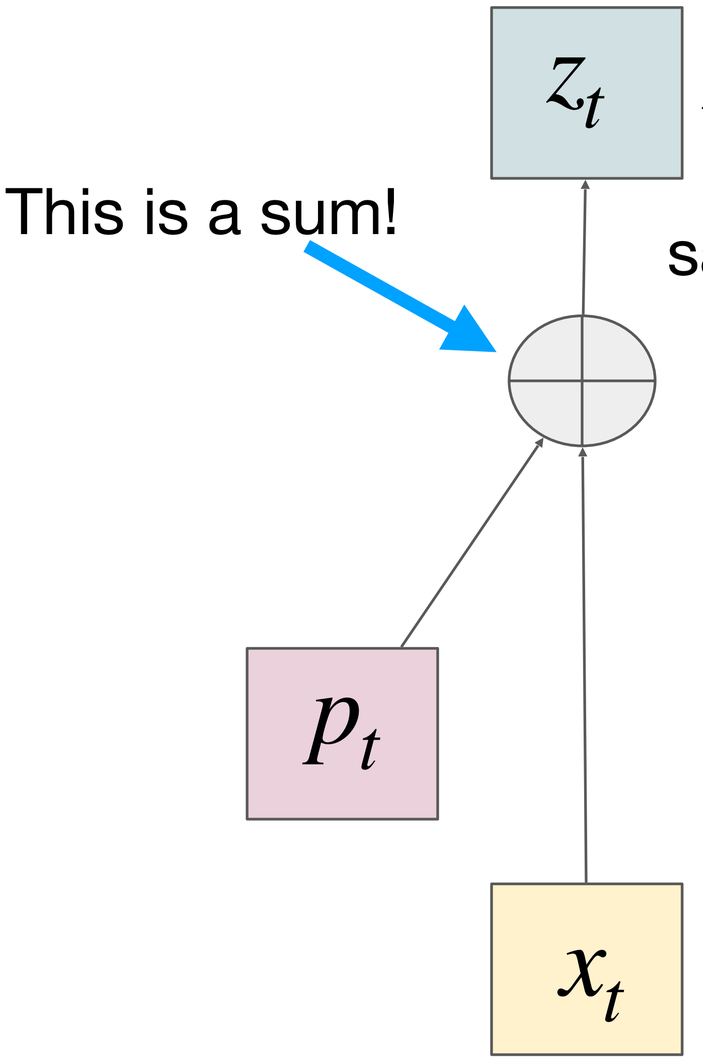
$$z_t = x_t + p_t$$

$$q_t = W_Q z_t = W_Q x_t + W_Q p_t$$

$$k_s = W_K z_s = W_K x_s + W_K p_s$$

$$q_t \cdot k_s = (W_Q x_t) \cdot (W_K x_s) + (W_Q x_t) \cdot (W_K p_s) + (W_Q p_t) \cdot (W_K x_s) + (W_Q p_t) \cdot (W_K p_s)$$

Why a sum?



$$z_t = x_t + p_t$$

same dimensions
D=512

$$q_t = W_Q z_t = W_Q x_t + W_Q p_t$$

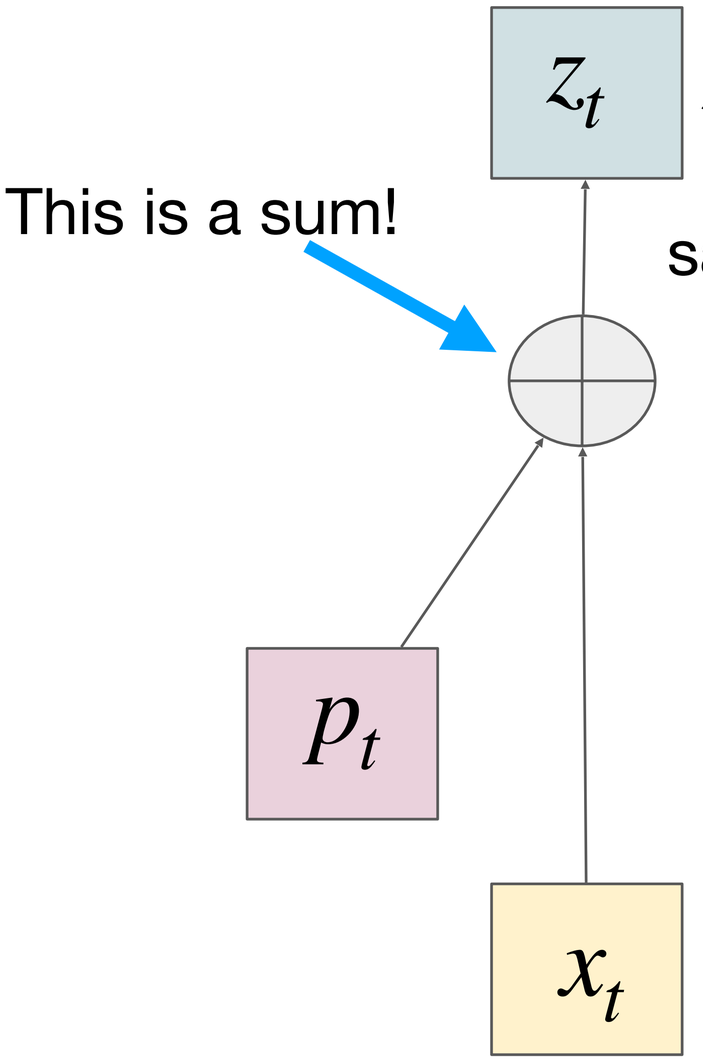
$$k_s = W_K z_s = W_K x_s + W_K p_s$$

$$q_t \cdot k_s = (W_Q x_t) \cdot (W_K x_s) + (W_Q x_t) \cdot (W_K p_s) + (W_Q p_t) \cdot (W_K x_s) + (W_Q p_t) \cdot (W_K p_s)$$

content-content

content-position

Why a sum?



$$z_t = x_t + p_t$$

same dimensions
D=512

$$q_t = W_Q z_t = W_Q x_t + W_Q p_t$$

$$k_s = W_K z_s = W_K x_s + W_K p_s$$

$$q_t \cdot k_s = (W_Q x_t) \cdot (W_K x_s) + (W_Q x_t) \cdot (W_K p_s) + (W_Q p_t) \cdot (W_K x_s) + (W_Q p_t) \cdot (W_K p_s)$$

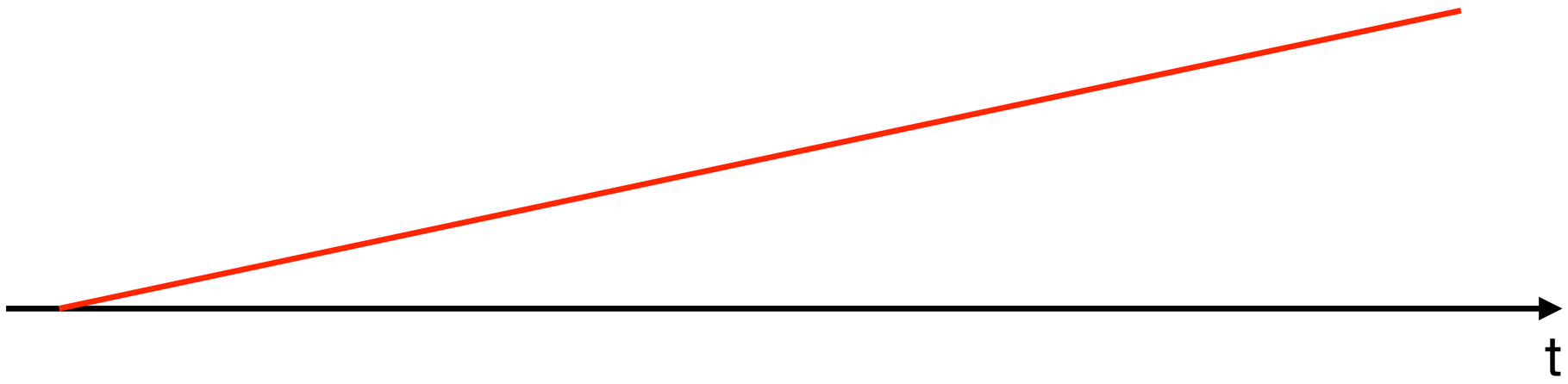
content-content

content-position

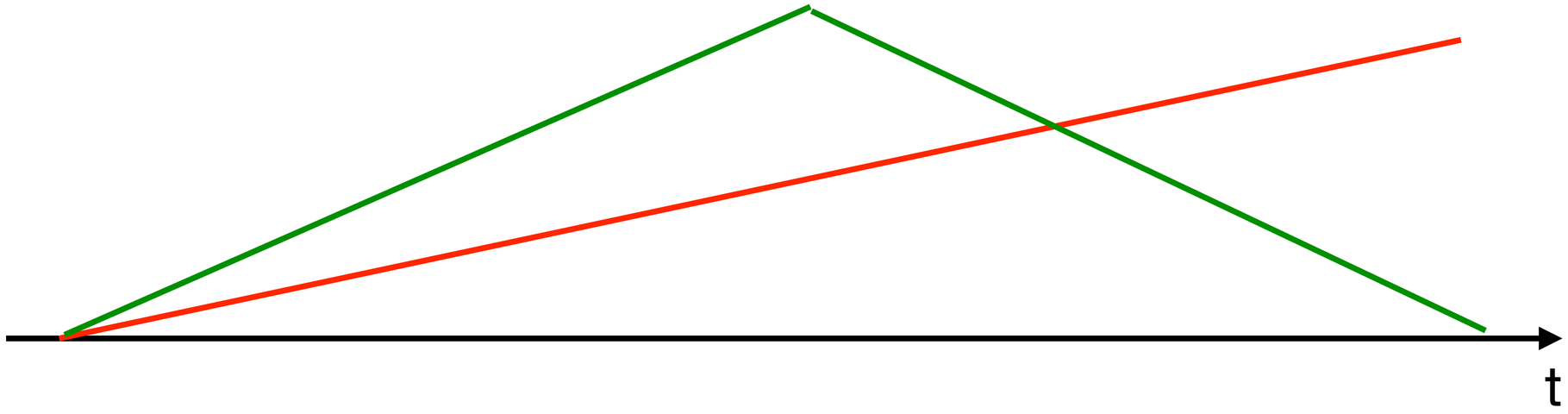
content-position

Options for position code

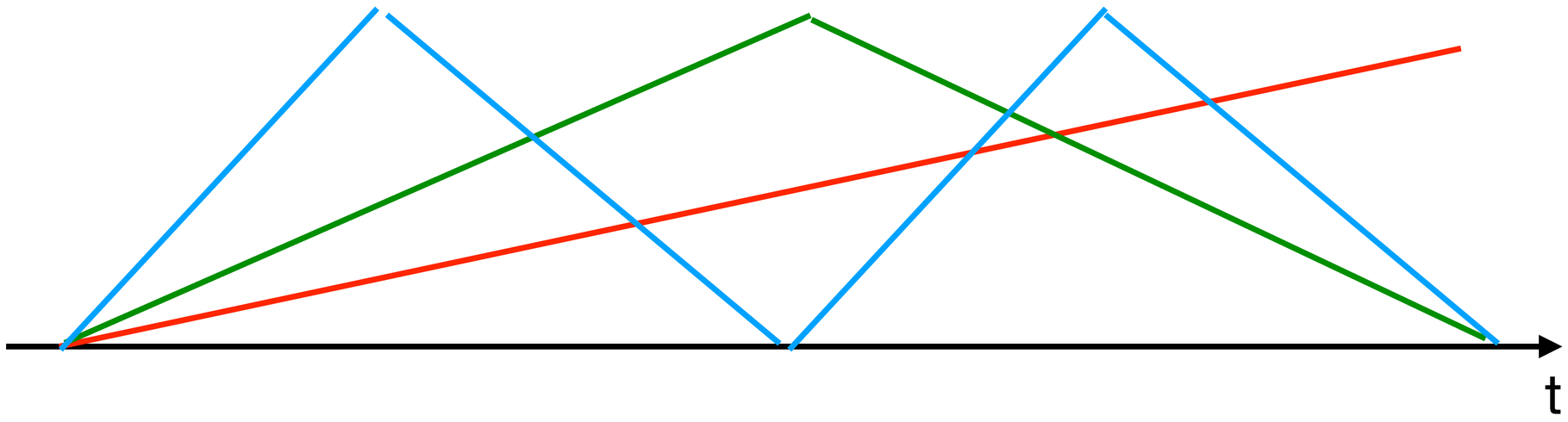
Options for position code



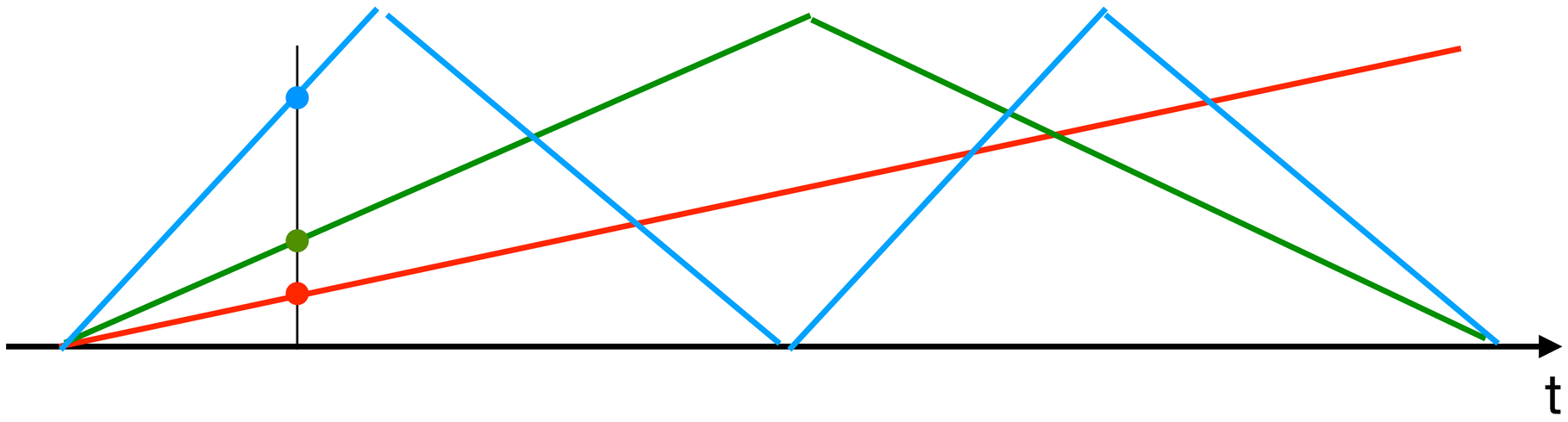
Options for position code



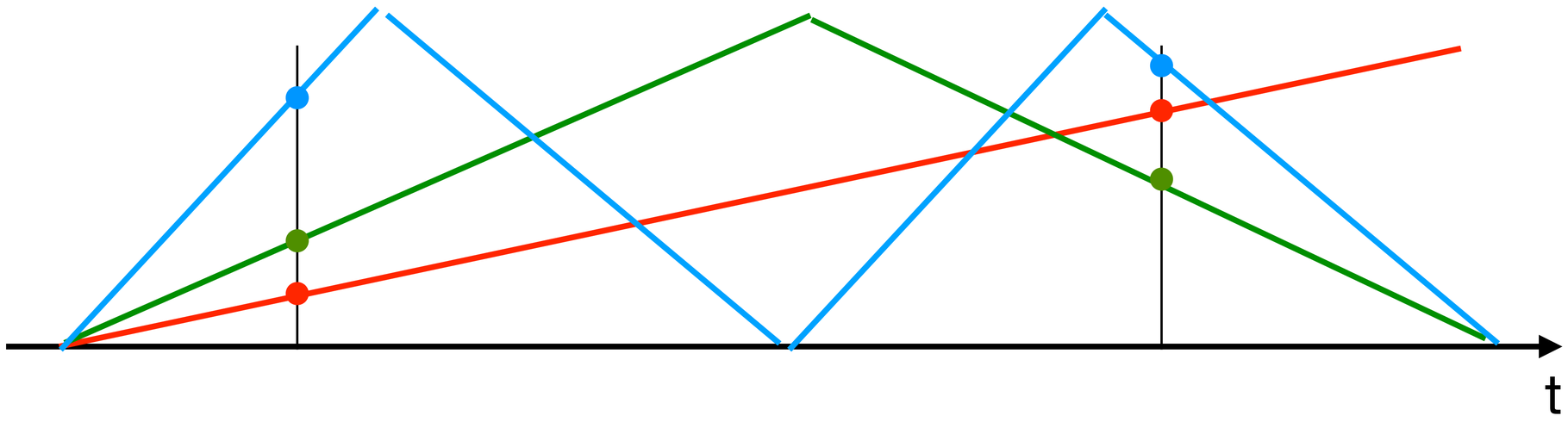
Options for position code



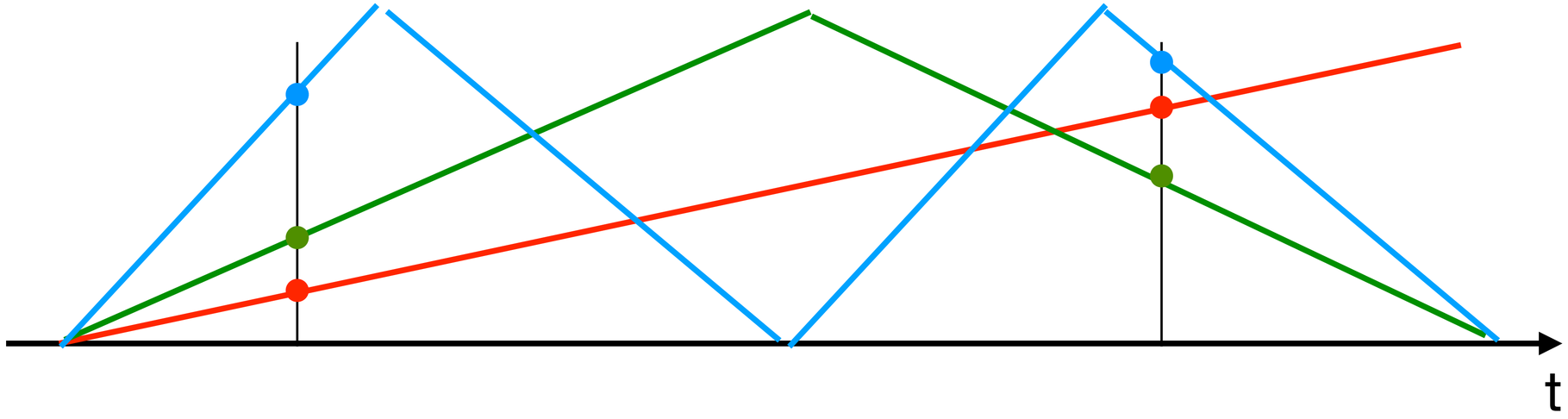
Options for position code



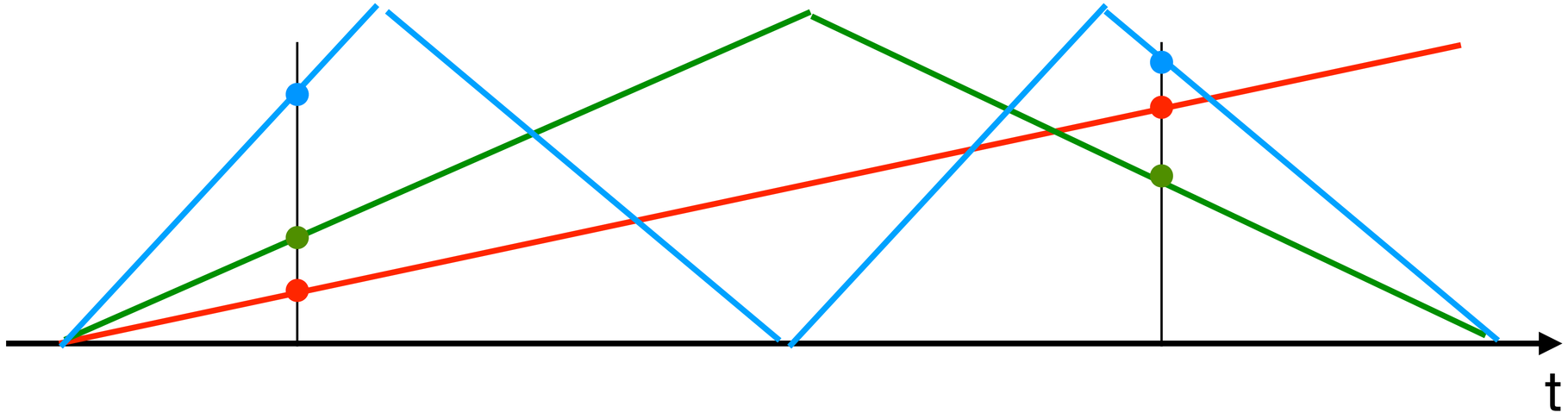
Options for position code



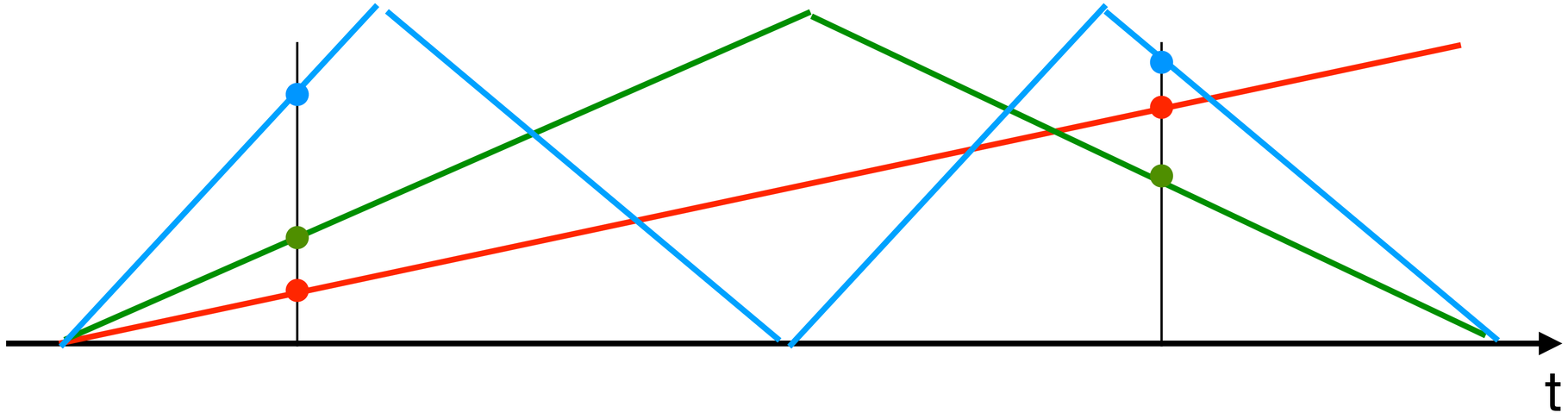
Options for position code

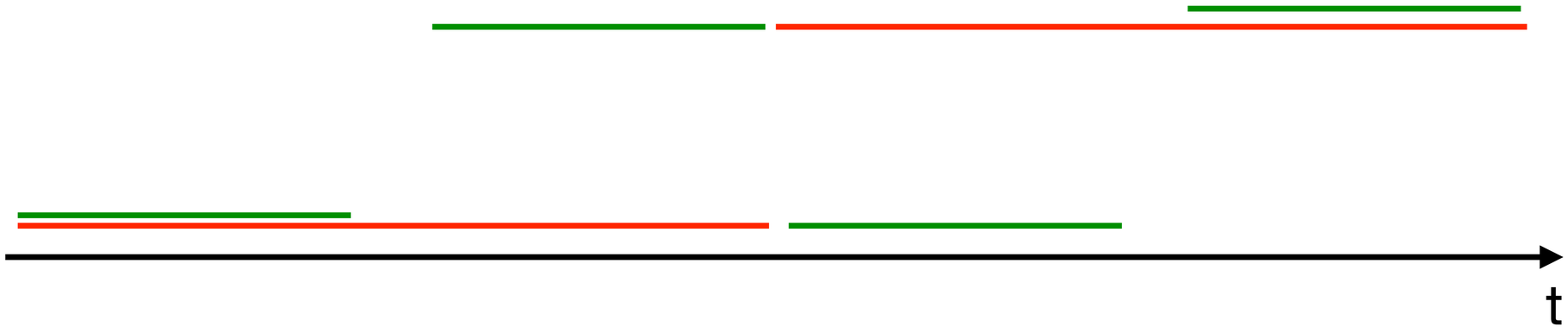


Options for position code

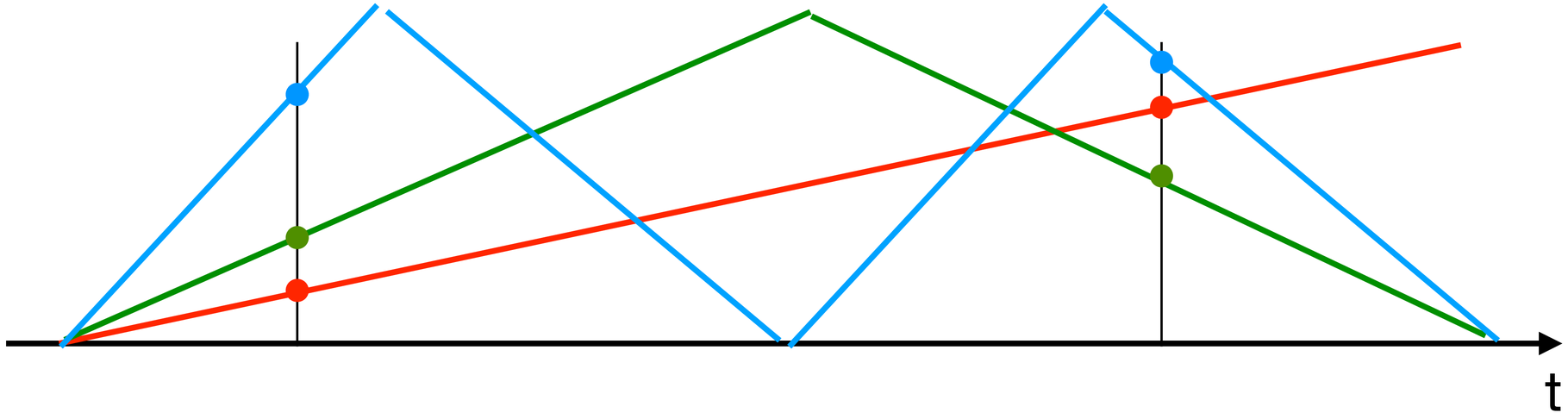


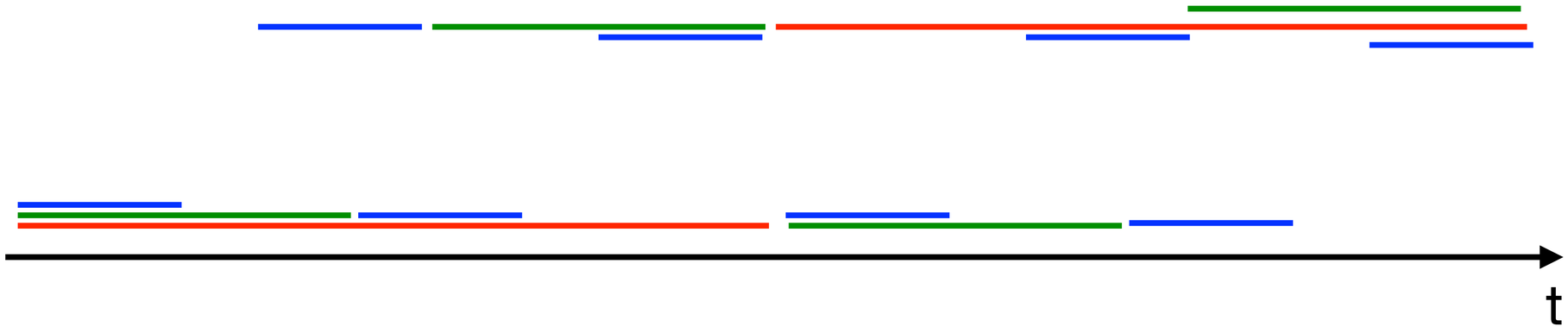
Options for position code





Options for position code





$$PE_{t,i} = \begin{bmatrix} \sin(\omega_i t) \\ \cos(\omega_i t) \end{bmatrix}$$

i=dimension

t=position

$$PE_{t,i} = \begin{bmatrix} \sin(\omega_i t) \\ \cos(\omega_i t) \end{bmatrix}$$

i=dimension

t=position

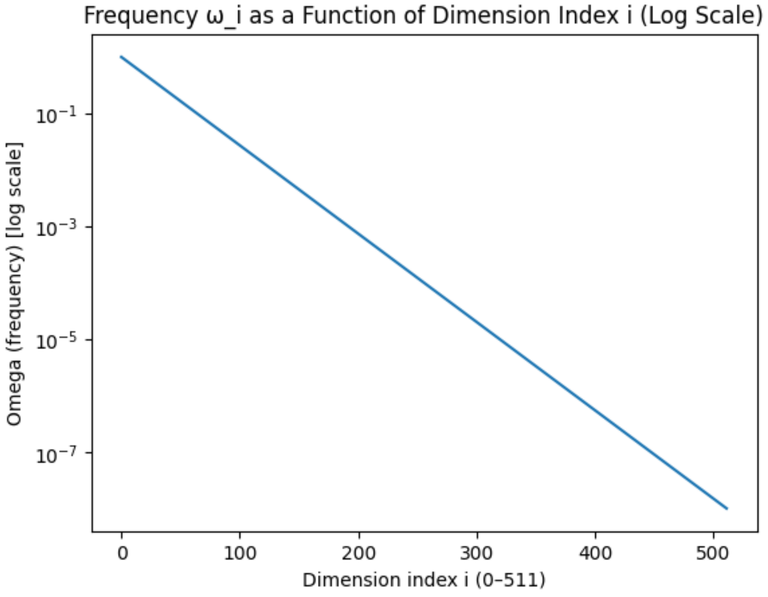
$$\omega_i = \frac{1}{10000^{2i/d}}$$

$$PE_{t,i} = \begin{bmatrix} \sin(\omega_i t) \\ \cos(\omega_i t) \end{bmatrix}$$

i=dimension

t=position

$$\omega_i = \frac{1}{10000^{2i/d}}$$



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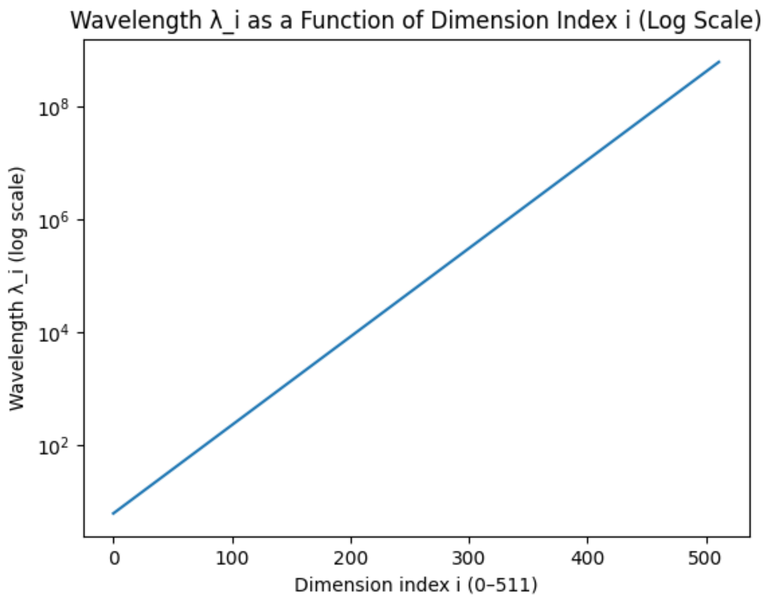

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i=dimension

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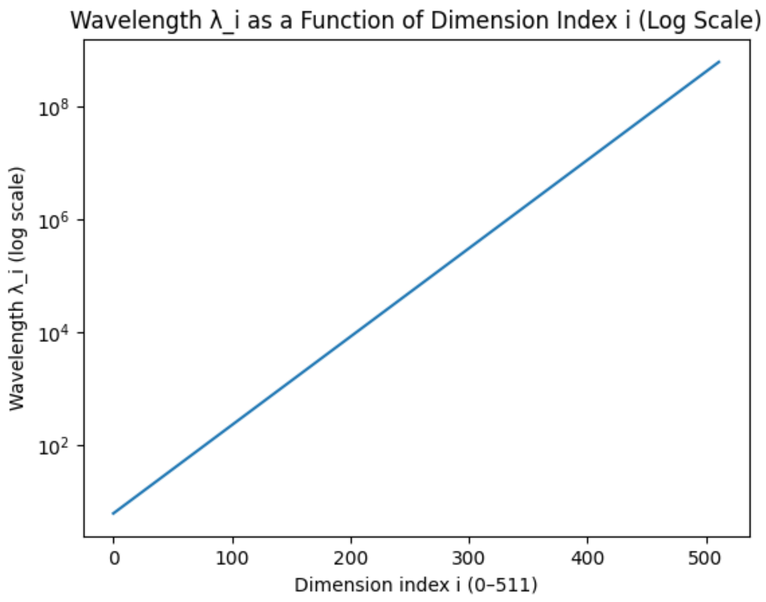


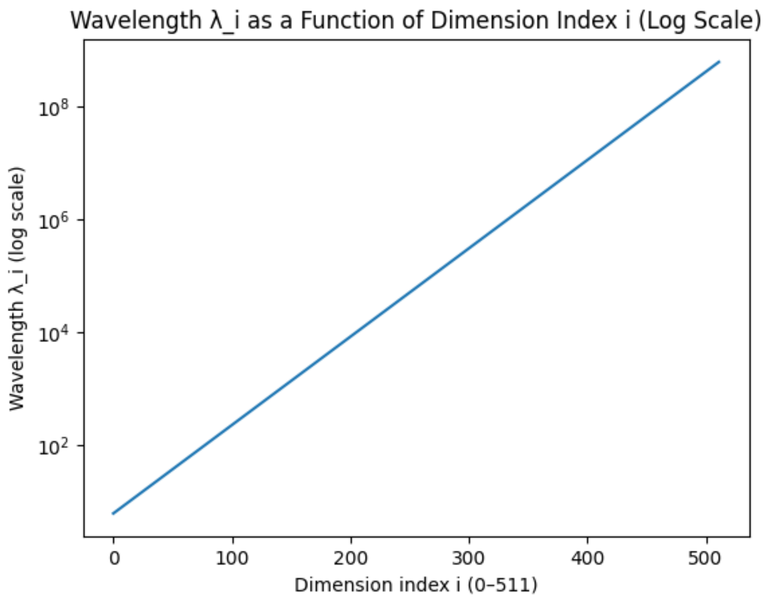
$$PE_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right)$$

$$PE_{t,i} = \begin{bmatrix} \sin(\omega_i t) \\ \cos(\omega_i t) \end{bmatrix}$$

i=dimension

t=position

$$\omega_i = \frac{1}{10000^{2i/d}}$$



$$PE_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right)$$

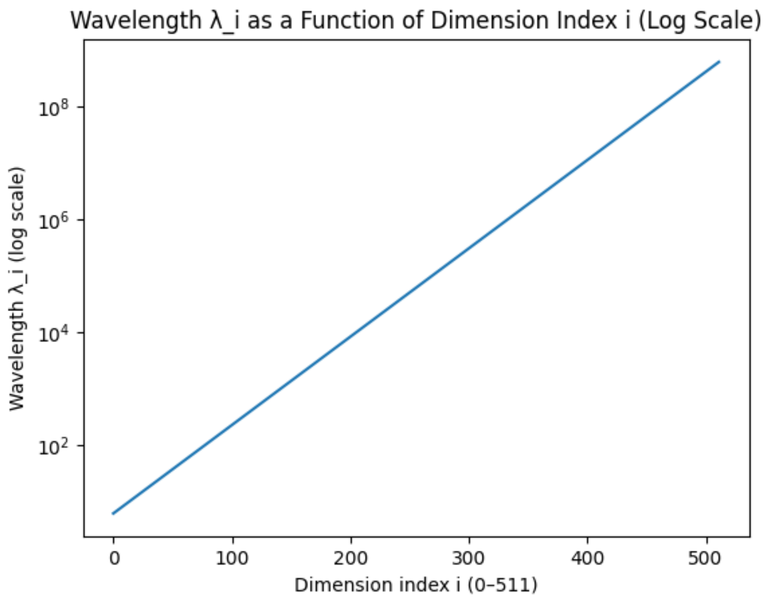
$$PE_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right)$$

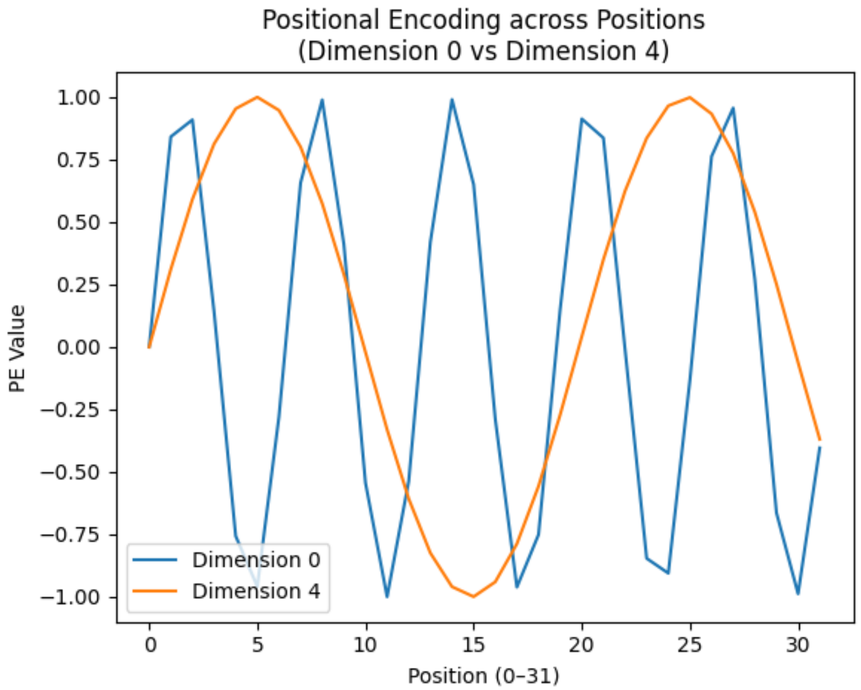
$$PE_{t,i} = \begin{bmatrix} \sin(\omega_i t) \\ \cos(\omega_i t) \end{bmatrix}$$

i=dimension

t=position

$$\omega_i = \frac{1}{10000^{2i/d}}$$





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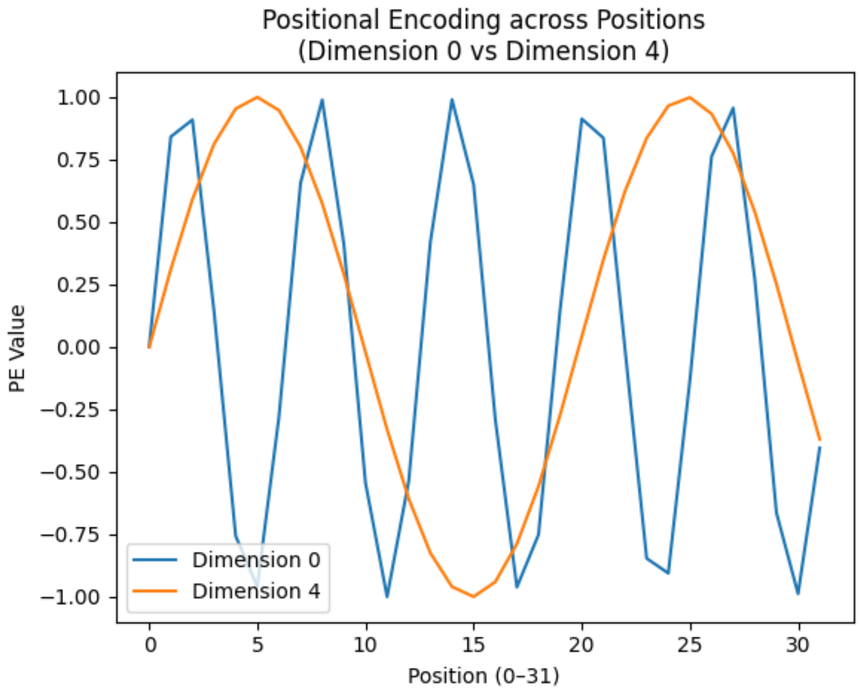
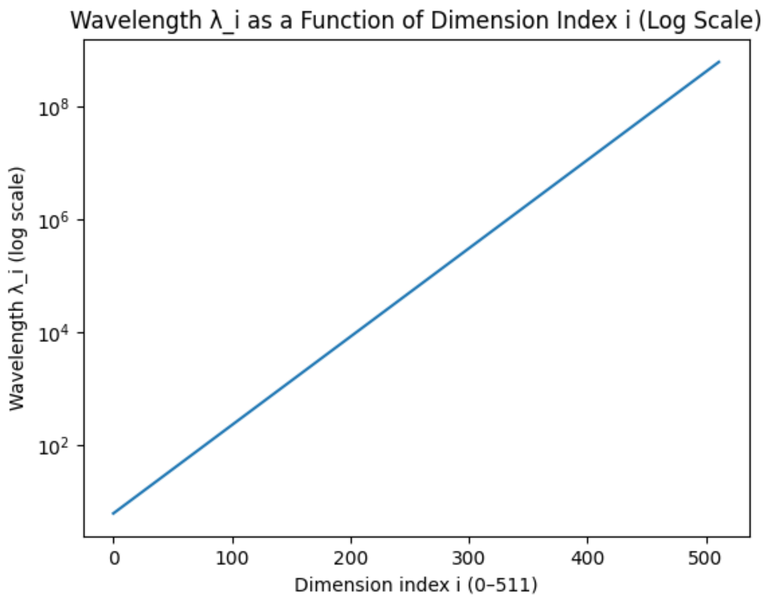
i=dimension

t=position

$$\omega_i = \frac{1}{10000^{2i/d}}$$



$$i = 0 \rightarrow \omega_i = \frac{1}{10000^0} = 1$$



$$PE_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right)$$

$$PE_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right)$$

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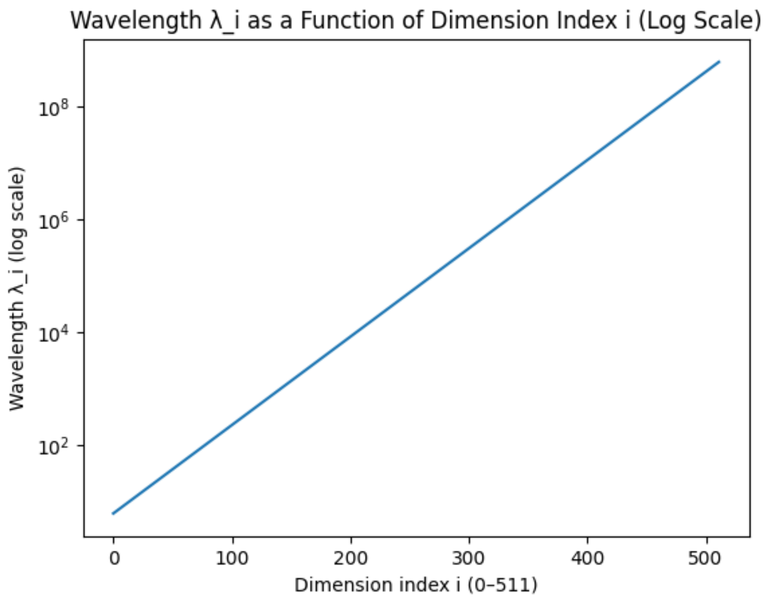
i=dimension

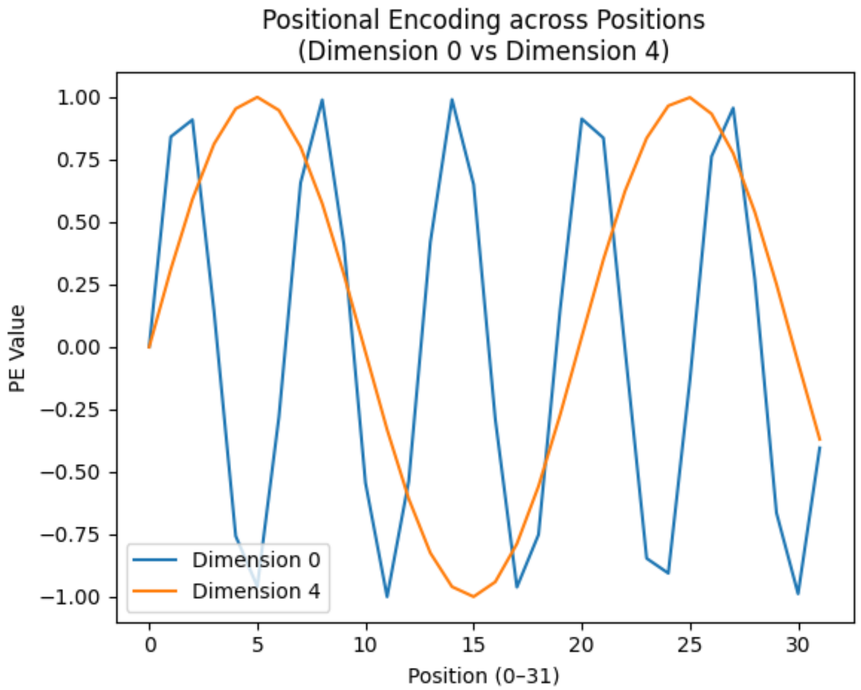
t=position

$$\omega_i = \frac{1}{10000^{2i/d}}$$



$$PE_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right)$$

$$PE_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right)$$



$$i = 0 \rightarrow \omega_i = \frac{1}{10000^0} = 1$$

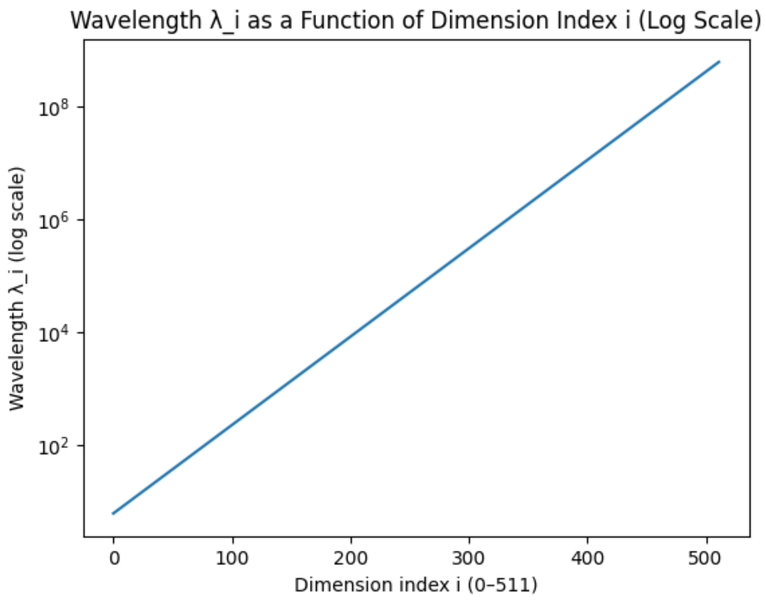
$$i = 4 \rightarrow \omega_i = \frac{1}{10000^{8/512}} = 0.86$$

$$PE_{t,i} = \begin{bmatrix} \sin(\omega_i t) \\ \cos(\omega_i t) \end{bmatrix}$$

i=dimension

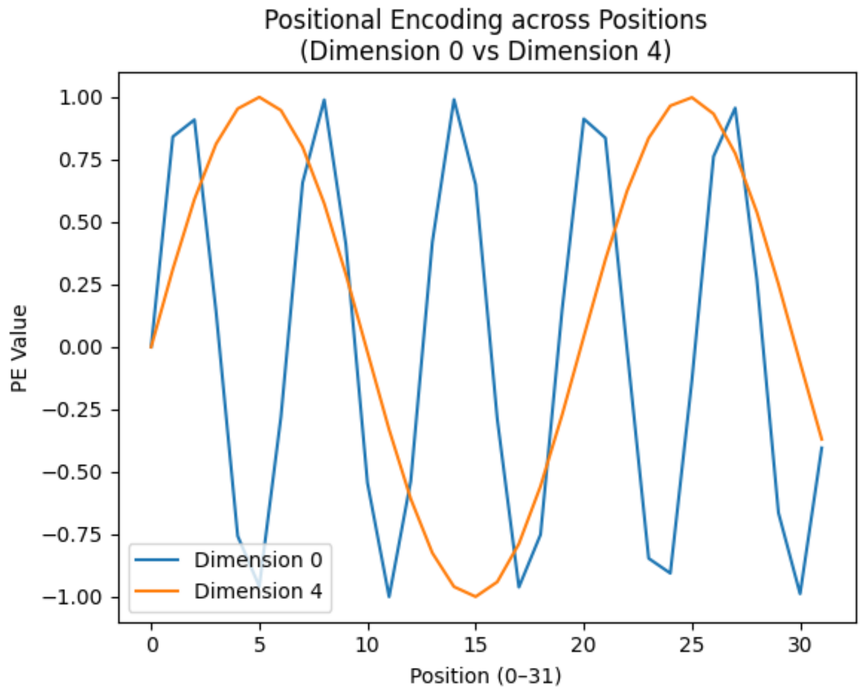
t=position

$$\omega_i = \frac{1}{10000^{2i/d}}$$



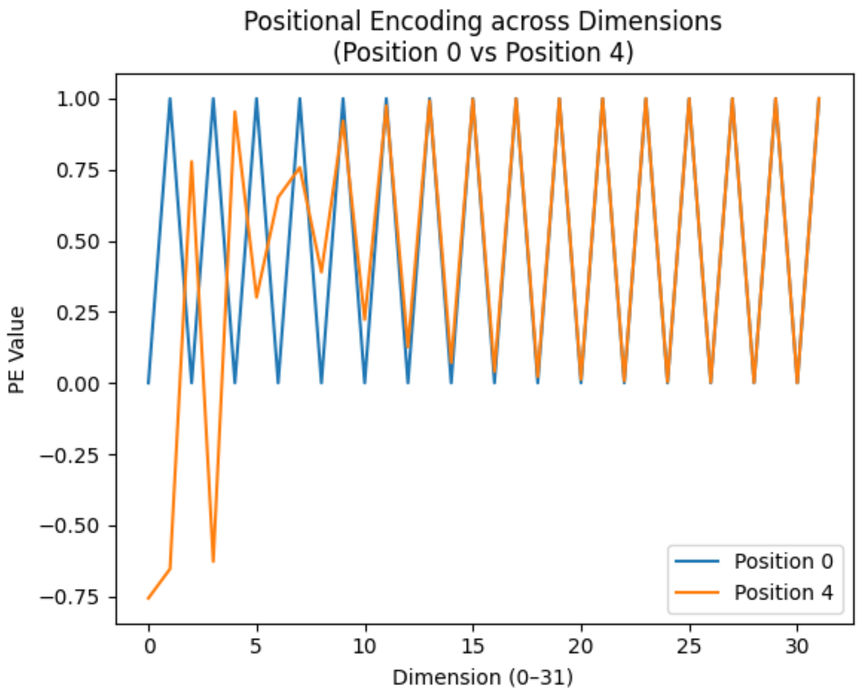
$$PE_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right)$$

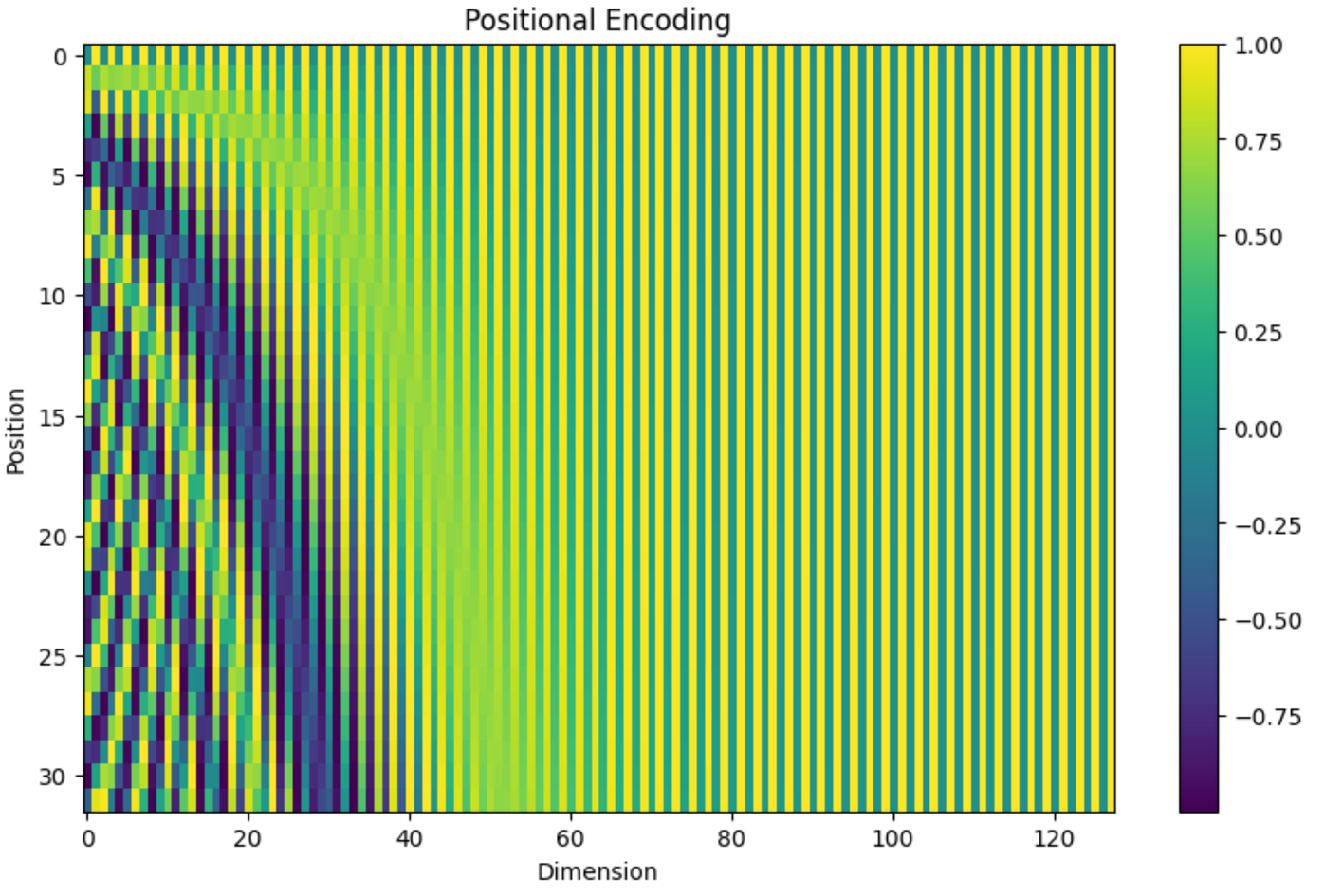
$$PE_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right)$$



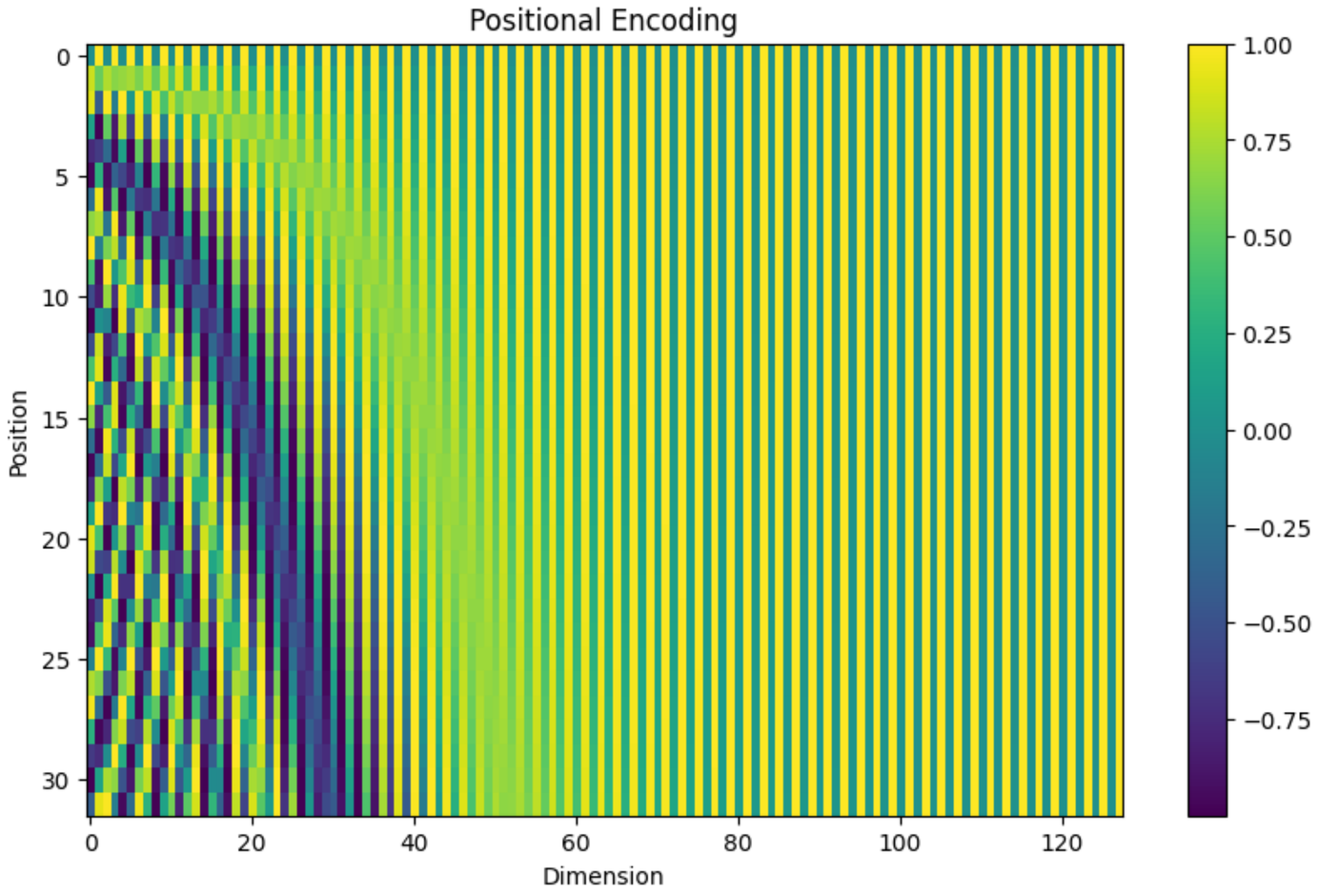
$$i = 0 \rightarrow \omega_i = \frac{1}{10000^0} = 1$$

$$i = 4 \rightarrow \omega_i = \frac{1}{10000^{8/512}} = 0.86$$



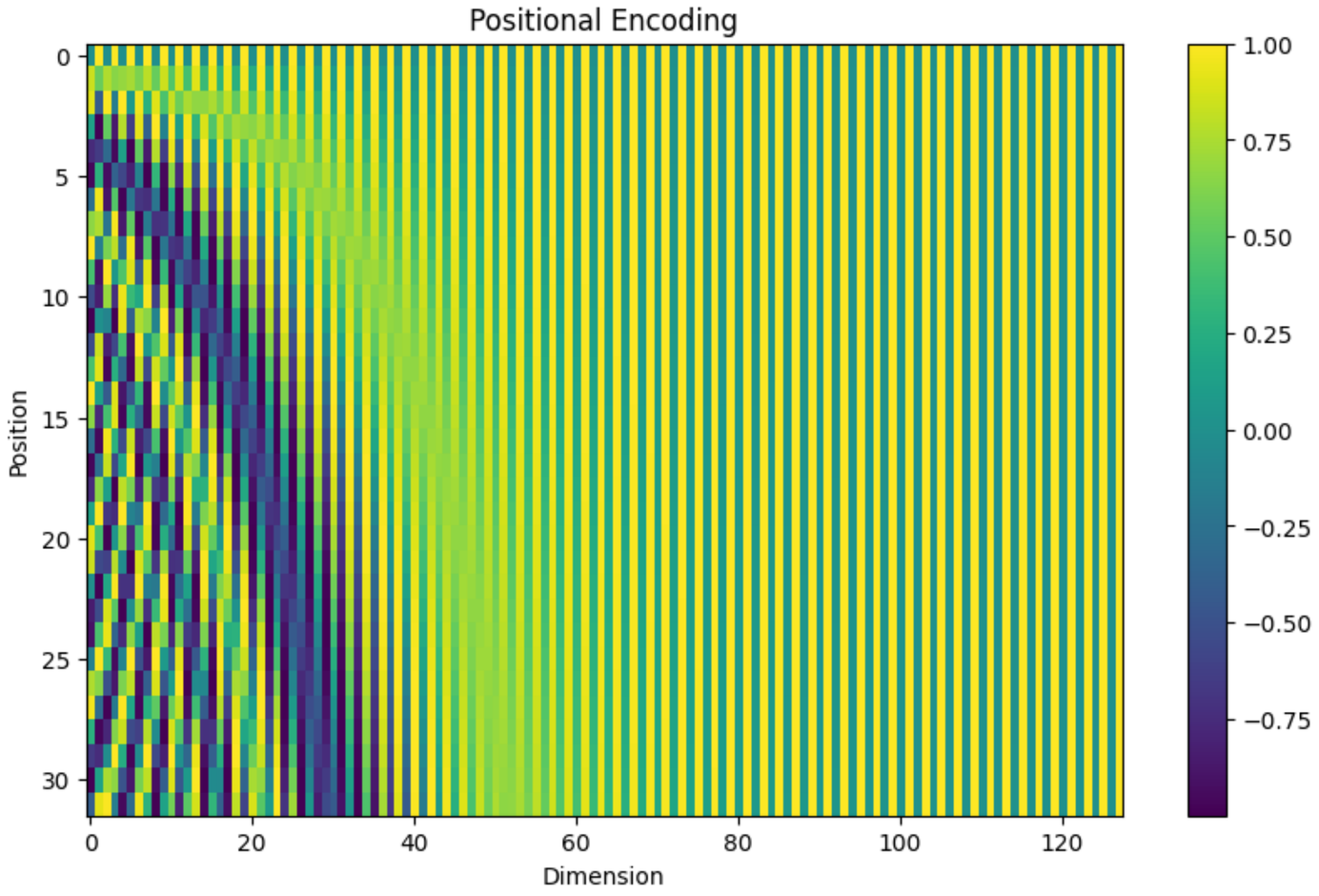

Positional Encoding
Position
Dimension
0
5
10
15
20
25
30
0
20
40
60
80
100
120
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

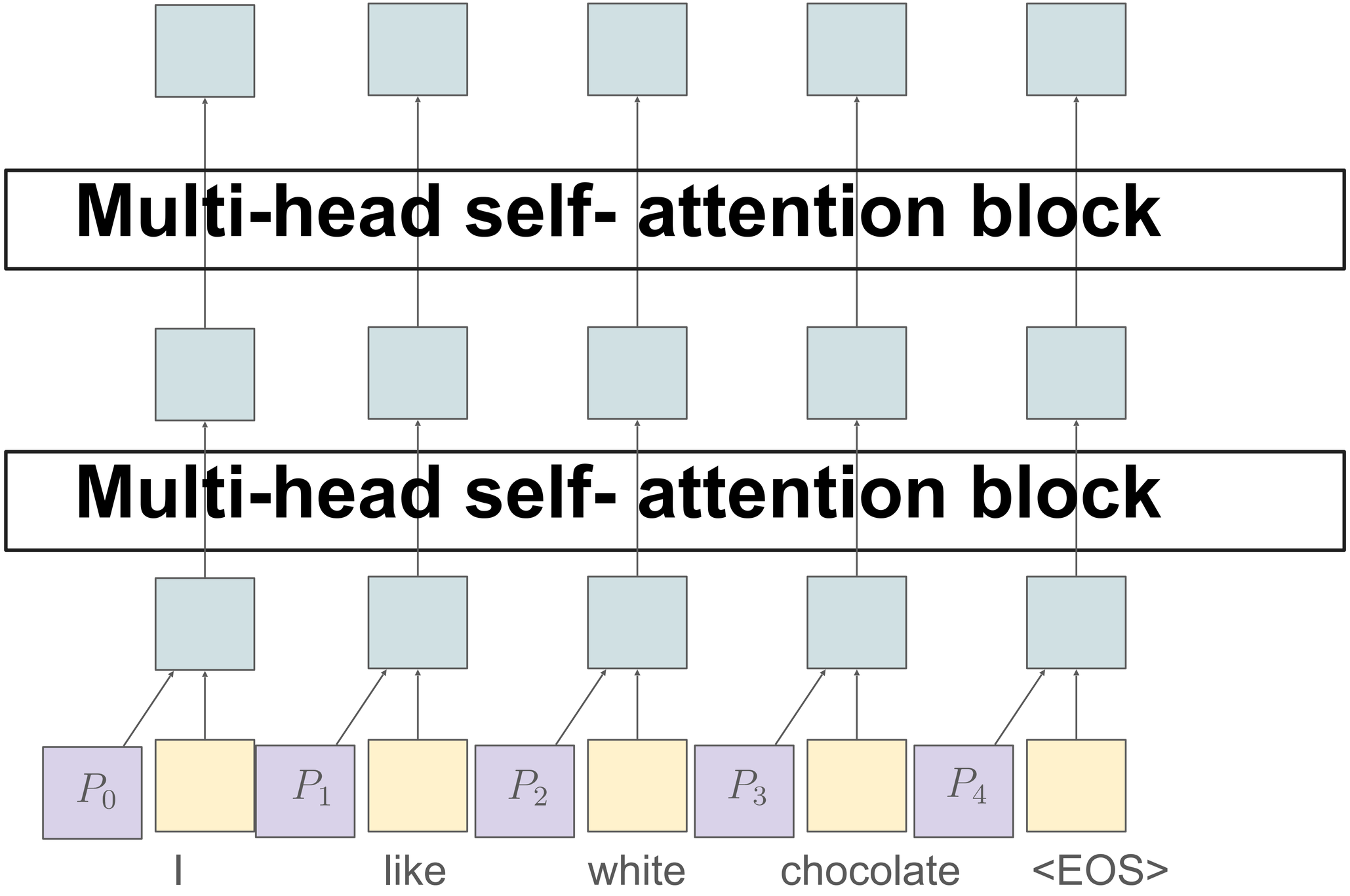
$$PE_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right)$$



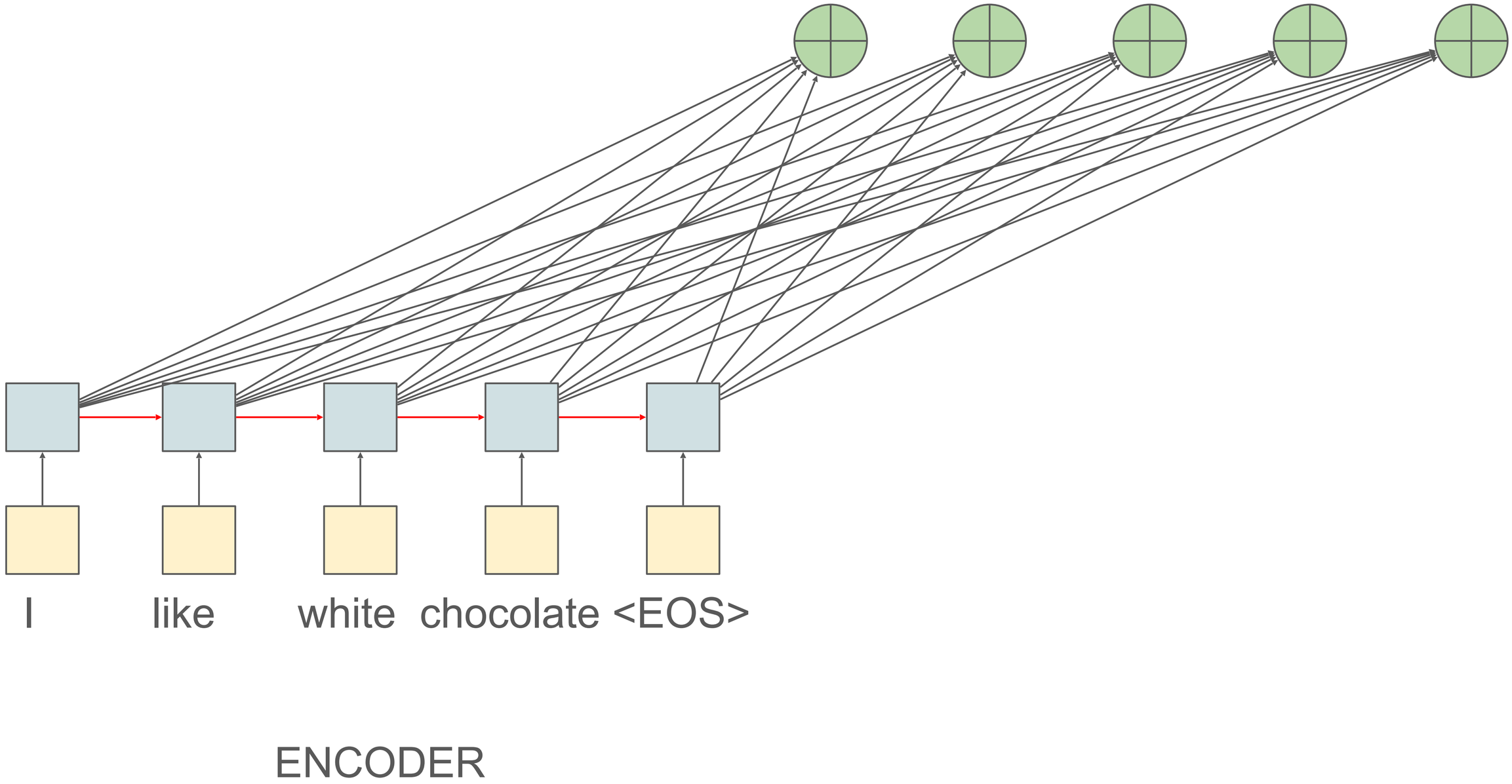
$$PE_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right)$$

$$PE_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right)$$

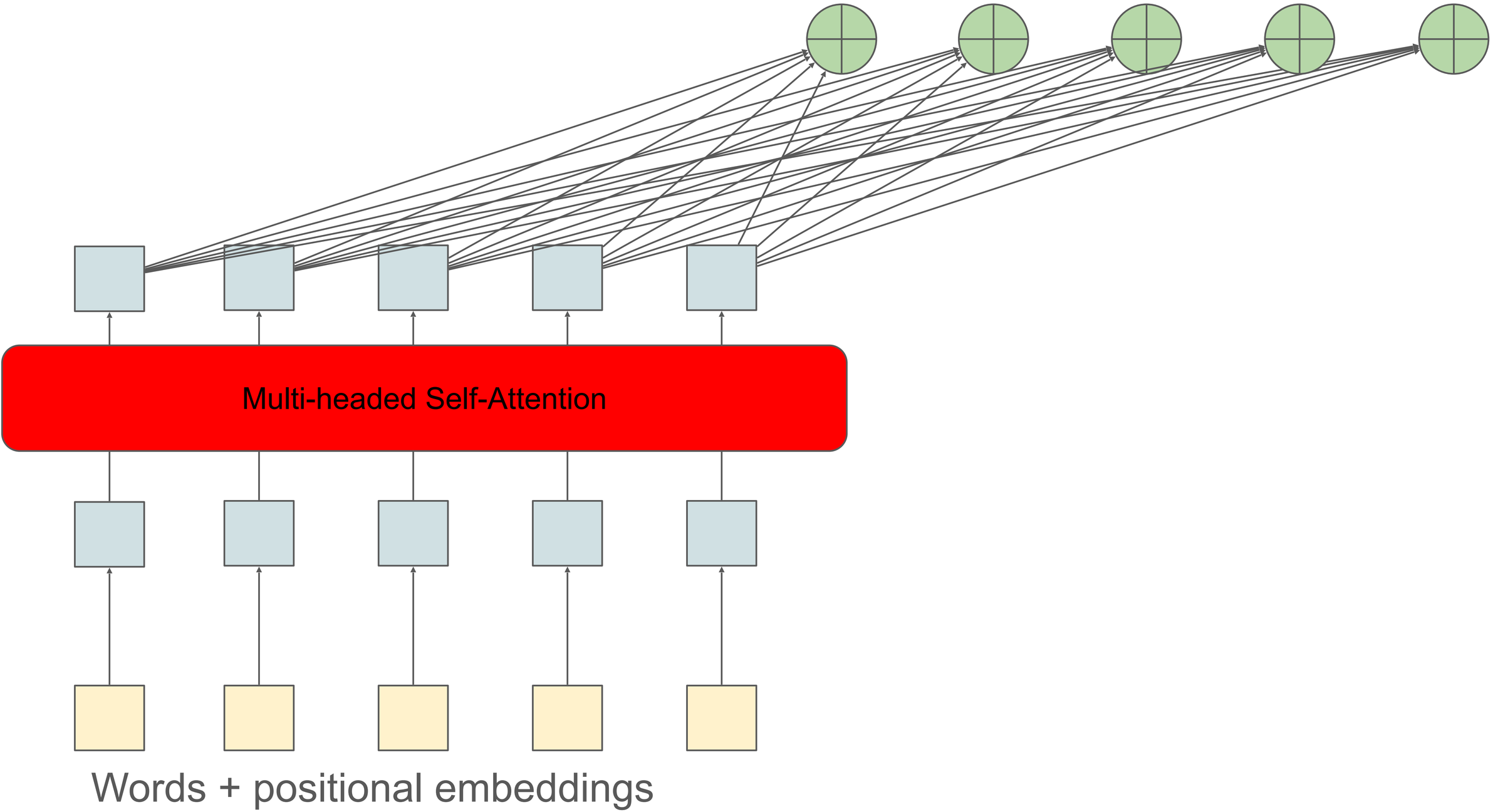


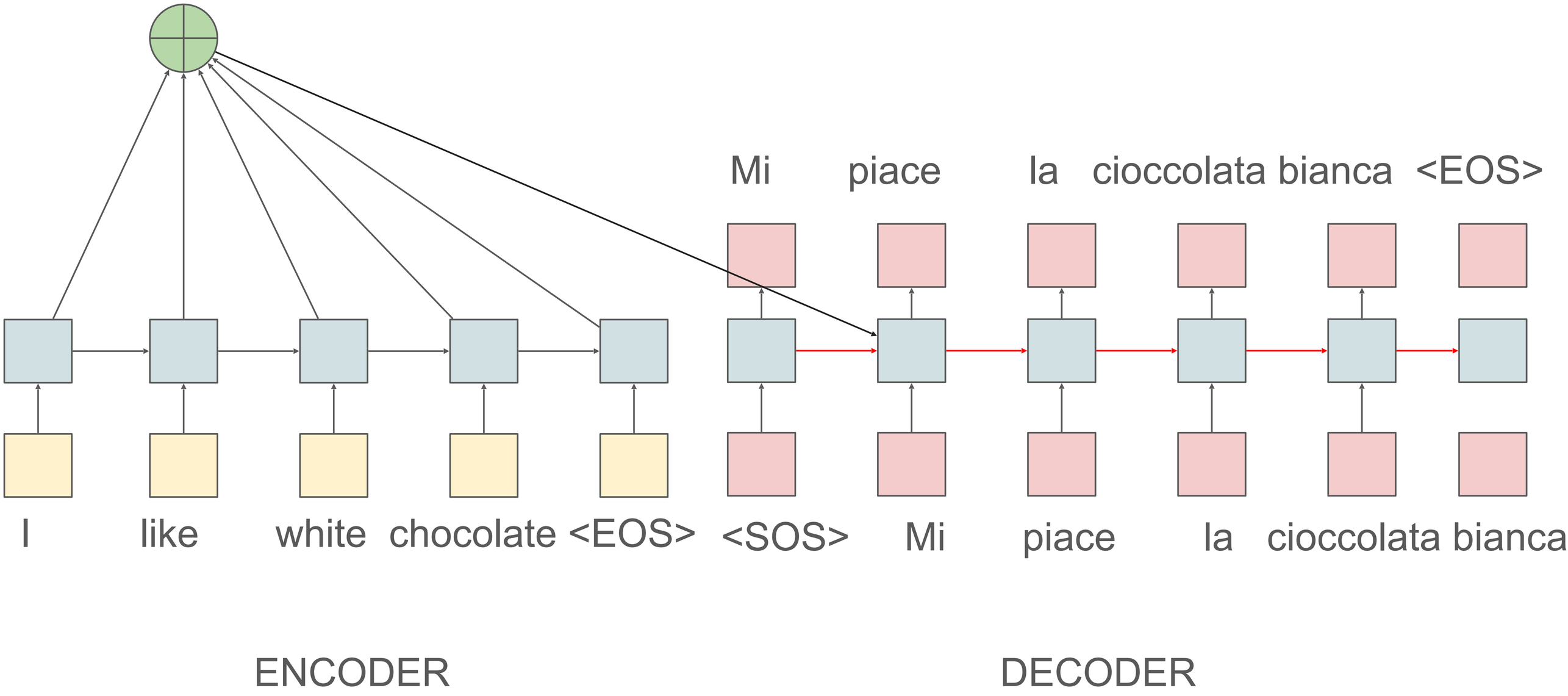

Multi-head self- attention block
Multi-head self- attention block
P_0
P_1
P_2
P_3
P_4
I
like
white
chocolate
<EOS>

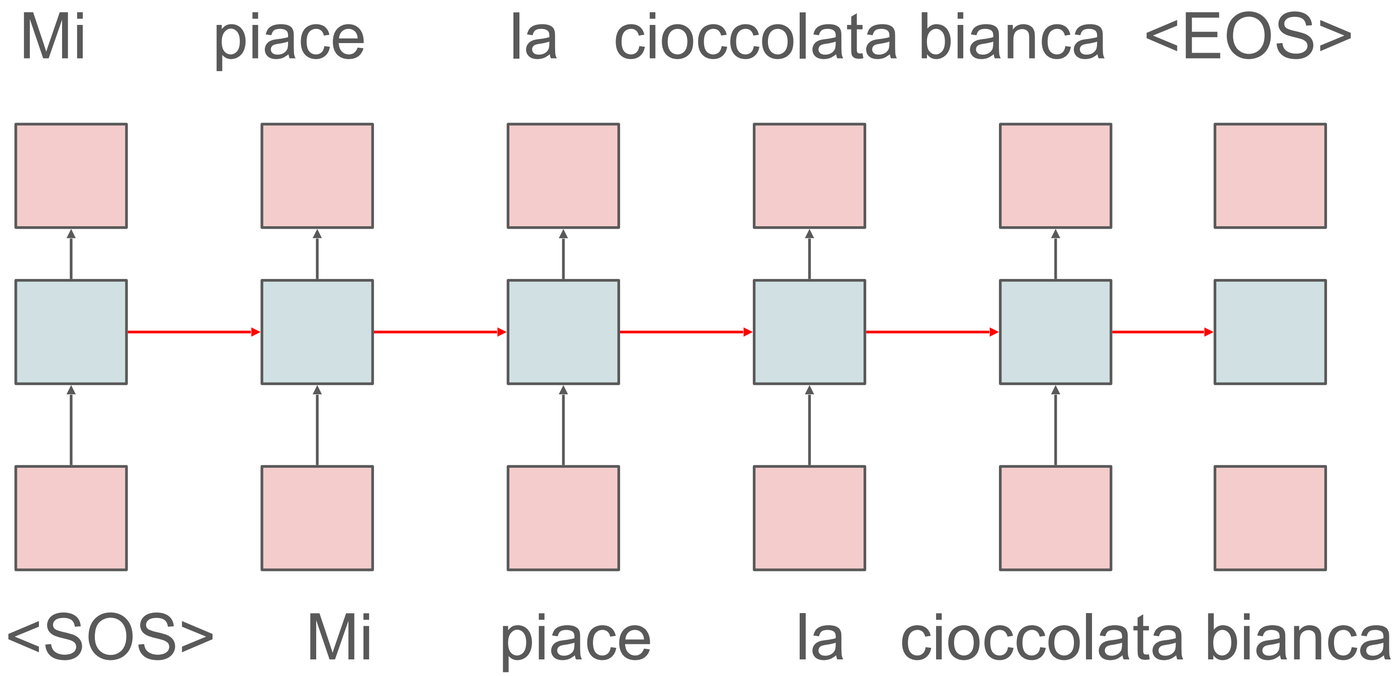
Decoder (Masked) self-attention



ENCODER


Multi-headed Self-Attention
Words + positional embeddings


Mi
piace
la
cioccolata
bianca
<EOS>
I
like
white
chocolate
<EOS>
<SOS>
Mi
piace
la
cioccolata
bianca
ENCODER
DECODER


Mi
piace
la
cioccolata
bianca
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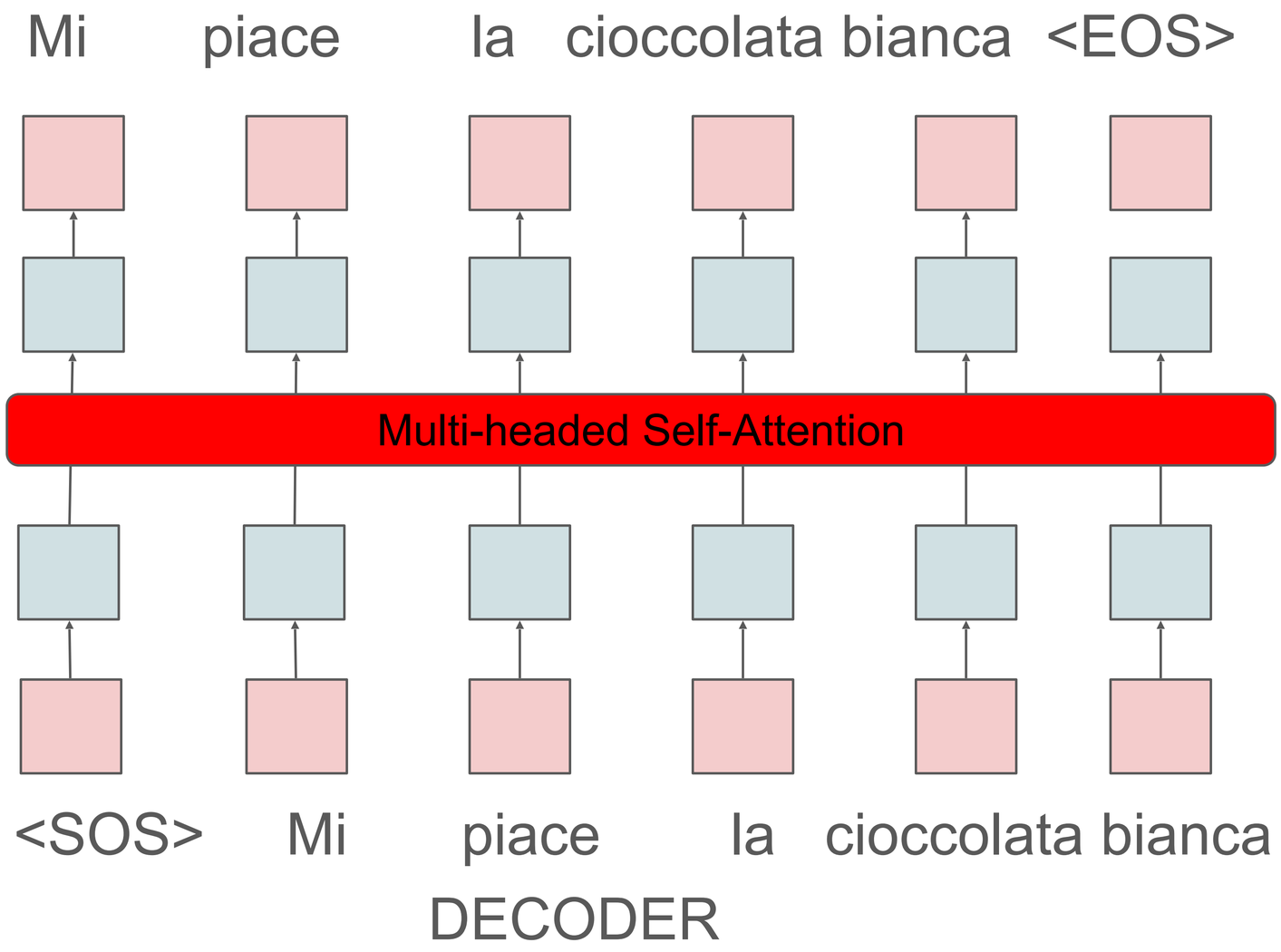
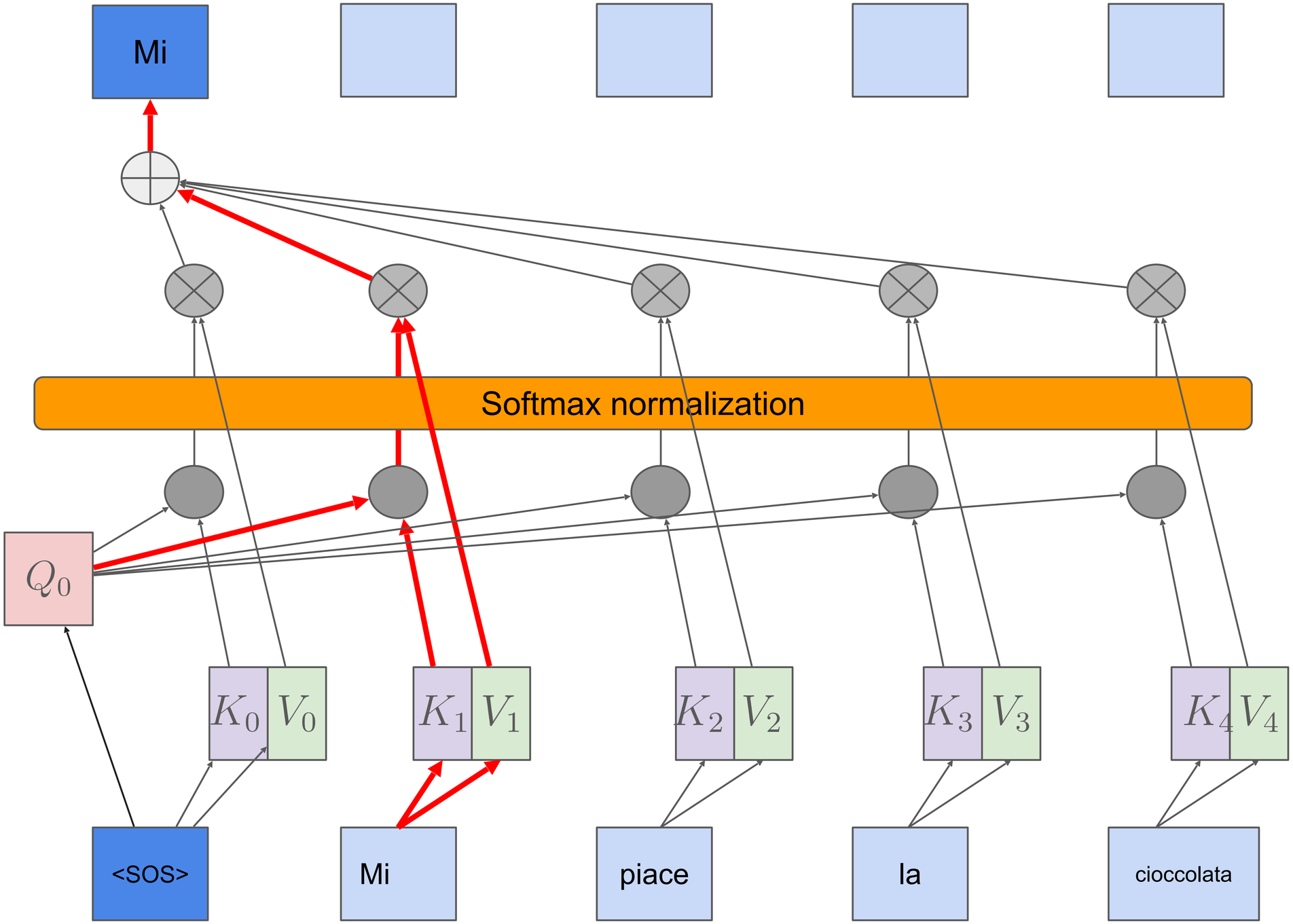
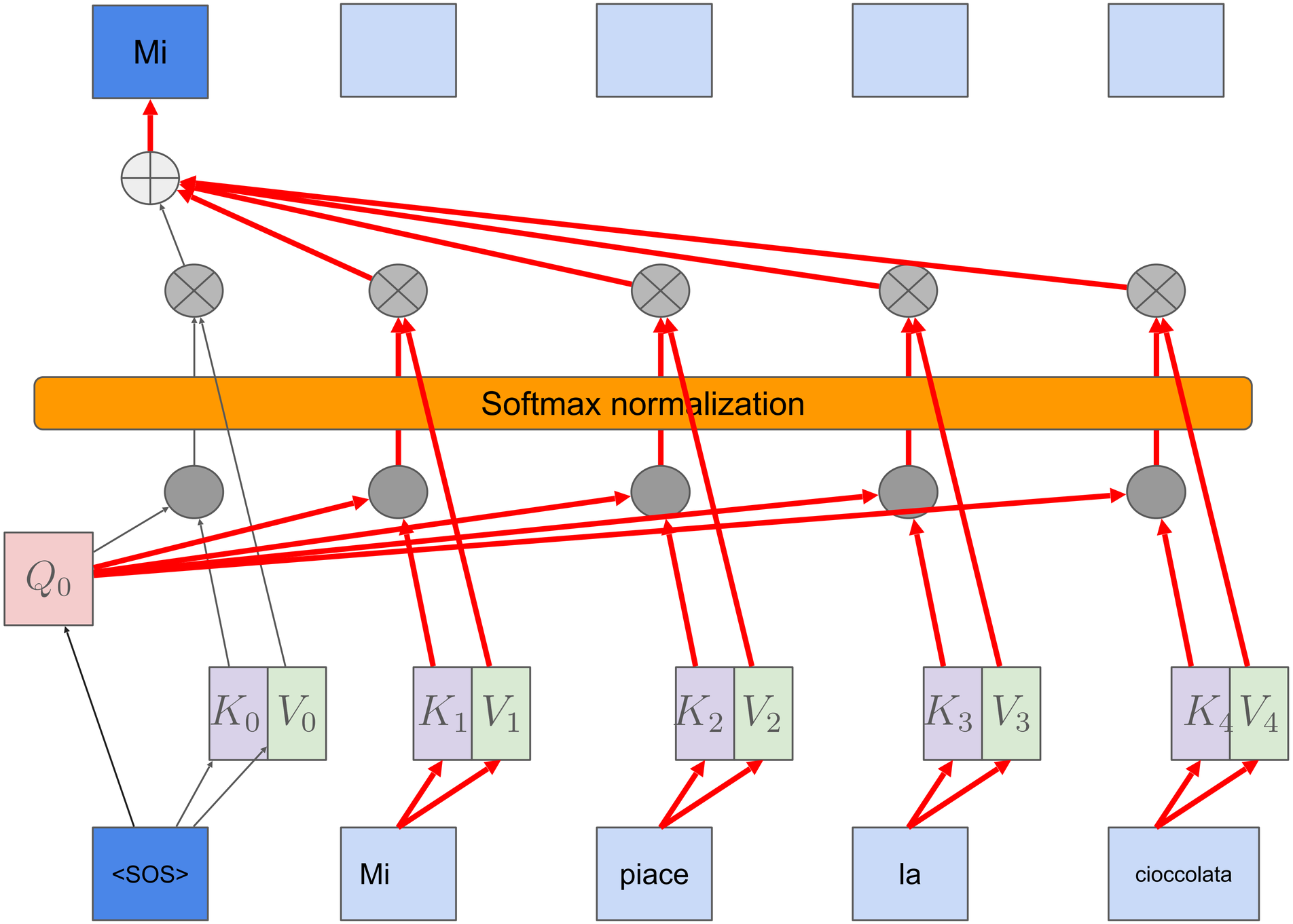
<SOS>
Mi
piace
la
cioccolata
bianca

DECODER


Mi
piace
la
cioccolata
bianca
<EOS>
Multi-headed Self-Attention
<SOS>
Mi
piace
la
cioccolata
bianca
DECODER


Mi
Softmax normalization
Q_0
K_0 V_0
K_1 V_1
K_2 V_2
K_3 V_3
K_4 V_4
<SOS>
Mi
piace
la
cioccolata


Mi
Softmax normalization
Q_0
K_0 V_0
K_1 V_1
K_2 V_2
K_3 V_3
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Mi
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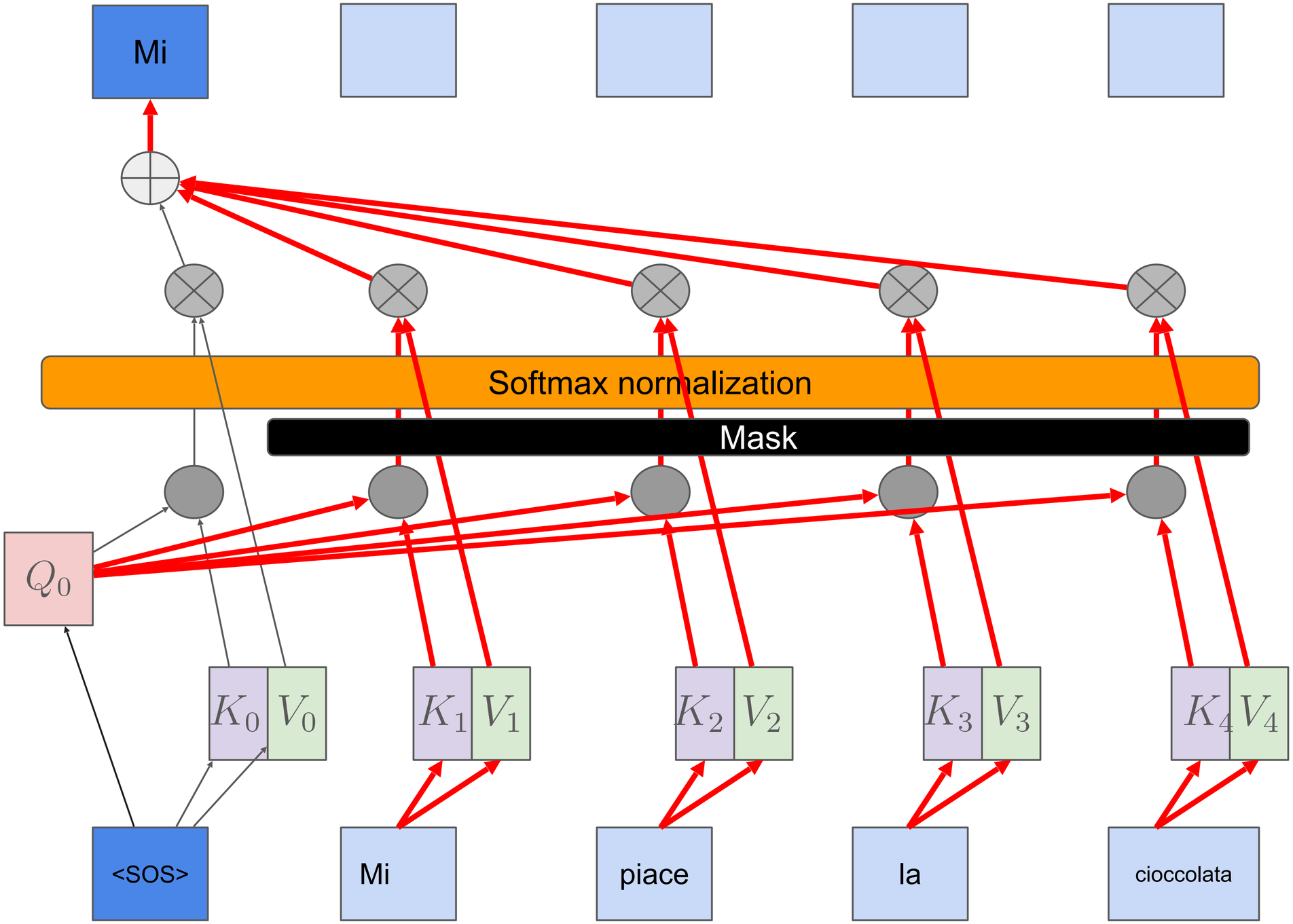
cioccolata


Mi
Softmax normalization
Mask
Q_0
K_0 V_0
K_1 V_1
K_2 V_2
K_3 V_3
K_4 V_4
<SOS>
Mi
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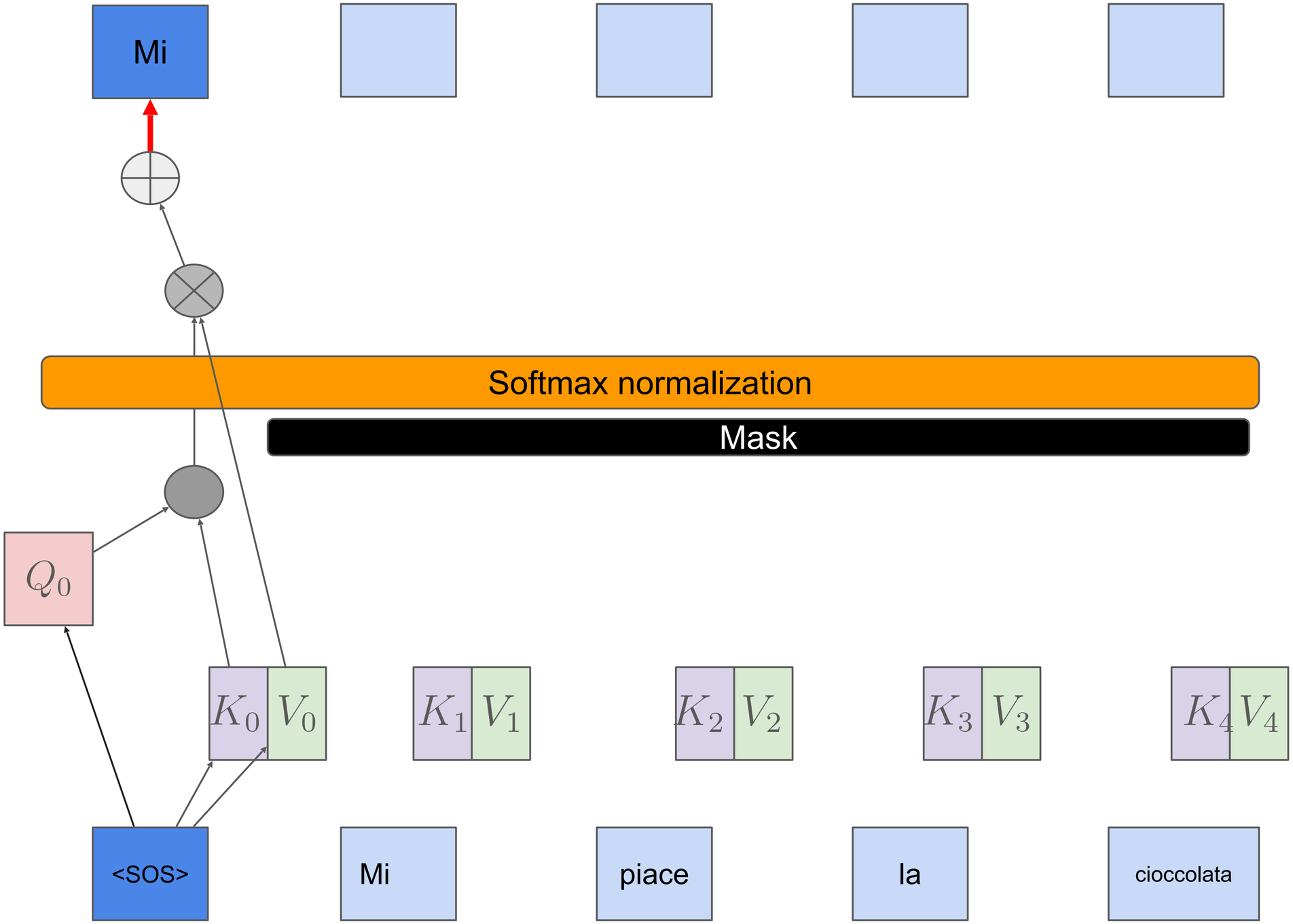
la
cioccolata


Mi
Softmax normalization
Mask
Q_0
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K_1 V_1
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K_3 V_3
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Mi
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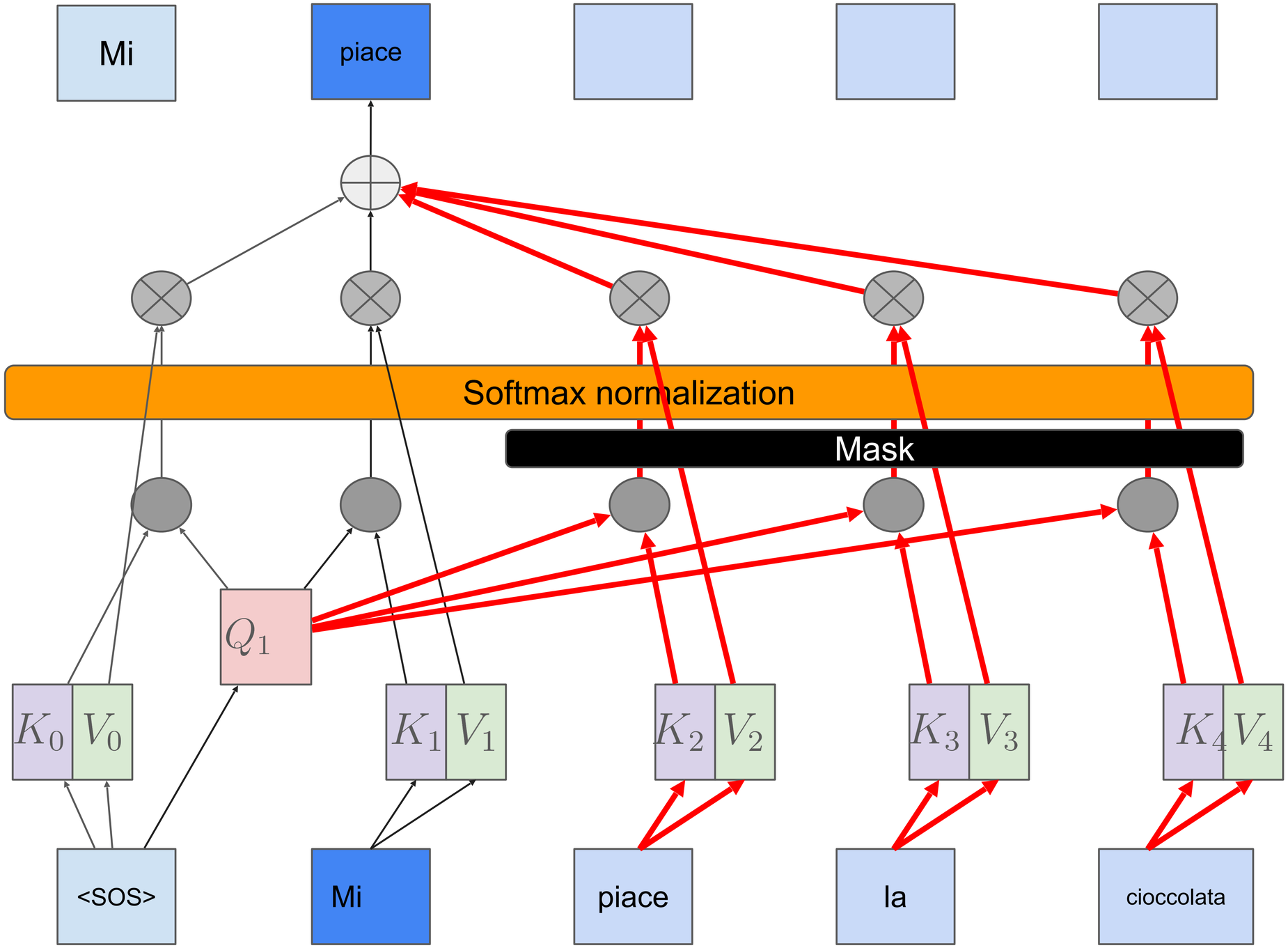
Mi
piace
Softmax normalization
Mask
Q_1
K_0 V_0
K_1 V_1
K_2 V_2
K_3 V_3
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<SOS>
Mi
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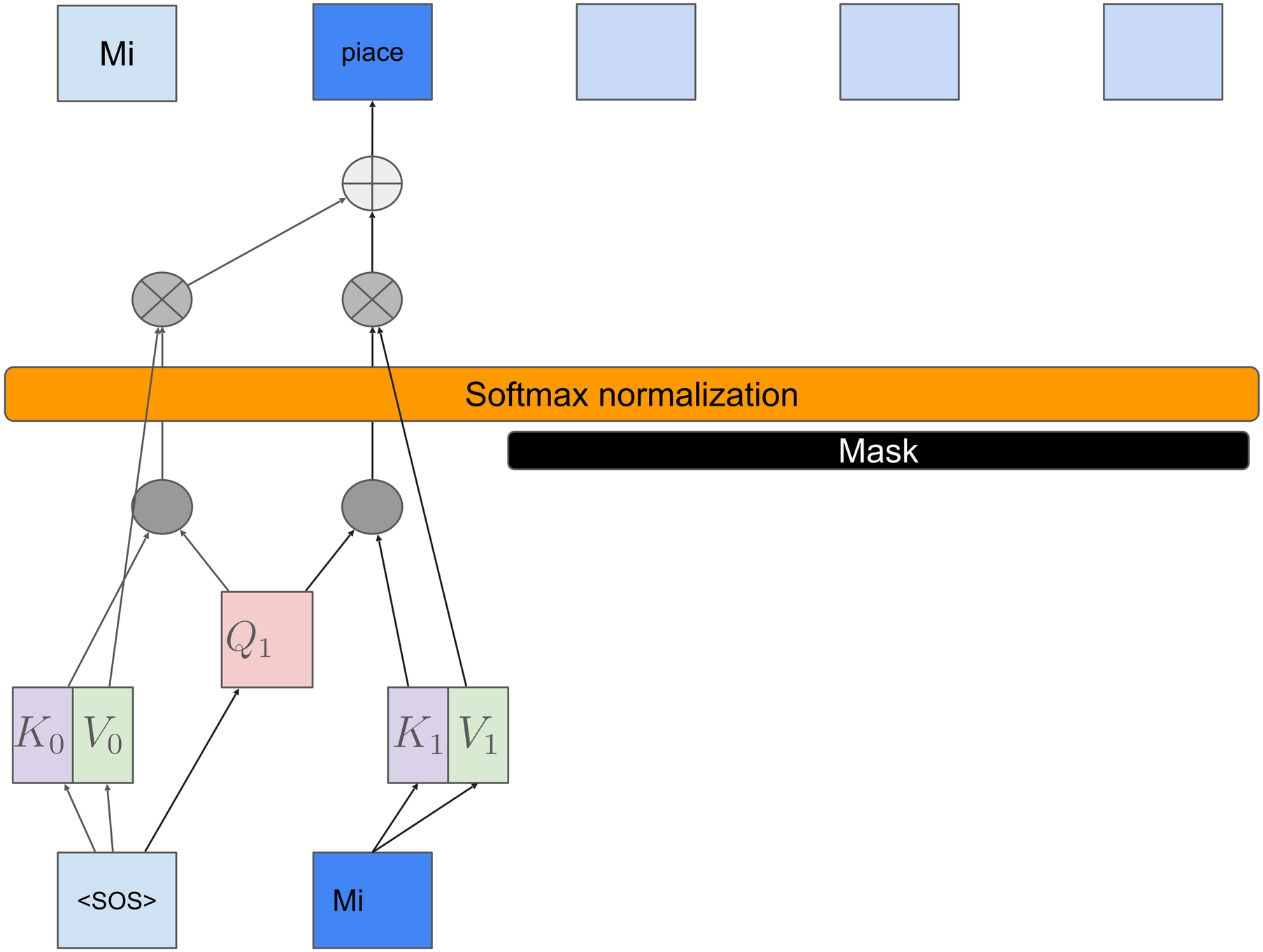
cioccolata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piace
Softmax normalization
Mask
Q_1
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V_0
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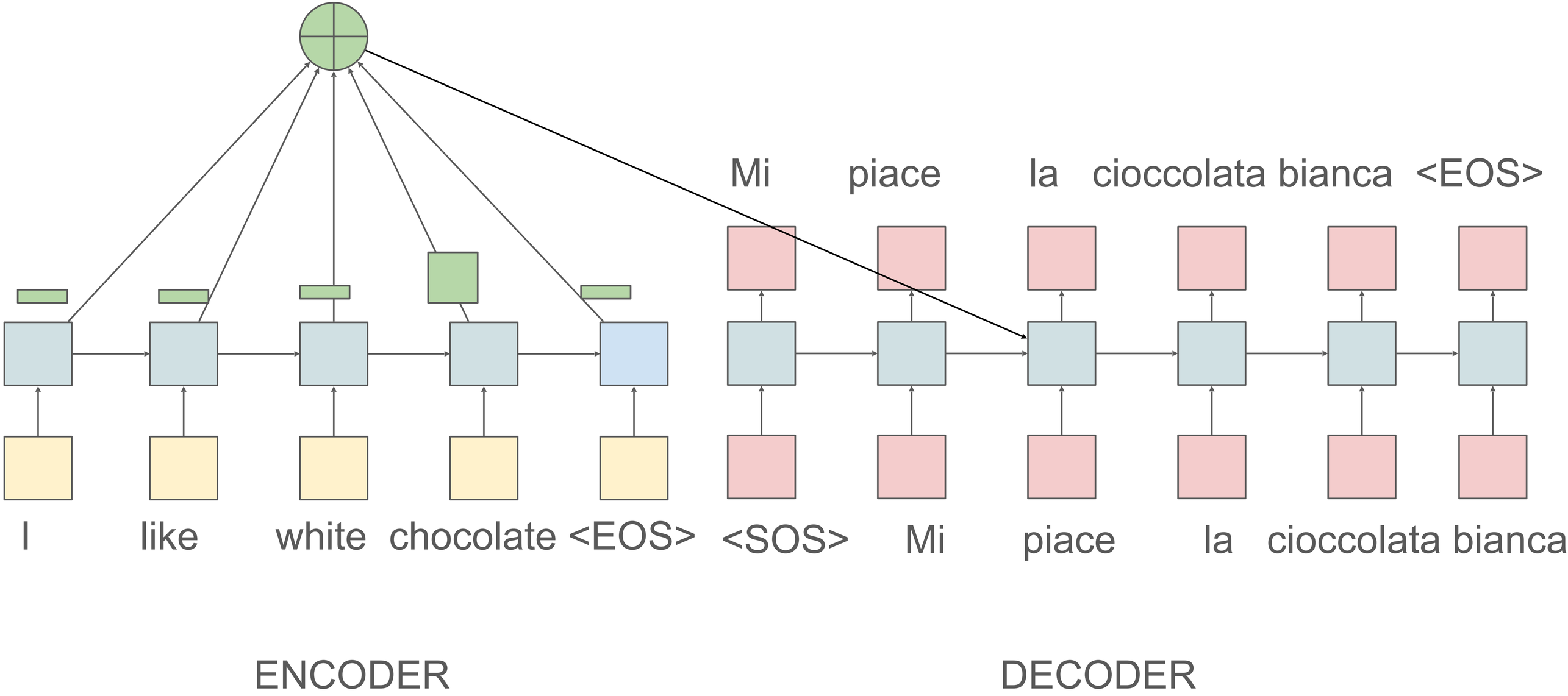
V_1
<SOS>
Mi


Mi
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cioccolata
bianca
<EOS>
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like
white
chocolate
<EOS>
<SOS>
Mi
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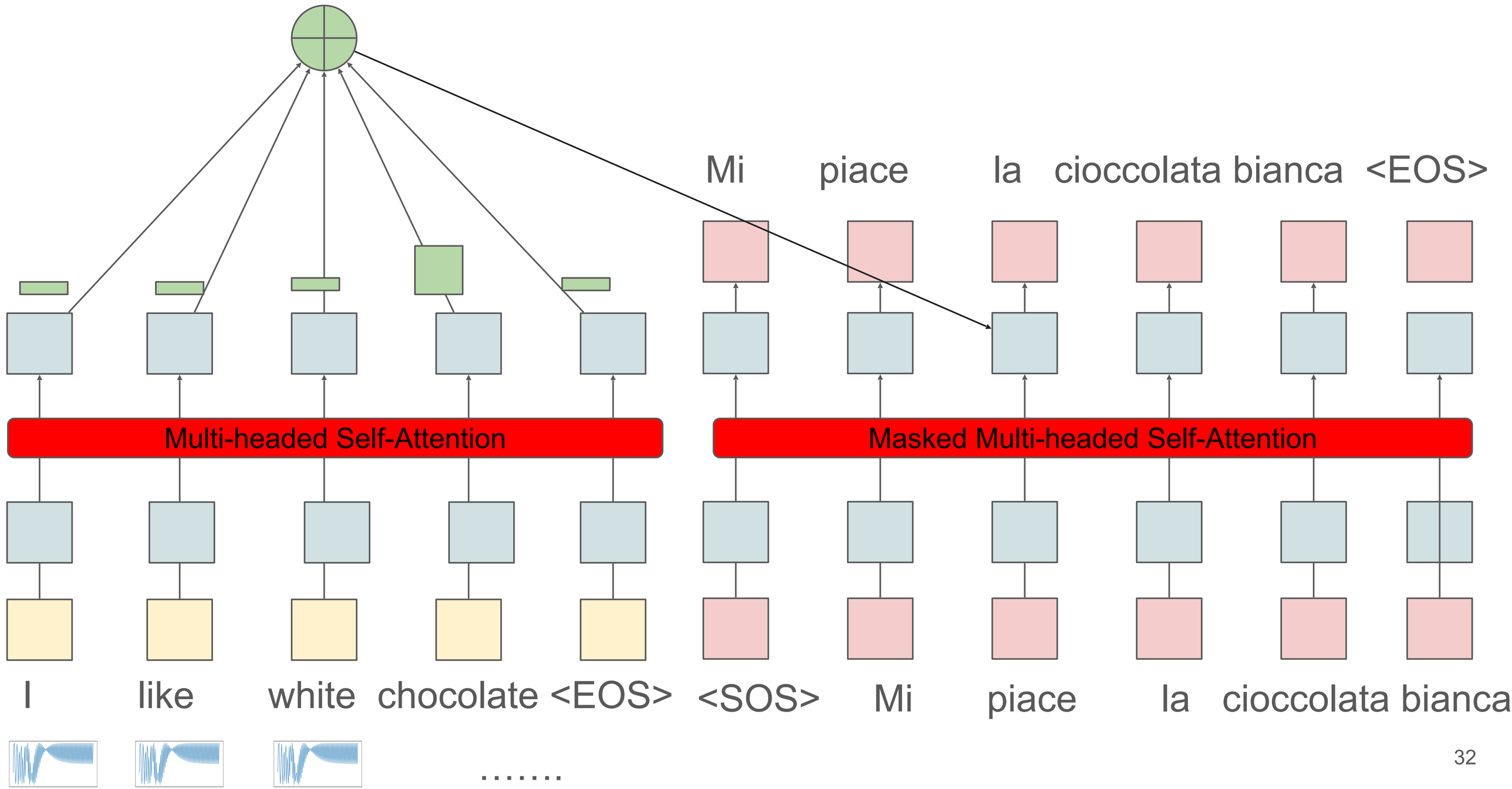
la
cioccolata
bianca
ENCODER
DECODER


Mi
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cioccolata
bianca
<EOS>
Multi-headed Self-Attention
Masked Multi-headed Self-Attention
I
like
white
chocolate
<EOS>
<SOS>
Mi
piace
la
cioccolata
bianca
.....

Transformers

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin* ‡
illia.polosukhin@gmail.com

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Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com

Noam Shazeer* Google Brain noam@google.com

Niki Parmar* Google Research nikip@google.com

Jakob Uszkoreit* Google Research usz@google.com

Llion Jones* Google Research llion@google.com

Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

Łukasz Kaiser* Google Brain lukaszkaiser@google.com

Illia Polosukhin* ‡ illia.polosukhin@gmail.com

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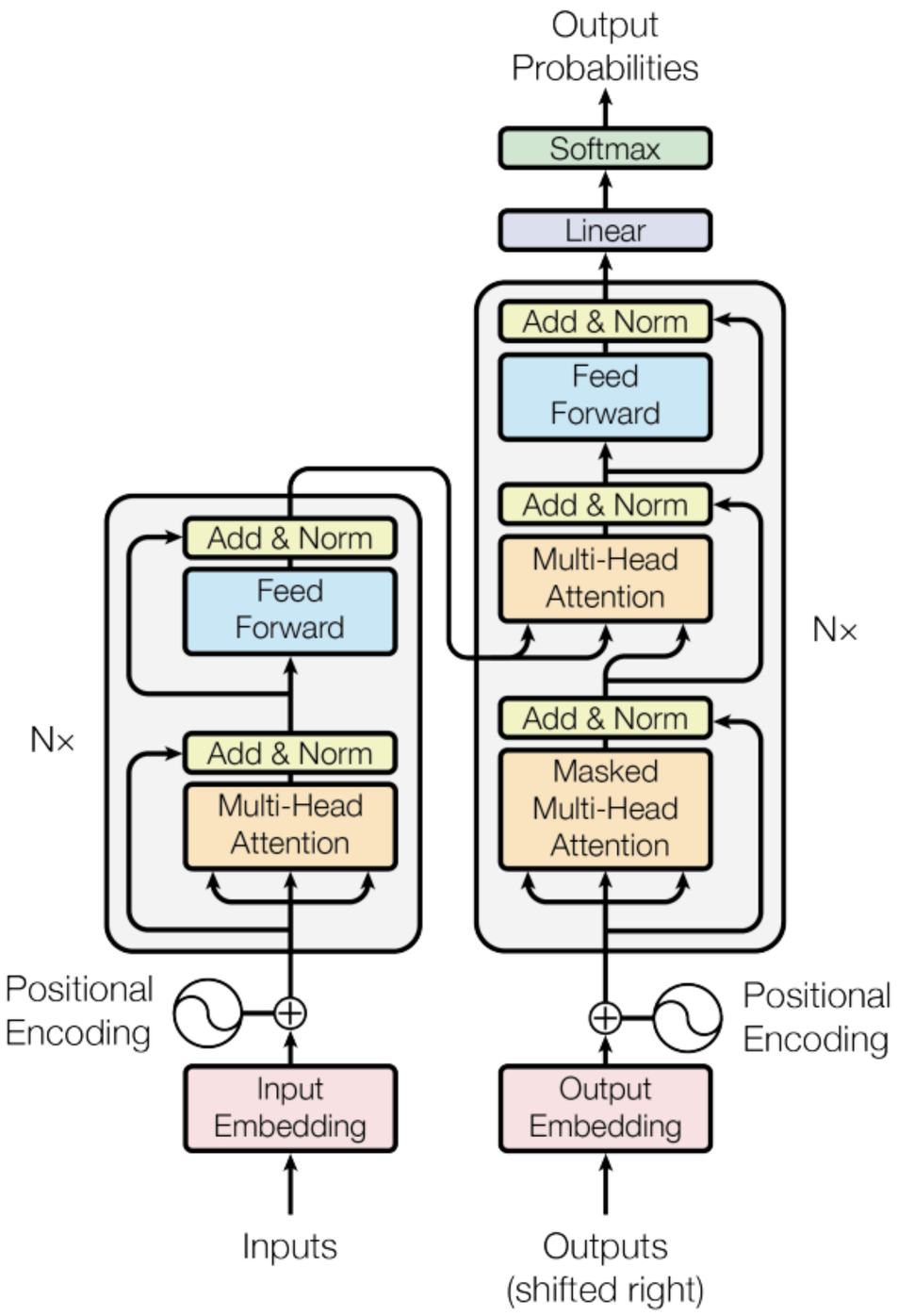
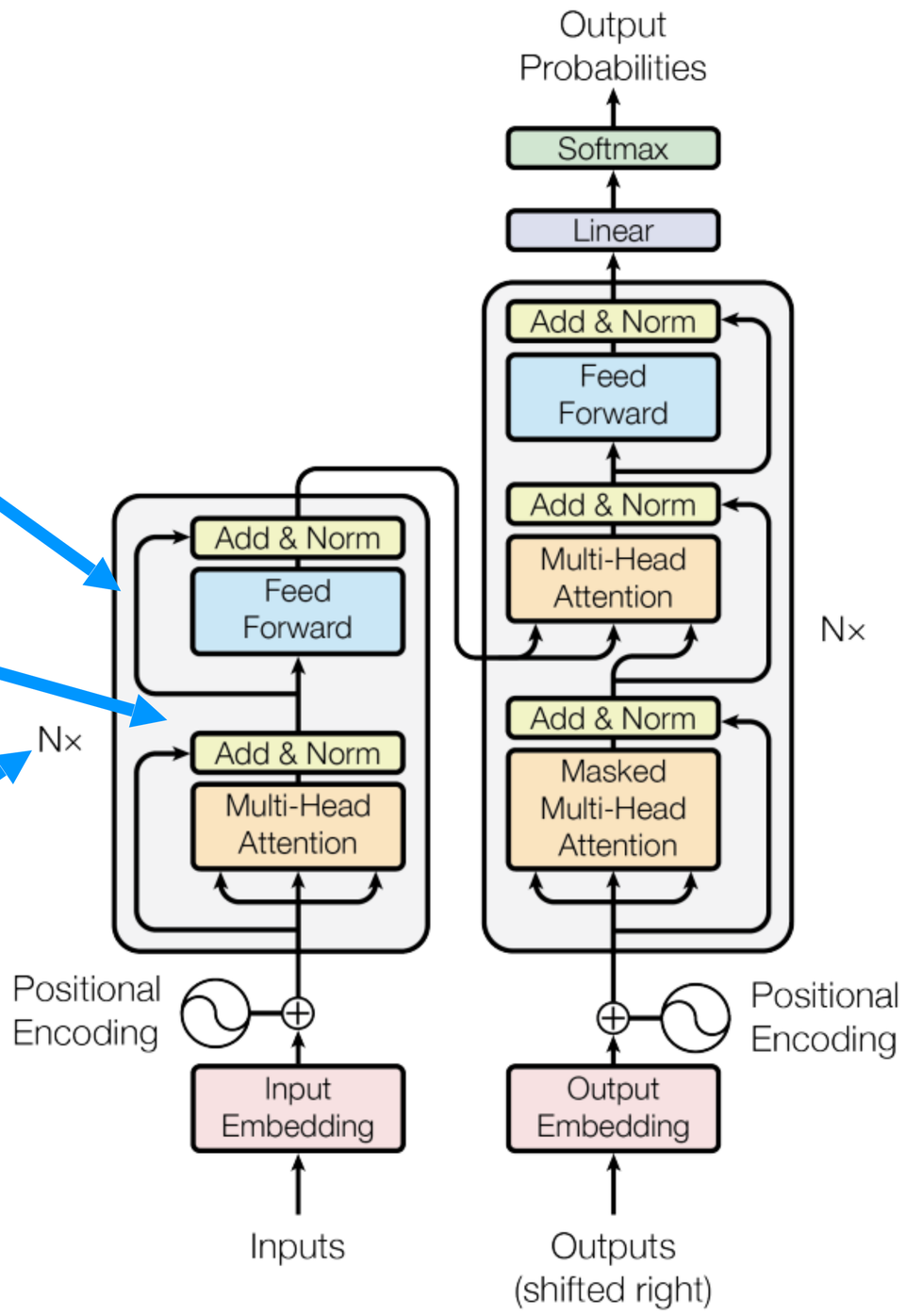


Residual connection
(See resNets)

Layer normalization

Repeated blocks